

Absolute Purity for Motivic Cohomology in Dimensions at Most Three

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Abstract

We prove absolute purity for the intrinsic motivic coefficient $H = H\mathbf{Z}^{\mathbf{A}^1} = s_0 KGL$. Integral point purity holds for regular noetherian local rings with fraction fields of characteristic zero in dimension at most two, and in dimension at most three under excellence, with the corresponding flat regular-pair statements. For an invertible prime ℓ , purity modulo every ℓ^m holds in local or ambient dimension at most $\ell + 2$. In every finite dimension, integral purity also holds for local algebras regular over a mixed-characteristic DVR and for flat regular $\mathbf{Z}_{(p)}$ -pairs with regular special fibres. The proofs combine filtered Gysin maps and Adams operations, valuative comparison and descent, and detection by smooth function fields.

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1 Main results and conventions

Let

$$H_T = H\mathbf{Z}_T^{\mathbb{A}^1} = s_0 KGL_T.$$

This is the Cartesian oriented absolute coefficient of (Bachmann et al., 2025, arXiv v1, Definition 4.15, Theorem 9.3(1),(2), and Corollary 9.7), with the corrected comparison arguments in section A. For a regular closed immersion $i : Z \hookrightarrow X$ with normal bundle N_i , we study the fundamental-class transformation

$$\mathrm{pur}_i(H_X) : \mathrm{Th}(-N_i) \otimes H_Z \longrightarrow i^! H_X. \quad (1)$$

For constant codimension c , orientation identifies its source with $H_Z(-c)[-2c]$. The local and ambient dimensions in the following statements are Krull dimensions.

All derived complexes are cohomologically indexed: $H^j(C[a]) = H^{j+a}(C)$, and their associated spectra satisfy $\pi_i = H^{-i}$. Put $T = \Sigma^{2,1}\mathbf{1}$ and write

$$\begin{aligned} A(q)(Y) &= \mathbf{Z}(q)^{\mathbb{A}^1}(Y), & A_n(q) &= A(q) \otimes^{\mathbf{L}} \mathbf{Z}/n, \\ A(q)(Y)[2q] &= \mathrm{map}(\mathbf{1}_Y, T^q \otimes H_Y). \end{aligned}$$

Negative-weight values are zero. For a presheaf P and a closed subset $Z \subset Y$ with quasi-compact complement, set

$$P_Z(Y) = \mathrm{fib}(P(Y) \longrightarrow P(Y \setminus Z)).$$

Finite coefficients commute with these supported fibres. For a general $E \in \mathbf{SH}(Y)$, $E(U)(q)$ denotes its mapping spectrum on a smooth test U , with the Tate twist alone: $E(U)(q) = \mathrm{map}(\Sigma^\infty U_+, E(q))$. Thus $T^q E = E(q)[2q]$, and no cohomological shift $[2q]$ is included in $E(U)(q)$. Continuous local test schemes are affine inverse limits of smooth test schemes; their values are the corresponding filtered colimits. Only the coefficient-module values are written as complexes in $D(\mathbf{Z})$. All schemes evaluated by these presheaves are qcqs.

For a regular local ring R , the condition $\mathrm{char} \mathrm{Frac}(R) = 0$ is equivalent to $\mathrm{char} R = 0$: the ring is a domain (The Stacks Project Authors, 2026, Tag 00NP) and embeds into its fraction field. The residue field is allowed to have positive characteristic.

Theorem A (Integral purity). *Let R be a regular noetherian local ring of dimension d , with closed point s and residue field κ , and assume that $\mathrm{char} \mathrm{Frac}(R) = 0$. If $d \leq 2$, or if $d \leq 3$ and R is excellent, then*

$$H_\kappa(-d)[-2d] \xrightarrow{\sim} s^! H_R \quad \text{in } \mathbf{SH}(\kappa).$$

In particular, after choosing regular parameters, its value in every weight q is the equivalence

$$A(q-d)(\kappa)[-2d] \xrightarrow{\sim} A(q)_s(R).$$

For a regular closed immersion $i : Z \hookrightarrow X$ between regular noetherian schemes flat over $\mathbf{Z}$, $\mathrm{pur}_i(H_X)$ is an equivalence if $\dim X \leq 2$, or if $\dim X \leq 3$ and both schemes are excellent.

Theorem B (Prime-power purity). *Let R be a regular noetherian local ring of dimension d , with closed point s and residue field κ , and let ℓ be a prime invertible in R . If $d \leq \ell + 2$, then for every $m \geq 1$ the point-purity transformation for H_R/ℓ^m is an equivalence. Thus, for every weight q ,*

$$A_{\ell^m}(q-d)(\kappa)[-2d] \xrightarrow{\sim} A_{\ell^m}(q)_s(R).$$

For a regular closed immersion $i : Z \hookrightarrow X$ between regular noetherian schemes, if ℓ is invertible on X and $\dim X \leq \ell + 2$, then

$$\mathrm{Th}(-N_i) \otimes (H_Z/\ell^m) \xrightarrow{\sim} i^!(H_X/\ell^m) \quad (m \geq 1).$$

These assertions impose no excellence, flatness or residue-field perfection hypothesis.

Theorem C (Purity over a regular DVR map). *Let R be a regular noetherian local ring of finite dimension d , with closed point s and residue field κ . Suppose there is a local regular ring map $V \rightarrow R$ from a mixed-characteristic DVR of residue characteristic p . Then*

$$H_\kappa(-d)[-2d] \xrightarrow{\sim} s^! H_R.$$

Its integral value in every weight is $A(q-d)(\kappa)[-2d] \simeq A(q)_s(R)$, and

$$A_{p^m}(q)_s(R) \simeq W_m \Omega_{\kappa, \log}^{q-d}[-(q+d)] \quad (m \geq 1),$$

with zero value for $q < d$. In particular this applies to every unramified regular local ring of mixed characteristic $(0, p)$, that is, with $p \notin \mathfrak{m}_R^2$.

For a regular closed immersion $i : Z \hookrightarrow X$ between regular noetherian $\mathbf{Z}_{(p)}$ -schemes of finite Krull dimension, assume that both schemes are flat over $\mathbf{Z}_{(p)}$ and that both special fibres are regular. Then $\mathrm{pur}_i(H_X)$ is an integral equivalence. No dimension bound or excellence assumption is imposed.

Here a regular ring map is flat with geometrically regular fibres. In Theorem A, positive residue characteristic p is nonzero in R ; flatness supplies this condition for its pair statement. Theorem C also allows DVR bases ramified over $\mathbf{Z}_{(p)}$.

Gabber's étale absolute purity theorem has no dimension bound for invertible finite coefficients (Riou, 2014, Théorème 3.1.1). The motivic comparison with étale cohomology controls only a truncation. We use it to bound the Gysin defect, then use the filtered K -theory Gysin map and exact Adams eigenweights to eliminate that defect. At the residue prime, polynomial invariance on F -smooth valuation rings supplies comparison with the $\mathbb{A}^1$ -invariant theory. Regular surfaces admit a quadratic-transform argument without excellence; in dimension three, resolution supplies the bounded valuation models needed for two distinct descent arguments. Detection by smooth function fields then passes from values to objects and, by composition, to regular pairs. With $F_p(q) = \mathbf{Z}/p(q)^{\mathrm{cdh}}$, Theorem 4.4 also gives the independent comparison $F_p(q)(Y) \simeq A_p(q)(Y)$ for every excellent noetherian scheme Y of dimension at most three, every prime p , and every $q \geq 0$. It allows singularities, arbitrary residue fields and arbitrary ramification; its dimension-three proof invokes Cossart–Piltant's resolution theorem. Theorem 4.2 supplies the regular-surface comparison without excellence. The endpoint Adams argument and these residue-characteristic comparisons are the main mechanisms developed below. Levine proves integral cycle-complex localization for finite-type schemes over regular noetherian one-dimensional bases (Levine, 2001, preprint, Theorem 0.7 and Section 6). Bouis and Kundu prove a $\mathbf{Z}/p^m$ generic-fibre equivalence for ind-smooth schemes over perfectoid valuation rings of residue characteristic p (Bouis and Kundu, 2025, arXiv v2, Corollary 5.9 and Remark 5.10) in their non- $\mathbb{A}^1$ -invariant motivic theory, distinct from H . Fundamental-class DVR purity for Spitzweck's coefficient is proved in the mixed-characteristic DVR setup of Frankland–Spitzweck's Section 4 (Frankland and Spitzweck, 2026, arXiv v4, Section 4, Proposition 8.3 and Corollary 8.4). For every mixed-characteristic DVR, the internal dimension-one case of Theorem A supplies this input; Section 6 extends it to regular DVR maps by Popescu approximation and continuity.

We use the six operations, the exactness and naturality of purity in its coefficient, compatibility with twists, composition, transverse base change and smooth exceptional base change of (Déglise et al., 2021b, arXiv v3, Paragraphs 2.1.9–2.1.10 and 4.3.1, Proposition 2.5.4(i),(ii)(a), Theorem 3.3.2(iii), Proposition 4.3.10(i), and Remark 4.3.8(i)). The composition input uses the strict double deformation and Thom identifications proved in proposition D.3. For continuous tests these compatibilities are applied on finite-type models and passed to the limit (Hoyois, 2015, arXiv v5, Lemma A.7(1)).

Comparison with Spitzweck’s coefficient. Let $a_X : X \rightarrow \mathrm{Spec} \mathbf{Z}$ and put $M_X^{\mathrm{Spi}} = a_X^* M_{\mathbf{Z}}^{\mathrm{Spi}}$. The integral comparison over $\mathbf{Z}$ follows from Bachmann’s effective-cover theorem (Bachmann, 2025, arXiv v2, Theorem 1.1(1), Example 1.3 and Lemma 2.1), as recorded in (Bachmann et al., 2025, arXiv v1, Remark 4.14). Together with its Theorem 9.3(2), this gives

$$M_X^{\mathrm{Spi}} \xrightarrow{\sim} a_X^* s_0 KGL_{\mathbf{Z}} \xrightarrow{\sim} s_0 KGL_X = H_X$$

for every qcqs scheme X , including all local tests used below. Spitzweck’s construction directly over a mixed-characteristic Dedekind domain agrees with this absolute pullback by (Spitzweck, 2018, Theorem 8.25, p. 88; arXiv v3, Theorem 9.19). These are integral comparisons of coefficient objects; finite reductions transport them by exactness.

For any coefficient morphism $u_X : M_X \rightarrow H_X$, naturality of the intrinsic fundamental-class transformation gives the square

$$\begin{array}{ccc} \mathrm{Th}(-N_i) \otimes i^* M_X & \xrightarrow{\mathrm{pur}_i(M_X)} & i^! M_X \\ \downarrow \mathrm{id} \otimes i^* u_X & & \downarrow i^! u_X \\ \mathrm{Th}(-N_i) \otimes i^* H_X & \xrightarrow{\mathrm{pur}_i(H_X)} & i^! H_X. \end{array}$$

This is (Déglise et al., 2021b, arXiv v3, Paragraphs 2.5.1 and 4.3.1). If u_X is an equivalence, both vertical arrows are equivalences, so intrinsic purity transports without an independent purity-compatibility hypothesis. Equality of particular oriented Tate formulas additionally requires alignment of the chosen Thom isomorphisms. An E_∞ comparison is needed to identify coefficient algebras or their module categories; no such upgrade of the cited object comparison is asserted here. Other independently constructed coefficients require their own integral recognition on the bases in question, compatible with their Cartesian identifications. Rational or prime-to-characteristic comparison alone does not supply it.

2 Filtered purity and invertible prime powers

The absolute KGL spectrum has effective covers f_j , slices s_j and Bott identifications

$$s_q KGL_Y = T^q \otimes H_Y, \quad T \otimes f_j KGL_Y \simeq f_{j+1} KGL_Y, \quad g^* f_j KGL_Y \simeq f_j KGL_{Y'}.$$

Its values form the natural filtration

$$\begin{aligned} \mathrm{Fil}^j KH(Y) &= \mathrm{map}(\mathbf{1}_Y, f_j KGL_Y), \\ \mathrm{Fil}^j KH(Y) &= KH(Y) \quad (j \leq 0), \\ \mathrm{gr}^q KH(Y) &= A(q)(Y)[2q]. \end{aligned}$$

The supported filtration is $\mathrm{Fil}^j KH_Z(Y) = \mathrm{fib}(\mathrm{Fil}^j KH(Y) \rightarrow \mathrm{Fil}^j KH(Y \setminus Z))$. These assertions are (Bachmann et al., 2025, arXiv v1, Section 2.1, (4.7), Remark 4.20, Theorem 4.39(1), and Theorem 9.3(1)), with the comparison and connectivity deductions in

proposition A.1. The filtration is finitary, and $\mathrm{Fil}^j KH(Y)$ is $(j - a)$ -connective when $\dim_v Y \leq a$ (Bachmann et al., 2025, arXiv v1, Theorem 1.1(1),(3) and Corollary 9.5). For locally noetherian schemes, valuative and Krull dimensions agree (Elmanto et al., 2021, arXiv v3, Proposition 2.3.2(9)). The graded-piece connectivity therefore gives

$$A(q)(Y) \in D^{\leq q+a} \quad (\dim_v Y \leq a). \quad (2)$$

Indeed $\mathrm{gr}^q KH(Y)$ is $(q - a)$ -connective. Reduction by finite coefficients preserves this upper bound.

Fix a prime ℓ and, on $\mathbf{Z}[1/\ell]$ -schemes, put

$$G_\ell(q)(Y) = R\Gamma_{\acute{\mathrm{e}}\mathrm{t}}(Y, \mu_\ell^{\otimes q}), \quad T_\ell(q) = L_{\mathrm{cdh}} \tau^{>q} G_\ell(q).$$

The multiplicative comparison is

$$A_\ell(q) = L_{\mathrm{cdh}} \tau^{\leq q} G_\ell(q), \quad b : A_\ell(q) \longrightarrow G_\ell(q)$$

by (Bachmann et al., 2025, arXiv v1, Theorem 1.1(4) and Proposition 6.8). Étale cohomology with the fixed torsion coefficient $\mu_\ell^{\otimes q}$ satisfies v-descent by (Bhatt and Mathew, 2021, arXiv v4, Proposition 5.2). Nisnevich and abstract blow-up covers are v-covers, so this gives the required cdh descent with the original comparison map. Consequently $A_\ell(q) \rightarrow G_\ell(q) \rightarrow T_\ell(q)$ is a cofiber sequence. Limits and filtered colimits preserve the lower bound, so

$$T_\ell(q)(Y) \in D^{\geq q+1}. \quad (3)$$

Every cdh covering sieve of a field is maximal. For every field L with ℓ invertible,

$$A_\ell(q)(L) = \tau^{\leq q} G_\ell(q)(L). \quad (4)$$

We use Gabber–Riou étale purity with its fundamental class and arbitrary-base-change compatibility (Riou, 2014, Définitions 2.1.1 and 2.3.1, Proposition 2.2.3.1, and Théorème 3.1.1); excellence is not required.

Lemma 2.1 (Coefficient algebras and supported module actions). *For every qcqs Y , H_Y is an E_∞ algebra over $c(H\mathbf{Z})$, where $c : \mathrm{Sp} \rightarrow \mathbf{SH}(Y)$ is the constant symmetric monoidal functor. Hence for every prime ℓ , including 2,*

$$H_Y/\ell \simeq H_Y \otimes_{c(H\mathbf{Z})} c(H\mathbf{F}_\ell)$$

is an E_∞ ring. The analogous derived coefficient algebra exists over $c(H\mathbf{Z}/\ell^m)$ for every $m \geq 1$.

For a regular closed immersion $i : Z \rightarrow Y$ of qcqs schemes, pur_i is a morphism of i^ -bimodules. For a cartesian motivic E_∞ ring E it therefore lifts to the purity transformation on its module categories. In particular the purity maps for KGL , H , and the derived coefficient algebras H/n are linear over their pullbacks to Z .*

Let Y be qcqs, E an E_∞ ring in $\mathbf{SH}(Y)$, and let $i : Z \rightarrow Y$, $j : U \rightarrow Y$ be complementary closed and open immersions. Then $i_ i^! E \rightarrow E \rightarrow j_* j^* E$ is a fiber sequence of E -modules. On values it is the fiber sequence of $E(Y)$ -modules*

$$E_Z(Y) \longrightarrow E(Y) \longrightarrow E(U).$$

The $i^ E$ -action on $i^! E$ for which purity is linear corresponds to the cup product with supports, namely the $E(Y)$ -module structure on $\mathrm{fib}(E(Y) \rightarrow E(U))$. For assertions on homotopy groups, a homotopy-commutative ring suffices.*

Proof. Construction 4.35 of Bachmann–Elmanto–Morrow supplies the E_∞ comparison $R\Gamma_{\text{Nis}}(-, \mathbf{Z}) \rightarrow \mathbf{Z}(0)^{\mathbb{A}^1}$ on smooth schemes over a regular noetherian base (Bachmann et al., 2025, Construction 4.35 and Remark 4.37). Over $\text{Spec } \mathbf{Z}$, evaluation gives an E_∞ map $H\mathbf{Z} \rightarrow \text{map}(\mathbf{1}, s_0 KGL_{\mathbf{Z}})$. The symmetric monoidal adjunction for c and its lax monoidal right adjoint lifts to algebras, giving $c(H\mathbf{Z}) \rightarrow H\mathbf{Z}$. The zero slice is the absolute E_∞ algebra specified by (Bachmann et al., 2025, Theorem 1.6(1),(2)); pulling back this map along $Y \rightarrow \text{Spec } \mathbf{Z}$ gives the assertion. Relative tensor with $c(H\mathbf{Z}/n)$ computes the cofiber of n on the $H\mathbf{Z}$ -module and remains an E_∞ algebra. In particular the argument uses no ring structure on $\mathbf{S}/2$. On represented values this is the derived coefficient structure of (Bachmann et al., 2025, Remark 4.38).

Linearity follows from (Frankland and Spitzweck, 2026, arXiv version 4, Lemma 8.1 and Corollary 8.2); the cartesian property of the spectra used here is part of the absolute construction in section 2 and the coefficient construction above.

The projection formula makes $i_* = i_!$ linear over $\mathbf{SH}(Y)$. The linear adjunction $i_* \dashv i^!$ lifts to the adjunction on modules over i^*E and E (Lurie, 2017, Corollary 7.3.2.7 and Example 7.3.2.8). The lax action on the right adjoint is adjoint to the projection-formula map formed with the counit; this is exactly the exchange map $i^!E \otimes i^*E \rightarrow i^!(E \otimes E)$ (Lurie, 2017, Remark 7.3.2.9) and (Déglise et al., 2021b, arXiv version 3, Paragraph 2.1.10). Thus $i^!E$ is an i^*E -module, its pushforward is an E -module, and the counit $i_*i^!E \rightarrow E$ is E -linear. The object j_*j^*E is an E -algebra through its unit. The composite of these two maps is null already in modules, since

$$\text{Map}_{\text{Mod}_E}(i_*i^!E, j_*j^*E) \simeq \text{Map}_{\text{Mod}_{j^*E}}(j^*i_*i^!E, j^*E)$$

and $j^*i_* = 0$. Hence the counit factors through the module fiber of $E \rightarrow j_*j^*E$. Its underlying factorization is the localization equivalence, so it is an equivalence in modules.

The functor $\text{map}(\mathbf{1}_Y, -)$ is exact and lax monoidal, and sends this sequence to $E(Y)$ -modules. Uniqueness of the module fiber identifies its action with the cup product with supports. The projection formula identifies the value of the action on $i^!E$ with this action on $i_*i^!E$. $\square$

Cup products of two supports are formal for Zariski sheaves of E_∞ rings: $P_Z \otimes P_W \rightarrow P_{Z \cap W}$. A multiplicative comparison preserves these products. We apply this to the graded coefficient values and to the comparison b .

Lemma 2.2 (Filtered Gysin maps and connectivity). *Let $i : Z \rightarrow Y$ be a regular closed immersion of codimension c between regular noetherian schemes of finite dimension, with normal bundle N . The maps*

$$\Gamma_j = \text{pur}_i(f_j KGL_Y) : \text{Th}(-N) \otimes i^*f_j KGL_Y \longrightarrow i^!f_j KGL_Y$$

form a map of $\mathbf{Z}$ -indexed filtered objects. With the Thom identification of the source, this gives

$$\text{Gys}^j : \text{Fil}^{j-c} KH(Z) \longrightarrow \text{Fil}^j KH_Z(Y).$$

In particular a specified trivialization of N gives the identification by T^{-c} and Bott periodicity. Without such an identification the first display is the intrinsic, Thom-twisted statement. Its graded pieces are

$$A(q-c)(Z)[-2c] \longrightarrow A(q)_Z(Y),$$

shifted by $[2q]$. The same assertions hold after every finite coefficient $\mathbf{S}/n$, without requiring n to be invertible on Y .

Let $Y = \operatorname{Spec} R$ for a regular noetherian local ring of dimension d , with residue field κ , and $Z = s$ its closed point. A regular system of parameters trivializes N . Then

$$\operatorname{pur}_s(KGL_Y) \circ \beta^{-d} = \pm \eta'_s,$$

where η'_s is the Cisinski–Déglise class; consequently Gys^j is an integral equivalence for $j \leq 0$, and remains an equivalence after every finite coefficient.

For $d \geq 1$, put integrally,

$$\Psi^j = \operatorname{fib}(\operatorname{Fil}^{j-d} KH(\kappa) \xrightarrow{\operatorname{Gys}^j} \operatorname{Fil}^j KH_s(Y)).$$

Then Ψ^j is $(j - d - 1)$ -connective, so Ψ^{q+1} is $(q - d)$ -connective and $\Sigma \Psi^{q+1}$ is $(q + 1 - d)$ -connective. The same bounds hold for the filtered fiber Ψ_n after every finite coefficient $\mathbf{S}/n$, without an invertibility hypothesis on n . Moreover $\Psi^j = 0$ and $\Psi_n^j = 0$ for $j \leq 0$.

Proof. Naturality of (1) along $f_{j+1} \rightarrow f_j$ makes Γ filtered. Base change of the covers is section 2. A trivialized $\operatorname{Th}(-N)$ is T^{-c} , so Bott periodicity identifies its tensor with $f_j KGL_Z$ with $f_{j-c} KGL_Z$. Localization identifies the value of $i^! f_j KGL_Y$ at Z with the supported filtration (Déglise et al., 2021b, arXiv version 3, Proposition 2.2.10). Taking cofibers in the filtration and using exactness of purity gives $\operatorname{pur}_i(s_q KGL_Y)$; the slice formula and compatibility with T^q give the stated graded Gysin maps. These are exact constructions, so finite coefficients commute with them.

The bottom equivalence. Both maps $KGL_\kappa \rightarrow i^! KGL_Y$ are KGL_κ -linear. Such a map is determined by the image of 1 in

$$\pi_0 \operatorname{map}(\mathbf{1}_\kappa, i^! KGL_Y) = K_0^s(Y) = K_0(\kappa) = \mathbf{Z}.$$

The Cisinski–Déglise class takes 1 to $[\mathcal{O}_s]$, and is an equivalence for a closed immersion of regular noetherian finite-dimensional schemes; on supported K -theory it is dévissage. The proper-retraction and deformation argument proving this input is proposition D.1, with the normalization of (Cisinski and Déglise, 2019, arXiv version 4, Definition 13.5.4). For $t = (t_1, \dots, t_d)$, compare the graph $\Gamma_t : Y \rightarrow \mathbb{A}_Y^d$ with the zero section 0_Y . The pullback of 0_Y is s , and the square is Tor-independent because the Koszul complex of t resolves $\mathcal{O}_s$. Transverse base change gives $\eta_s = \Gamma_t^* \eta_{0_Y}$. Smooth-pair purity makes the class of 0_Y a generator of $K_0^{0_Y}(\mathbb{A}_Y^d) = K_0(Y) = \mathbf{Z}$; it is $\pm[\operatorname{Koszul}(T)]$. Pullback gives $\pm[\operatorname{Koszul}(t)] = \pm[\mathcal{O}_s]$, as required. For $j \leq 0$, both value filtrations are the unfiltered spectra. Finally $K = KH$ for regular noetherian rings (Weibel, 2013, Chapter IV, Corollary 12.3.1), so the value map is the dévissage equivalence up to sign.

Connectivity. Put $U = Y \setminus \{s\}$. Its dimension is $d - 1$: the upper bound follows by appending the closed point to a chain; the lower bound follows from a maximal prime chain in the regular local ring. The filtration on U is the value filtration of section 2, including when U is not affine. Thus $\operatorname{Fil}^j KH(Y)$ is $(j - d)$ -connective and $\operatorname{Fil}^j KH(U)$ is $(j - d + 1)$ -connective. Their fiber is $(j - d)$ -connective. The source $\operatorname{Fil}^{j-d} KH(\kappa)$ is also $(j - d)$ -connective. The fiber of a map between spectra with this connectivity is $(j - d - 1)$ -connective. Finite coefficient cofibers preserve all these bounds and commute with fibers. The bottom vanishing follows from the preceding dévissage equivalence. $\square$

For a regular noetherian local ring R of dimension d , write $X = \operatorname{Spec} R$, s for its closed point, κ for its residue field, and $U = X \setminus \{s\}$. Use regular parameters for the Thom identification and set

$$\Delta(q) = \operatorname{cofib}(A(q - d)(\kappa)[-2d] \xrightarrow{\operatorname{pur}_s} A(q)_s(R)), \quad \Delta_n(q) = \Delta(q)/n. \quad (5)$$

Linearity over $c(H\mathbf{Z})$ and exactness of purity make $\Delta(q)$ an object of $D(\mathbf{Z})$, and identify $\Delta_n(q)$ with the Gysin cofiber for $A_n(q)$. Exactness of the filtered construction gives

$$\mathrm{gr}^q \Psi_n = \Delta_n(q)[2q - 1].$$

Global filtered statements retain the Thom twist; the parameter identification is used only at regular local points.

Lemma 2.3 (Supported comparison and the étale fundamental class). *For every closed pair $Z \subset Y$ of qcqs $\mathbf{Z}[1/\ell]$ -schemes, the comparison $A_\ell(q)_Z(Y) \rightarrow G_\ell(q)_Z(Y)$ is an isomorphism on H^n for $n \leq q$, and injective for $n = q + 1$.*

Let R be regular noetherian local of dimension $d \geq 1$, with ℓ invertible. Then there is a unit $a \in (\mathbf{Z}/\ell)^\times$, independent of q , such that on every H^n the composite

$$A_\ell(q - d)(\kappa)[-2d] \xrightarrow{\mathrm{pur}_s(q)} A_\ell(q)_s(X) \xrightarrow{b} G_\ell(q)_s(X) \simeq G_\ell(q - d)(\kappa)[-2d]$$

is a times the truncation map of (4). The last equivalence is Gabber–Riou absolute purity. This assertion concerns maps on cohomology groups; an identity of morphisms in the derived category is not asserted.

Proof. By (3), both $T_\ell(q)(Y)$ and $T_\ell(q)(Y \setminus Z)$ lie in $D^{\geq q+1}$. Their fiber also lies in $D^{\geq q+1}$. Apply the long exact sequence to $A_\ell(q)_Z(Y) \rightarrow G_\ell(q)_Z(Y) \rightarrow T_\ell(q)_Z(Y)$.

Let $\eta_s = \mathrm{pur}_s(1) \in H^{2d}A_\ell(d)_s(X)$. By lemma 2.1, the coefficient spectrum is an E_∞ ring, including for $\ell = 2$. Purity is linear over its pullback to s , and lemma 2.1 identifies that action with the cup product with supports. Consequently, for any lift $\tilde{x} \in H^m A_\ell(w)(X)$ of $x \in H^m A_\ell(w)(\kappa)$,

$$\mathrm{pur}_s(x) = \tilde{x} \cup \eta_s, \quad b(\mathrm{pur}_s(x)) = b(\tilde{x}) \cup b(\eta_s).$$

The second equality uses the multiplicativity of b with supports. If X is henselian, every such x has a lift: for $m \leq w$ both groups are étale cohomology and henselian rigidity applies (The Stacks Project Authors, 2026, Tag 09ZI); the target is zero for $m > w$.

We first normalize $b(\eta_s)$ over the original regular local base X . Put $V = \mathbb{A}_X^d$ and consider its zero section 0_X . For the rank-one zero section in $\mathbb{A}_X^1$, smooth-pair purity and pullback invariance of H identify

$$H^2 A_\ell(1)_{0_X}(\mathbb{A}_X^1) = H^0 A_\ell(0)(X) = \mathbf{Z}/\ell.$$

The fundamental class η_0 generates this line. The comparison is injective in degree $2 = 1 + 1$ by the first assertion; therefore $b(\eta_0) = u \mathrm{Cl}_0$ for a unit u , where Cl_0 is the étale divisor class. Here and below the smooth-pair input is (Déglise et al., 2021b, arXiv version 3, Proposition 4.3.10(i), Remark 4.3.8(i), and Remark 2.5.2), and the pullback input is (Bachmann et al., 2025, Theorem 1.6(2)).

Smooth-pair purity also makes

$$L_A = H^{2d} A_\ell(d)_{0_X}(V) = \mathbf{Z}/\ell$$

a line generated by η_{0_X} . Étale absolute purity makes $L_{\mathrm{et}} = H_{0_X, \mathrm{et}}^{2d}(V, \mu_\ell^{\otimes d})$ a line generated by Cl_{0_X} . For the coordinate projections $\mathrm{pr}_h : V \rightarrow \mathbb{A}_X^1$, the product

$$c = \mathrm{pr}_1^* \eta_0 \cup \cdots \cup \mathrm{pr}_d^* \eta_0$$

has support 0_X and lies in L_A . Its image is $u^d \mathrm{pr}_1^* \mathrm{Cl}_0 \cup \cdots \cup \mathrm{pr}_d^* \mathrm{Cl}_0$, which is nonzero. Indeed cup product with each coordinate divisor class is the Gysin map for that divisor,

and is an isomorphism by absolute purity; iterating it sends $1 \in H^0(X, \mathbf{Z}/\ell)$ to this product. Thus $b : L_A \rightarrow L_{\text{et}}$ is a nonzero map between one-dimensional $\mathbf{Z}/\ell$ -modules, hence an isomorphism. Write $b(\eta_{0_X}) = v \text{Cl}_{0_X}$ with v a unit.

Pull back by the graph $\Gamma_t : X \rightarrow V$. Its square with the zero section is Tor-independent. Transverse base change for the motivic fundamental class, and base change for Riou's generalized class, give

$$\eta_s = \Gamma_t^* \eta_{0_X}, \quad \text{Cl}_s = \Gamma_t^* \text{Cl}_{0_X}, \quad b(\eta_s) = v \text{Cl}_s.$$

The conormal bundles agree in this pullback. This proves the required normalization without a product formula for motivic fundamental classes.

For nonhenselian R , use affine étale neighborhoods (X', s') of (X, s) with $\kappa(s') = \kappa$. A principal shrinking, by prime avoidance, makes $X' \times_X s = s'$. These squares are cartesian and Tor-independent, and $X' \rightarrow X$ is a morphism of the type to which purity base change applies (Déglise et al., 2021b, arXiv version 3, Theorem 3.3.2(iii) and Proposition 2.5.4(ii)). Nisnevich excision identifies their supported A_ℓ and G values with those on X . Finitariness and étale continuity identify the colimit of these values with the supported values on the henselization (Bachmann et al., 2025, Theorem 1.1(3)). A lift of x on the henselization consequently comes from a lift on one X' . Linearity, multiplicativity, and the normalization of $\eta_{s'}$ apply on that X' , and descend through excision to X . This uses no purity base-change assertion for the henselization map itself.

Finally cup product with Cl_s is the étale absolute-purity isomorphism. The comparison at κ is the truncation map by (4); take $a = v$. $\square$

Proposition 2.4 (The prime-to-characteristic defect window). *Let R be regular noetherian local of dimension $d \geq 1$, and let ℓ be a prime invertible in R . For every integer q ,*

$$\Delta_\ell(q) \in D^{[q+2, q+d]}.$$

More generally $\Delta(q)/\ell^m$ has this window for every $m \geq 1$. The lower bound uses the prime-to-characteristic comparison and étale absolute purity; the integral upper bound $\Delta(q) \in D^{\leq q+d}$ uses the value filtration. No henselian or excellence assumption and no dimension bound is used.

Proof. Gabber's purity identifies $G_\ell(q)_s(X)$ with $G_\ell(q-d)(\kappa)[-2d]$. Put $S = A_\ell(q-d)(\kappa)[-2d]$. By (4), $S = \tau^{\leq q+d} G_\ell(q)_s(X)$. By lemma 2.3, $b \circ \text{pur}_s(q)$ is an isomorphism on H^n for $n \leq q+d$ and injective in every degree. The supported comparison is an isomorphism through q and injective at $q+1$ by lemma 2.3. At $q+1 \leq q+d$ the composite is surjective, so the supported comparison is surjective as well, hence an isomorphism. Thus $\text{pur}_s(q)$ is an isomorphism through $q+1$. It is injective at $q+2$: if $d \geq 2$ this follows from the isomorphism of the composite; if $d = 1$ its source in that degree is zero. The long exact sequence gives $\Delta_\ell(q) \in D^{\geq q+2}$.

Integrally, (2) gives $A(q)(X) \in D^{\leq q+d}$ and $A(q)(U) \in D^{\leq q+d-1}$. Their fiber is in $D^{\leq q+d}$. Also $A(q-d)(\kappa)[-2d] \in D^{\leq q+d}$. Their cofiber therefore has the same upper bound. Finite coefficient cofibers preserve it. The lower bound for ℓ^m follows from the case $m = 1$ by the exact coefficient sequences $\Delta/\ell^{m-1} \rightarrow \Delta/\ell^m \rightarrow \Delta/\ell$, since $D^{\geq q+2}$ is extension-closed. $\square$

Proposition 2.5 (Absolute Adams operations and exact eigenweights). *For every integer $k \geq 1$ there is an E_∞ endomorphism*

$$\psi_{\mathbf{Z}}^k : KGL_{\mathbf{Z}}[1/k] \longrightarrow KGL_{\mathbf{Z}}[1/k]$$

such that, for the unit u and multiplication β by the Bott element,

$$\psi^k \circ u \simeq u, \quad \psi^k \circ \beta \simeq k\beta \circ \Sigma^{2,1}\psi^k. \quad (6)$$

Pullback gives these relations over every qcqs scheme, naturally under further pullback. The localization $[1/k]$ is in coefficients; k need not be invertible on the scheme.

For a regular noetherian local point and every prime $\rho \nmid k$, the operation acts on the filtered Gysin fibre Ψ_{ρ^m} for all $m \geq 1$. On its weight- q graded piece it acts by exactly k^q . Its spectral-sequence pages are $\mathbf{Z}/\rho^m$ -modules and the differentials are linear. For a differential from weight q to weight $q + r$,

$$(k^r - 1)d_r = 0. \quad (7)$$

The prime ρ need not be invertible on the local ring.

Proof. We use the absolute Adams endomorphism of $KGL[1/k]$, natural over every qcqs scheme, as formulated in (Elmanto and Morrow, 2026, arXiv v2, Appendix B.1).¹ Its unit and Bott actions can be read from the motivic Snaith presentation, which identifies KGL with the Bott localization of $\Sigma_+^\infty B\mathbf{G}_m$ over a noetherian base of finite Krull dimension (Gepner and Snaith, 2010, arXiv v3, Theorem 4.17 and Section 2.1). Delooping $x \mapsto x^k$ on $\mathbf{G}_m$ gives an E_∞ self-map of that presentation and sends its Bott element to $k\beta$. After inverting k in coefficients it extends across the Bott localization. The map is unital and multiplicative, which gives (6). The Bott convention $1 - [\mathcal{O}(-1)]$ agrees with $[\mathcal{O}(1)] - 1$ because the square of the latter is zero in $K_0(\mathbf{P}^1)$ (Bachmann et al., 2025, Definition 4.2). Finally KGL is absolute (Bachmann et al., 2025, Remark 4.3), and pullback is symmetric monoidal and exact.

On every spectrum modulo ρ^m , multiplication by k is invertible, so reduction agrees with reduction after coefficient localization $[1/k]$. Covers and slices commute with filtered colimits and cofibers. The zero slice of the unit is an equivalence (Bachmann et al., 2025, arXiv v1, Corollary 9.7). The unit relation therefore gives $s_0(\psi^k) = \text{id}$. The Bott relation and $s_q(T^q \otimes E) = T^q \otimes s_0 E$ give

$$s_q(\psi^k) = k^q \text{id} \quad \text{on } s_q(KGL[1/k]). \quad (8)$$

Naturality of purity makes

$$E \longmapsto \text{fib}(\text{map}(\mathbf{1}_s, \text{Th}(-N) \otimes i^* f_j E) / \rho^m \longrightarrow \text{map}(\mathbf{1}_s, i^! f_j E) / \rho^m)$$

an exact functor to filtered spectra. Its weight- q graded functor is an exact functor of $s_q E$. Applying it to the single homotopy (8) proves the scalar k^q on $\text{gr}^q \Psi_{\rho^m}$. In the untwisted Bott identification the source action is $k^d \psi^k$, whose weight- $q - d$ scalar is $k^d k^{q-d} = k^q$.

The integral Gysin cofiber lies in $D(\mathbf{Z})$; its derived reductions have $\mathbf{Z}/\rho^m$ -module cohomology. These are the E_1 terms, and later pages are subquotients. Their additive differentials are therefore $\mathbf{Z}/\rho^m$ -linear. They commute with the filtered operation. Comparing scalars in weights q and $q + r$ and cancelling the unit k^q proves (7). No ring structure on a Moore spectrum is used. $\square$

¹For the Snaith construction, compare (Bachmann and Hopkins, 2020, arXiv v4, Sections 3.3.1–3.3.2 and Lemmas 3.10–3.11). Their Section 3 is written under the standing hypothesis $1/2 \in S$; the general qcqs formulation used here is the one in Elmanto–Morrow. The KGL construction uses Snaith’s presentation over $\text{Spec } \mathbf{Z}$ and the Bott calculation in $K_0(\mathbf{P}^1)$; it inverts k in the coefficient spectrum and requires no inversion of 2 on the base.

Lemma 2.6 (The prime-power Adams window). *Let ℓ be prime, let N be an integer, and let $B_q \in D(\mathbf{Z})$ for $q \geq 0$. Suppose for $m = 1, 2$ there is a filtered spectrum $\mathcal{T}_m$ with $\mathcal{T}_m^0 = 0$, connectivity tending to infinity with its filtration index, and*

$$\mathrm{gr}^q \mathcal{T}_m = (B_q/\ell^m)[2q - \epsilon], \quad B_q/\ell^m \in D^{[q+2, q+N]},$$

where ϵ is one fixed integer, equal to 0 or 1. Assume the Adams relations (7) for all k prime to ℓ . Then $B_q/\ell = 0$ for every q if $N \leq \ell + 2$. If $N \leq \ell + 1$, only the hypotheses for $m = 1$ are needed. No compatibility between the two filtrations is required beyond their displayed graded identifications with reductions of the same B_q .

Proof. Write $e = n - q$ for the excess of a class in $H^n(B_q/\ell^m)$. Its homotopy degree is $i = 2q - \epsilon - n = q - \epsilon - e$. Thus only the weights $q \in [i + \epsilon + 2, i + \epsilon + N]$ contribute to a fixed degree. The finite window together with the increasing connectivity gives strong convergence of the filtered spectral sequence to $\pi_* \mathcal{T}_m^0 = 0$. For a fixed degree, truncate at a stage where the tail has zero homotopy in that degree and its neighboring degree. The finite filtration spectral sequence then computes that degree, and the finite window makes its pages stabilize as the truncation increases. A differential d_r increases weight by r and decreases homotopy degree by 1, so it increases excess by $r + 1$. A nonzero differential requires $1 \leq r \leq N - 3$.

If $N \leq \ell + 1$, choose k primitive modulo ℓ . Then $k^r - 1$ is a unit modulo ℓ for $1 \leq r \leq N - 3 \leq \ell - 2$. For $m = 1$ all differentials vanish, so $E_1 = E_\infty = 0$ and $B_q/\ell = 0$. This also covers $N \leq 3$, where there is no possible differential.

Now let $N = \ell + 2$. Choose k primitive modulo ℓ^2 if ℓ is odd, and choose $k = 3$ if $\ell = 2$. For $1 \leq r \leq \ell - 2$, $k^r - 1$ is prime to ℓ , while $k^{\ell-1} - 1 = \ell u$ with u prime to ℓ . Thus, for $m = 1, 2$, the only possible differential is $d_{\ell-1}$ and $\ell d_{\ell-1} = 0$. It runs from excess 2 in weight q to excess N in weight $q + \ell - 1$. Strong convergence to zero gives, for all q and $m = 1, 2$,

$$H^{q+e}(B_q/\ell^m) = 0 \quad (3 \leq e \leq N - 1), \quad (9)$$

$$d_{\ell-1} : H^{q+2}(B_q/\ell^m) \xrightarrow{\sim} H^{(q+\ell-1)+N}(B_{q+\ell-1}/\ell^m), \quad (10)$$

$$H^{q+N}(B_q/\ell^m) = 0 \quad (q < \ell - 1), \quad (11)$$

$$\ell H^{q+2}(B_q/\ell^m) = 0. \quad (12)$$

The isomorphism holds because the lower groups have no incoming differentials and the upper groups have no outgoing differentials. The last line uses its injectivity and $\ell d_{\ell-1} = 0$.

Fix q and put $G_n = H^n(B_q)$. The universal-coefficient sequences are

$$0 \longrightarrow G_n/\ell^m \longrightarrow H^n(B_q/\ell^m) \longrightarrow G_{n+1}[\ell^m] \longrightarrow 0. \quad (13)$$

At $n = q + 1$, $m = 1$, the window gives $G_{q+2}[\ell] = 0$. At $n = q + 2$, $m = 2$, (12) gives $\ell(G_{q+2}/\ell^2) = 0$. For $g \in G_{q+2}$ write $\ell g = \ell^2 h$; injectivity of multiplication by ℓ gives $g = \ell h$. Hence $G_{q+2} = \ell G_{q+2}$. The $m = 2$ exponent bound on the quotient term also gives $\ell G_{q+3}[\ell^2] = 0$. At $n = q + 3$, $m = 1$, (9) gives $G_{q+3}/\ell = 0$. If $\ell x = 0$, write $x = \ell y$; then $\ell^2 y = 0$, so the exponent bound gives $\ell y = 0$ and $x = 0$. Thus $G_{q+3}[\ell] = 0$. Finally (13) at $n = q + 2$, $m = 1$ gives $H^{q+2}(B_q/\ell) = 0$.

The top groups in weights at least $\ell - 1$ vanish by (10); the smaller weights vanish by (11). The intermediate groups already vanish. The whole window is therefore zero for $m = 1$. $\square$

At the endpoint this argument uses the reductions modulo ℓ and ℓ^2 ; no higher prime-power input is required.

Proof of Theorem B on values. The dimension-zero case is the identity. For $d \geq 1$, apply lemma 2.6 to

$$B_q = \Delta(q), \quad \epsilon = 1, \quad \mathcal{T}_m = \Psi_{\ell^m}, \quad N = d.$$

The bottom equivalence and increasing connectivity are lemma 2.2, the windows are proposition 2.4, and the Adams relations are proposition 2.5. Thus $\Delta(q)/\ell = 0$ for every $q \geq 0$. Negative weights vanish. The coefficient sequences $\Delta(q)/\ell^{m-1} \rightarrow \Delta(q)/\ell^m \rightarrow \Delta(q)/\ell$ then give vanishing for every m . This proves the value assertion. The passage to objects and regular pairs is proved in section 5. $\square$

Proposition 2.7 (Rational purity). *Let $i : Z \rightarrow Y$ be a regular closed immersion between regular noetherian schemes of finite dimension. Then*

$$\mathrm{pur}_i(H_Y \otimes \mathbf{Q}) : \mathrm{Th}(-N_i) \otimes i^*(H_Y \otimes \mathbf{Q}) \xrightarrow{\sim} i^!(H_Y \otimes \mathbf{Q}).$$

The integral purity cofiber for H therefore has zero rationalization. Its mapping spectrum from any compact test object has torsion homotopy groups. No finite-type hypothesis over $\mathrm{Spec} \mathbf{Z}$ is imposed.

Proof. Rationalization commutes with i^* and the Thom twist. It commutes with $i^!$ because i_* preserves compact objects, so its right adjoint preserves filtered colimits; we use the purity colimit and retract formalism of (Déglise et al., 2021b, arXiv version 3, Remark 4.3.14). The rational zero slice is the weight-zero Adams summand

$$H_{Y, \mathbf{Q}} \simeq KGL_Y^{(0)} = H\mathbf{Q}_Y$$

of $KGL_{Y, \mathbf{Q}}$ (Bachmann et al., 2025, Corollary 4.52(2), Definition 4.53, and Remark 4.54), using the rationalization and Adams-operator corrections in proposition A.1. The KGL purity transformation is invertible for these regular schemes by proposition D.1; its match with the motivic fundamental-class purity map is the comparison in (Déglise et al., 2021b, arXiv version 3, Paragraph 4.4.3 and footnote 12); the corresponding orientation statement is (Déglise, 2018, Example 4.1.9(1) and Proposition 4.2.2). The cited comparison applies to regular noetherian finite-dimensional schemes without a finite-type hypothesis over $\mathbf{Z}$. Purity is natural in its argument, so the rational retract $KGL_Y^{(0)}$ is pure. Exactness then gives zero rationalization of the integral cofiber. Compact test objects commute with this rationalization, and a spectrum has zero rationalization exactly when all its homotopy groups are torsion. $\square$

3 Valutive polynomial invariance

Fix a prime p and $q \geq 0$. Keep $A(q)$ integral, and set

$$\begin{aligned} M_p(q) &= \mathbf{Z}/p(q)^{\mathrm{mot}}, \\ F_p(q) &= L_{\mathrm{cdh}} M_p(q) = \mathbf{Z}(q)^{\mathrm{cdh}}/p, \\ A_p(q) &= L_{\mathbb{A}^1} M_p(q) = L_{\mathbb{A}^1} F_p(q). \end{aligned}$$

Here M_p is Bouis's theory; the finite-coefficient identifications and localization units are (Bouis, 2024, arXiv v3, Definition 3.12, Remark 3.25, and Theorem 6.5) and (Bachmann et al., 2025, arXiv v1, Theorem 1.1(9)). Their truncated syntomic route and finite-coefficient localization argument are recorded in section A. Put

$$\mathcal{S}_p(q) = \mathbf{Z}/p(q)^{\mathrm{syn}}, \quad \mathcal{E}_p(q)(X) = R\Gamma_{\mathrm{ét}}(X[1/p], \mu_p^{\otimes q}),$$

$$\begin{aligned}
Q_p(q) &= \operatorname{cofib}(F_p(q) \rightarrow A_p(q)), \\
\mathcal{N}_p(q) &= \operatorname{cofib}(M_p(q) \rightarrow F_p(q)), \\
\mathcal{D}_p(q)(X) &= \operatorname{cofib}(M_p(q)(X) \rightarrow M_p(q)(\mathbb{A}_X^1)).
\end{aligned}$$

These are cofibers, whereas all supported values are fibers. The canonical comparison $F_p \rightarrow \mathcal{E}_p$ factors through A_p compatibly with its localization unit (Bachmann et al., 2025, arXiv v1, (8.10), Theorem 7.16, and Theorem 1.13).

A valuation ring has rank equal to its Krull dimension; fields have rank zero. Mixed characteristic $(0, p)$ means that its fraction field has characteristic zero and p lies in its maximal ideal. F-smoothness is relative to p in the sense of (Bhatt and Mathew, 2023, arXiv v2, Definition 4.1). It presupposes p -quasisyntomicity: for a p -torsionfree ring, the p -completed cotangent complex has Tor-amplitude $[-1, 0]$. Its divided Frobenius fibers have p -complete Tor-amplitude at least $i + 2$, and its Nygaard filtration is complete. These conditions depend only on derived p -completion. Regular noetherian rings are F-smooth. F-smoothness passes to filtered colimits and localizations, and to a p -completely flat algebra B over T when $T/p \rightarrow B/p$ is smooth (Bhatt and Mathew, 2023, arXiv v2, Propositions 4.6, 4.9, 4.10, and Theorem 4.15). For a p -torsionfree F-smooth ring T ,

$$\operatorname{fib}(\mathcal{S}_p(q)(T) \rightarrow \mathcal{E}_p(q)(T)) \in D^{\geq q+1}. \quad (14)$$

This is (Bhatt and Mathew, 2023, arXiv v2, Proposition 5.2). The analogous bound for the fiber of $M_p \rightarrow \mathcal{E}_p$ is (Bouis, 2024, arXiv v3, Corollary 5.12); neither bound requires noetherianity.

Proposition 3.1 (Comparison, descent and Milnor excision). *For every $q \geq 0$, $Q_p(q)$ is a finitary cdh sheaf with values in $D(\mathbf{Z}/p)$ and satisfies Milnor excision. It vanishes on qcqs schemes over $\mathbf{F}_p$ or $\mathbf{Q}$, and on qcqs $\mathbf{Z}[1/p]$ -schemes of finite valuative dimension. In particular it vanishes on every ring in which p is invertible, evaluated as an affine scheme.*

On every qcqs $\mathbf{Z}[1/p]$ -scheme, $M_p(q) \rightarrow F_p(q)$ is an equivalence. On every qcqs $\mathbf{Q}$ -scheme, therefore, $M_p(q) \simeq F_p(q) \simeq A_p(q)$ and $\mathcal{D}_p(q) = 0$.

Proof. F_p and A_p are finitary cdh sheaves by (Bachmann et al., 2025, arXiv v1, Theorems 1.1(3) and 7.12(4)); a cofiber of a map of sheaves in a stable category is again a sheaf. The equal-characteristic vanishing follows from that source's Corollary 1.11. When p is invertible, the hypothesis $\operatorname{Val}(-, p, j)$ there is vacuous; Corollary 9.4 and Remark 8.2 give the stated finite-valuative-dimensional vanishing. Every noetherian scheme of finite Krull dimension has finite valuative dimension by (Elmanto et al., 2021, arXiv v3, Proposition 2.3.2(9)). An arbitrary $\mathbf{Z}[1/p]$ -algebra is the filtered union of its finitely generated subalgebras. Finitarity therefore proves the assertion for arbitrary affine schemes where p is a unit.

Milnor excision for $\mathbf{Z}(q)^{\operatorname{cdh}}$ is (Bachmann et al., 2025, arXiv v1, Theorem 7.21). For the intrinsic value presheaf, Milnor excision follows from proposition C.2: the finite-rank valuation calculation passes to arbitrary rank by finitariness, and then (Elmanto et al., 2021, arXiv v3, Theorem 3.3.4, implication (1) $\Rightarrow$ (2)) applies over $\operatorname{Spec} \mathbf{Z}$. Reduction modulo p and taking the cofiber of the original localization unit $F_p \rightarrow A_p$ preserve cartesian squares. Consequently, for any prime P of a valuation ring V ,

$$Q_p(q)(V) \simeq Q_p(q)(V_P) \times_{Q_p(q)(\kappa(P))} Q_p(q)(V/P). \quad (15)$$

The underlying ring square is the Milnor square of (Elmanto et al., 2021, arXiv v3, equation (3.3.1)).

The cartesian square of (Bouis, 2024, arXiv v3, Theorem B) compares $M_p \rightarrow F_p$ with the BMS syntomic theory and its cdh sheafification. That theory depends only on the derived p -completion (Bouis, 2024, arXiv v3, Remark 2.16), so both BMS terms vanish where p is invertible. The square proves the first claim. The equal-characteristic vanishing already proved, together with the polynomial invariance of A_p , proves the remaining claims. $\square$

Lemma 3.2 (Injectivity at valuation-local generic points). *Let Λ be a valuation ring, let R' be the henselization of an essentially smooth local Λ -algebra, and let s be the image of its closed point in $\mathrm{Spec} \Lambda$. The fibre of $\mathrm{Spec} R'$ over s is irreducible; write η for its generic point. Suppose G is a finitary Nisnevich sheaf of spectra on qcqs Λ -schemes and, for every qcqs T , the two maps*

$$j_T^*, \pi_T^* \infty_T^* : \pi_* G(\mathbf{P}_T^1) \longrightarrow \pi_* G(\mathbb{A}_T^1) \quad \text{agree.} \quad (16)$$

Here j_T is the standard affine chart, ∞_T its complementary section, and $\pi_T : \mathbb{A}_T^1 \rightarrow T$ the projection. Then $\pi_n G(R') \rightarrow \pi_n G(R'_\eta)$ is injective for every integer n .

In this setting, put $\Lambda' = (\Lambda_s)^h$. Then R'_η is a valuation ring with the value group of Λ' and is ind-smooth over Λ' . In particular it has rank one if Λ' has rank one, and is a field if Λ' is a field.

Let T be local and let $a \in \mathfrak{m}_T$ satisfy $V(a) = \{\mathfrak{m}_T\}$ set-theoretically. For a finitary Nisnevich sheaf G of spectra or complexes,

$$G(T) \simeq G(T[1/a]) \times_{G(T^h[1/a])} G(T^h).$$

If T is a rank-one valuation ring and $a \neq 0$, the two localized rings here are $\mathrm{Frac} T$ and $\mathrm{Frac} T^h$.

Proof. The irreducibility and the reduction replacing Λ by $(\Lambda_s)^h$ are the first claim and first reduction of (Bouis and Kundu, 2025, arXiv v2, Theorem 3.2). Thus assume that Λ is henselian and s is its closed point. We give the remaining argument at the level of fibre spectra.

Let $\sigma \in \pi_n G(R')$ vanish after localization at η . Finitarity gives $f \notin \eta$ such that $\sigma|_{R'_f} = 0$. The support sequence gives a lift of σ to $\pi_n G_{V(f)}(R')$. Expressing R' as the filtered colimit of affine Nisnevich neighbourhoods of a smooth affine finite-presentation model (X, x) , and using that filtered colimits commute with fibres, descends this lift to $\sigma_0 \in \pi_n G_Z(X)$ with $Z = V(f_0)$.

After shrinking X , its closed fibre is irreducible. The flat map from the fibre of $\mathrm{Spec} R'$ has image containing the generizations of x , so its generic point maps to the generic point of this closed fibre. As $f(\eta) \neq 0$, Z_s has positive codimension in X_s . The presentation lemma (Bouis and Kundu, 2025, arXiv v2, Corollary 2.2) gives a Nisnevich neighbourhood $X' \rightarrow X$, a smooth scheme T over a Nisnevich neighbourhood of $\mathrm{Spec} \Lambda$, and an étale map $\psi : X' \rightarrow \mathbb{A}_T^1$ such that $Z' = Z \times_X X'$ maps isomorphically to a closed subscheme $Z'' \subset \mathbb{A}_T^1$ finite over T and $Z' = X' \times_{\mathbb{A}_T^1} Z''$. Since Λ is henselian, X' can be used as a Nisnevich neighbourhood of x , with the same henselized local ring. The finite T -scheme Z'' is closed also in $\mathbf{P}_T^1$ and misses infinity.

Nisnevich excision, first for ψ and then for the open immersion $\mathbb{A}_T^1 \subset \mathbf{P}_T^1$, gives

$$G_{Z''}(\mathbf{P}_T^1) \simeq G_{Z''}(\mathbb{A}_T^1) \simeq G_{Z'}(X').$$

This uses excision on spectra, not a claim that the kernels of maps of individual homotopy presheaves satisfy excision. Let τ correspond to $\sigma_0|_{X'}$, and let $\tilde{\tau}$ be its image in $\pi_n G(\mathbf{P}_T^1)$.

Restriction to infinity is zero because the class is supported on Z'' . By (16), its restriction to $\mathbb{A}_T^1$ is zero. Pulling back to X' and then to $\text{Spec } R'$ shows $\sigma = 0$.

The generic local ring. Smoothness gives étale coordinates over $\mathbf{A}_{\Lambda'}^r$ near the point under consideration. The local ring at the generic point of the closed fibre of the affine space is

$$G_0 = \Lambda'[t_1, \dots, t_r]_{\mathfrak{m}_{\Lambda'} \Lambda'[t]}.$$

It is the Gauss valuation ring. To verify this directly, extend the valuation to polynomials by the minimum of the finitely many coefficient values. Divide a polynomial by a coefficient of smallest value; its reduction is a nonzero residue polynomial. Products of those reductions are nonzero, so the minimum rule is multiplicative. For a rational function of nonnegative value, dividing numerator and denominator by a coefficient of minimum value in the denominator makes the numerator integral and the denominator primitive. It therefore lies in the displayed localization. Conversely that localization has nonnegative value. The coefficient field already realizes all values, proving the claim about the value group.

At each affine Nisnevich neighbourhood of x , the relevant generic local ring is a localization of an étale G_0 -algebra at a prime over its maximal ideal. It is a weakly unramified extension of valuation rings by (The Stacks Project Authors, 2026, Tags 0ASJ and 0ASG). The transition maps are local and flat, hence injective. Their filtered colimit is R'_η , which is therefore a valuation ring with this same value group. The localizations and étale maps are ind-smooth over Λ' .

Henselization excision. Present T^h by pointed affine étale neighbourhoods of $\text{Spec } T$. Their closed fibres are finite discrete schemes. By localizing at an element nonzero at the distinguished point and zero at the other closed-fibre points, arrange that the fibre over the reduced closed point of $\text{Spec } T$ is exactly $\text{Spec } \kappa(T)$. Each resulting étale neighbourhood and the open $\text{Spec } T[1/a]$ form a Nisnevich square. Pass its cartesian value square to the filtered colimit. For a rank-one valuation ring the only primes are zero and the maximal ideal, and its henselization has the same value group by (The Stacks Project Authors, 2026, Tag 0ASK). $\square$

Lemma 3.3 (The rank-one local upper bound). *Let Λ be a henselian valuation ring of rank at most one with fraction field of characteristic zero, let X be smooth over Λ , and let $R' = \mathcal{O}_{X,x}^h$. Then $M_p(i)(R') \in D^{\leq i}$ for every $i \geq 0$.*

Proof. The finite-coefficient projective-line formula with its natural Chern class, proved in proposition B.5, shows directly that $M_p(i)$ satisfies (16): the base summand restricts by projection, and the Chern class of $\mathcal{O}(1)$ restricts to that of a trivial line bundle on both the affine chart and infinity. Thus lemma 3.2 reduces the upper bound to a field or a possibly nonhenselian rank-one valuation ring V with characteristic-zero fraction field K .

At every henselian valuation ring W , including fields, one has

$$M_p(i)(W) \simeq \tau^{\leq i} \mathcal{S}_p(i)(W). \quad (17)$$

This is (Bouis, 2026, arXiv v1, Proposition 5.3); its proof uses (Bouis, 2024, arXiv v3, Theorem 5.10) and cdh-locality of henselian valuation rings. For a field of characteristic zero, its top group is naturally

$$H^i M_p(i)(K) = H_{\text{ét}}^i(K, \mu_p^{\otimes i}) = K_i^M(K)/p,$$

by (Bouis, 2026, arXiv v1, Proposition 5.5) and (Voevodsky, 2011, Theorem 6.16). The identification uses products of unit classes and is natural in K .

Let $K^h = \text{Frac } V^h$. The square of [lemma 3.2](#) gives the upper bound except possibly in degree $i + 1$, whose group is the cokernel of

$$H^i M_p(i)(K) \oplus H^i M_p(i)(V^h) \longrightarrow H^i M_p(i)(K^h).$$

For $i = 0$ the field restriction on constant $\mathbf{Z}/p$ is an isomorphism. For $i > 0$, it suffices to show that

$$K^\times \longrightarrow (K^h)^\times / ((K^h)^\times)^p \quad \text{is surjective.} \quad (18)$$

Here is the rank-one argument for that surjectivity. The extension $V \subset V^h$ preserves the value group by ([The Stacks Project Authors, 2026](#), Tag 0ASK); hence write any $z \in (K^h)^\times$ as bu , with $b \in K^\times$ and $u \in (V^h)^\times$. The completion of the rank-one valued field K is henselian: Newton iteration at a simple residue root has errors whose values at least double at each step, and positive multiples of a fixed value are cofinal in a rank-one value group. The sequence converges to a root in the complete field. The universal property embeds K^h in this completion, with its given valuation. Thus K is dense in K^h . Choose $u_0 \in V^\times$ so that $u/u_0 = 1 + y$, with $v(y) > 2v(p)$ if $p \in \mathfrak{m}_V$, and $v(y) > 0$ if p is a unit. If p is a unit, the simple root of $Z^p - (1 + y)$ modulo the maximal ideal lifts in V^h . Otherwise the polynomial

$$g(Z) = \frac{(1 + pZ)^p - 1 - y}{p^2} \in V^h[Z]$$

reduces to Z for odd p , and to $Z + Z^2$ for $p = 2$. In both cases zero is a simple residue root. The simple-root form of Hensel's lemma applies also to nonmonic polynomials ([The Stacks Project Authors, 2026](#), Tag 04GG). For its lift w we have $(1 + pw)^p = 1 + y$, proving (18). Multilinearity of Milnor symbols makes the map of top groups surjective, so the possible group in degree $i + 1$ vanishes. $\square$

Theorem 3.4 (Polynomial invariance over rank-one valuation rings). *Let V be a henselian rank-one F -smooth valuation ring of mixed characteristic $(0, p)$. For every $q \geq 0$ and $r \geq 0$, the pullback*

$$M_p(q)(V) \longrightarrow M_p(q)(V[t_1, \dots, t_r])$$

is an equivalence. Moreover $M_p(q)(V) \simeq F_p(q)(V) \simeq A_p(q)(V)$, so $Q_p(q)(V) = 0$. The same polynomial-invariance conclusion holds for $M_p(q)(X)$ for every ind-smooth qcqs V -scheme X .

Proof. The one-variable assertion on V . We first prove $\mathcal{D}_p(q)(V) = 0$. Write $T^i(X) = \text{Fil}_{\text{mot}}^i K(X)/p$. Its graded pieces are $M_p(i)(X)[2i]$ and $T^0(X) = K(X)/p$. For the tests V and $V[t]$ used here, completeness follows directly from the finite-coefficient pullback ([Bouis, 2024](#), arXiv v3, Proposition 3.26 and Remark 3.29), as proved in [lemma B.2](#). Their valuative dimensions are one and two, respectively. No integral or rational completeness theorem is used. Set

$$\Psi^i = \text{fib}(T^i(V) \longrightarrow T^i(V[t])).$$

Every valuation ring is K -regular by ([Kelly and Morrow, 2018](#), arXiv v1, Theorem 3.3(i)), so $\Psi^0 = 0$.

The upper bound of [lemma 3.3](#) at V and at all Nisnevich local rings of $V[t]$ gives $M_p(i)(V) \in D^{\leq i}$. The scheme $\text{Spec } V[t]$ has dimension at most two: its valuative dimension is two by ([Elmanto et al., 2021](#), arXiv v3, Proposition 2.3.2(7),(8)). Its Nisnevich topos has homotopy dimension at most two and is Postnikov complete by ([Clausen and Mathew, 2021](#), arXiv v3, Theorems 3.2 and 3.18). Nisnevich descent consequently gives $M_p(i)(V[t]) \in D^{\leq i+2}$.

In spectrum notation the two i th graded pieces are respectively i -connective and $(i - 2)$ -connective. This also gives

$$T^i(V) \in \mathrm{Sp}_{\geq i}, \quad T^i(V[t]) \in \mathrm{Sp}_{\geq i-2}, \quad \Psi^i \in \mathrm{Sp}_{\geq i-3}. \quad (19)$$

Indeed each finite quotient T^i/T^n has the corresponding lower connectivity bound; for fixed k its π_k and π_{k+1} stabilize as n increases because the next graded piece is $(n - 2)$ -connective. Completeness identifies T^i with the inverse limit of these quotients, and stabilization eliminates the possible $\lim^1$ loss. The last bound in (19) is the fibre bound, with a loss of one on the target.

Next, V and $V[t]$ are p -torsionfree, p -quasisyntomic and F -smooth by the stability results in section 3. The motivic comparison with $\mathcal{E}_p(q)$ is an isomorphism in degrees at most $q - 1$ and injective in degree q on both rings. Put $K = \mathrm{Frac} V = V[1/p]$ and $a : \mathrm{Spec} K[t] \rightarrow \mathrm{Spec} K$. This projection is the scheme of splittings of $0 \rightarrow \mathcal{O} \rightarrow \mathcal{O}^{\oplus 2} \rightarrow \mathcal{O} \rightarrow 0$. For $\Lambda = \mathbf{Z}/p$, the unit $\Lambda \rightarrow Ra_*\Lambda$ is an isomorphism by (Riou, 2014, Proposition 1.5, pp. 303–304). The sheaf $\mu_p^{\otimes q}$ is étale locally free of rank one over Λ , so the same statement holds after this twist. Taking derived global sections gives $\mathcal{E}_p(q)(V) \simeq \mathcal{E}_p(q)(V[t])$.

The pullback $M_p(q)(V) \rightarrow M_p(q)(V[t])$ is split by $t = 0$, so its cofiber cohomology is its degreewise cokernel. It vanishes below degree q by the comparison isomorphisms. In degree q , for $w \in H^q M_p(q)(V[t])$, the difference $w - \mathrm{pr}^*(w|_{t=0})$ maps to zero in $H^q \mathcal{E}_p(q)(V[t])$. Injectivity of the motivic comparison there makes that difference zero. Consequently

$$\mathcal{D}_p(q)(V) \in D^{\geq q+1}. \quad (20)$$

The graded fibre of Ψ is

$$\mathrm{gr}^q \Psi = \mathcal{D}_p(q)(V)[2q - 1], \quad \pi_k \mathrm{gr}^q \Psi = H^{2q-1-k} \mathcal{D}_p(q)(V) = 0 \quad (k \geq q - 1).$$

If $\Psi^q = 0$, the graded-fibre triangle identifies $\mathrm{gr}^q \Psi \simeq \Sigma \Psi^{q+1}$. By (19) the latter is $(q - 1)$ -connective. Its homotopy groups thus vanish both below $q - 1$ and at or above $q - 1$, so it is zero and $\Psi^{q+1} = 0$. Induction from $\Psi^0 = 0$ proves $\mathcal{D}_p(q)(V) = 0$ in every weight. This is an induction for all integers $q \geq 0$, not a finite test of the indices.

Smooth schemes and polynomial extensions. The defect $\mathcal{D}_p(q)$ is a finitary Nisnevich sheaf. The natural projective-line bundle formula for M_p induces one for $\mathcal{D}_p$ by taking cofibers of the formulas on X and $\mathbf{A}_X^1$. The Chern class on the second summand restricts to zero on the affine chart and on infinity; thus it satisfies (16).

Take a smooth finite-presentation V -scheme X and a Nisnevich local ring $R' = \mathcal{O}_{X,x}^h$. If x lies over the generic point, $\mathcal{D}_p(q)(R') = 0$ by proposition 3.1. Otherwise lemma 3.2 injects every homotopy group of $\mathcal{D}_p(q)(R')$ into the corresponding group of $\mathcal{D}_p(q)(V_\eta)$, where $V_\eta = R'_\eta$. By lemma 3.2, this is a rank-one valuation ring, ind-smooth over V . Its henselization remains rank one and ind-smooth, with characteristic-zero fraction field. It is p -torsionfree, p -quasisyntomic and F -smooth. Thus the rank-one argument above kills $\mathcal{D}_p(q)(V_\eta^h)$. The two fraction-field terms in lemma 3.2 vanish by proposition 3.1. That square kills $\mathcal{D}_p(q)(V_\eta)$, proving that all Nisnevich stalks on X vanish.

A smooth finite-presentation scheme over a rank-one valuation ring has finite Krull dimension, by the valuative dimension bound for finite-type morphisms in (Elmanto et al., 2021, arXiv v3, Proposition 2.3.5 and Corollary 2.3.3). The Nisnevich topos is therefore Postnikov complete by the Clausen–Mathew results cited above. Vanishing of all its stalks gives $\mathcal{D}_p(q)(X) = 0$. Finitarity extends this to ind-smooth schemes. Applying it successively to $\mathbf{A}_V^{r-1}$ proves the displayed assertion.

The presheaf-level $\mathbb{A}^1$ -localization is the Suslin realization

$$A_p(q)(V) = \operatorname{colim}_{[n] \in \Delta^{\text{op}}} M_p(q)(V \times \Delta^n), \quad \Delta_V^n \simeq \mathbf{A}_V^n.$$

This formula is used in (Bachmann et al., 2025, arXiv v1, proof of Corollary 9.5). All its augmentations are now equivalences, compatibly with the simplicial structure, so $M_p(q)(V) \rightarrow A_p(q)(V)$ is an equivalence. Henselian valuation rings are cdh-local, and a finitary presheaf retains its value there under cdh sheafification; see (Bouis, 2026, arXiv v1, proof of Proposition 5.3). Hence $M_p(q)(V) \rightarrow F_p(q)(V)$ is an equivalence. Compatibility of the localization units identifies their composite with $M_p(q)(V) \rightarrow A_p(q)(V)$, proving the result. $\square$

Proposition 3.5 (Reduction to the rank-one part at p). *Let V be a mixed-characteristic valuation ring, of arbitrary rank. Let $P = \sqrt{pV}$, let J be the union of its primes not containing p , and put*

$$T = V_P, \quad W_V = T/JT.$$

Then P is prime, J is the largest prime not containing p , and there is no prime strictly between J and P . Moreover

$$JT = \bigcap_{n \geq 1} p^n T,$$

and W_V is a rank-one mixed-characteristic valuation ring. If V is F -smooth, so is W_V . If V is henselian, then T and W_V are henselian.

For a henselian valuation ring V of any rank, $Q_p(q)(V) = 0$ if $p = 0$ or p is a unit in V . Otherwise

$$Q_p(q)(V) \simeq Q_p(q)(W_V).$$

Consequently $Q_p(q)(V) = 0$ in every weight whenever such a V is F -smooth, by theorem 3.4. There is no induction on the rank and no finite-rank restriction.

Proof. Ideals in a valuation ring are totally ordered. The radical of any proper ideal is prime: if a product belongs to that radical, comparison of the principal ideals of its two factors puts a power of one factor in the original ideal. This applies to pV . The union defining J is a prime ideal and omits p . Every prime omitting p is contained in J , and every prime containing p contains P , proving adjacency.

Use the valuation on T and put $I = \bigcap_n p^n T$. If $x, y \notin I$, some a, b satisfy $v(x) < av(p)$ and $v(y) < bv(p)$; then $v(xy) < (a+b)v(p)$, so I is prime. It does not contain p . If $\mathfrak{q}$ is a prime of T omitting p and $x \in \mathfrak{q}$ were outside I , then $p^n \in xT$ for some n , contrary to $p \notin \mathfrak{q}$. Thus I is the largest prime of T omitting p , namely JT . Quotienting T by this prime leaves exactly two primes. Its fraction field is $\kappa(J)$, of characteristic zero because $p \notin J$ and every integer prime to p was already a unit. This proves the rank and characteristic claims.

If V is F -smooth, its localization T is F -smooth. As $JT \subset p^n T$ for every n , $T/p^n \simeq W_V/p^n$. Both rings are p -torsionfree. The derived p -completions therefore agree. Their cotangent complexes modulo p agree over $\mathbf{F}_p$, so p -quasisyntomicity passes from T to W_V ; the defining prismatic conditions then pass because they depend only on this completion.

For the henselian assertion we prove the required localization statement directly, using the simple-root criterion (The Stacks Project Authors, 2026, Tag 04GG(2)). For any prime $\mathfrak{p}$ of a valuation ring V , one has $\mathfrak{p}V_{\mathfrak{p}} = \mathfrak{p}$ inside $\operatorname{Frac} V$: if $x \in \mathfrak{p}$ and $s \notin \mathfrak{p}$, comparison of xV and sV gives $x/s \in \mathfrak{p}$. Given $f \in V_{\mathfrak{p}}[X]$ with a simple residue root, translate a lift of that

root to zero and divide by $f'(0)$. The resulting polynomial is $h(X) = a_0 + X + \sum_{i \geq 2} a_i X^i$, with $a_0 \in \mathfrak{p}$. Choose $c \in V \setminus \mathfrak{p}$ clearing the finitely many denominators of the a_i . Then

$$g(X) = h(cX)/c = a_0/c + X + \sum_{i \geq 2} a_i c^{i-1} X^i \in V[X].$$

Here $g(0) \in \mathfrak{p}$ and $g'(0) = 1$. Henselianity of V gives $b \in \mathfrak{m}_V$ with $g(b) = 0$. The coefficient of b in this equation is $1 + \sum_{i \geq 2} a_i c^{i-1} b^{i-1} \in V^\times$, so $b \in \mathfrak{p}$. Thus cb gives the required root modulo $\mathfrak{p}$ in $V_{\mathfrak{p}}$, proving that every such localization is henselian. Apply this with $\mathfrak{p} = P$ to obtain henselianity of T . The same simple-root criterion passes to every quotient of a henselian local ring, and hence gives henselianity of $W_V = T/JT$. It also gives henselianity of every induced residue valuation $V/\mathfrak{p}$.

The first cases are [proposition 3.1](#). In the remaining case the Milnor square at P has two lower terms V/P and $\kappa(P)$ of characteristic p , where Q_p vanishes. It gives $Q_p(V) \simeq Q_p(V_P)$. The Milnor square for V_P at JV_P has localization V_J and residue field $\kappa(J)$, both $\mathbf{Q}$ -algebras, where Q_p also vanishes. This gives $Q_p(V_P) \simeq Q_p(W_V)$. Apply [theorem 3.4](#) when V is F -smooth. $\square$

4 Residue-characteristic comparison

We first prove the regular-surface comparison without excellence. The excellent dimension-three argument then uses regular models and two separate descent constructions: finite supports for Q_p , and bounded-below cdh detection for the étale comparison cofiber.

4.1 Regular surfaces

Lemma 4.1 (The noetherian support filtration). *Let G be a finitary Nisnevich sheaf of spectra on qcqs schemes, and let Y be a noetherian scheme of dimension at most d , with $d \geq 1$. Write*

$$h(y) = \dim \mathcal{O}_{Y,y}, \quad Y^{(r)} = \{y \in Y : h(y) = r\}, \quad X_y = \operatorname{Spec} \mathcal{O}_{Y,y}^h, \quad U_y = X_y \setminus \{y\}.$$

For $0 \leq r \leq d+1$, put

$$G^{(r)}(Y) = \operatorname{colim}_{Z \in \mathcal{Z}_r(Y)} G_Z(Y), \quad \mathcal{Z}_r(Y) = \{Z \subset Y \text{ closed} : h(z) \geq r \text{ for every } z \in Z\},$$

where supports are ordered by inclusion. There are natural equivalences

$$\operatorname{cofib}(G^{(r+1)}(Y) \longrightarrow G^{(r)}(Y)) \simeq \bigoplus_{y \in Y^{(r)}} G_{\{y\}}(X_y) \quad (0 \leq r \leq d). \quad (21)$$

Assume that

$$G_{\{y\}}(X_y) = 0 \quad (h(y) < d), \quad G(U_y) = 0 \quad (h(y) = d). \quad (22)$$

Then the map induced by finite closed supports is an equivalence

$$\bigoplus_{y \in Y^{(d)}} G(X_y) \xrightarrow{\sim} G(Y). \quad (23)$$

Every point of $Y^{(d)}$ is closed, and each homotopy class has finitely many nonzero components. Its component at y is its restriction to X_y . The lower-height condition in [\(22\)](#) follows, in particular, from $G(X_y) = G(U_y) = 0$ at those points. No excellence or catenarity assumption is made.

Proof. The support posets are filtered, since they are closed under finite unions. We have $G^{(0)}(Y) = G(Y)$ and $G^{(d+1)}(Y) = 0$. If z is a proper specialization of y , a prime chain ending at y can be extended by z ; hence $h(z) > h(y)$. This elementary inequality is the only dimension property needed for the filtration.

For closed supports $W \subset Z$, the definition of supported values gives a fibre sequence

$$G_W(Y) \longrightarrow G_Z(Y) \longrightarrow G_{Z \setminus W}(Y \setminus W).$$

Filtered colimits preserve this sequence. To calculate the layer at height r , one may therefore use pairs $W \in \mathcal{Z}_{r+1}(Y)$, $Z \in \mathcal{Z}_r(Y)$, with $W \subset Z$: replacing any pair by $(W, W \cup Z)$ gives this cofinal indexing.

A fixed Z has finitely many irreducible components. Include in W its components whose generic points have height greater than r , and the intersections of the remaining components. These subsets have height at least $r+1$. The remaining components of $Z \setminus W$ are disjoint closed supports in $Y \setminus W$, with generic points in $Y^{(r)}$. Excision splits their supported spectra as a finite direct sum. Enlarging W by proper closed subsets of these components removes all their non-generic points in the filtered limit. Zariski excision allows the ambient neighbourhood of each generic point to shrink independently. Finitarity then identifies the resulting value with

$$G_{\{y\}}(\mathrm{Spec} \mathcal{O}_{Y,y}).$$

Every $y \in Y^{(r)}$ occurs by using its closure as a support, and the filtered union of these finite sums is the direct sum over $Y^{(r)}$.

For completeness, the supported henselization identification uses pointed affine étale neighbourhoods of the local scheme $\mathrm{Spec} \mathcal{O}_{Y,y}$. Their closed fibres are finite discrete schemes. A principal shrinking keeps the distinguished point, with residue field $\kappa(y)$, and removes the other closed-fibre points. Nisnevich excision identifies its supported value with that on the original local scheme. Passing to the filtered limit gives $G_{\{y\}}(X_y)$. Complements are quasi-compact, and their inverse systems have affine transition maps, so finitarity applies also to these supported values. This proves (21) on spectra.

The vanishing assumptions kill every layer below d . At height d , $G_{\{y\}}(X_y) = G(X_y)$ because $G(U_y) = 0$, so the finite tower gives (23). A point of height d has no proper specialization in Y and is closed. A closed support consisting of such points is a finite set, by noetherianity. Excision gives the map in the displayed equivalence and identifies its components with the corresponding restrictions. Finally, homotopy groups commute with direct sums of spectra, which proves the assertion about homotopy classes. $\square$

In particular, if G vanishes on all excellent noetherian schemes of dimension less than d , the lemma applies to every excellent Y of dimension at most d . Its henselized local schemes are excellent, and their punctures have strictly smaller dimension. The same assertion holds without excellence when vanishing is known on the larger class of all lower-dimensional noetherian schemes.

For the surface comparison, M_p is a finitary Nisnevich sheaf (Bouis, 2024, arXiv v3, Theorem C), and $\mathcal{N}_p$ has the same properties. On a henselian valuation ring, the canonical map $M_p \rightarrow F_p$ is an equivalence, by cdh locality and finitarity (Bouis, 2026, arXiv v1, proof of Proposition 5.3). The regular blow-up formula for M_p holds for every regular closed immersion of qcqs schemes (Bouis, 2025, arXiv v1, Theorem 4.18).

Theorem 4.2 (Comparison on regular surfaces). *Let R be a henselian regular noetherian local ring of dimension at most two, with $0 \neq p \in \mathfrak{m}_R$. Then, for every $q \geq 0$, the canonical*

maps

$$M_p(q)(R) \xrightarrow{\sim} F_p(q)(R) \xrightarrow{\sim} A_p(q)(R)$$

are equivalences. Excellence, restrictions on the residue field, and restrictions on ramification are unnecessary. The first equivalence uses the valuation and regular blow-up inputs for M_p , and does not use [theorem 3.4](#) or rank reduction.

Proof. Fix p, q , and write G for either $\mathcal{N}_p(q)$ or $Q_p(q)$. Both are finitary Nisnevich sheaves. The preceding valuation comparison kills $\mathcal{N}_p$ on fields and henselian DVRs. For Q_p , fields are killed by [proposition 3.1](#). A henselian DVR in which $p = 0$ or p is a unit is killed by the same proposition. In the remaining case it is mixed characteristic and F-smooth, so [theorem 3.4](#) kills its Q_p -value. Thus G vanishes on fields and henselian DVRs.

Apply [lemma 4.1](#) with $d = 1$ to a regular noetherian scheme of dimension at most one. Its henselized local rings are fields or DVRs, and the puncture of a DVR is its fraction field. The hypotheses hold and every top local value is zero. Consequently

$$G(T) = 0 \quad \text{for every regular noetherian } T \text{ with } \dim T \leq 1. \quad (24)$$

On a regular noetherian scheme Y of dimension at most two, the lower-height local support spectra vanish for the same reason. The puncture of every height-two henselized local ring is regular noetherian of dimension at most one, so (24) kills it. Hence

$$G(Y) \simeq \bigoplus_{y \in Y^{(2)}} G(\mathcal{O}_{Y,y}^h). \quad (25)$$

In particular, for every two-dimensional regular noetherian local ring T , the natural map $G(T) \rightarrow G(T^h)$ is an equivalence: its puncture has zero value, and supported Nisnevich excision applies.

Let T be such a local ring with $0 \neq p \in \mathfrak{m}_T$, and let $B_T = \text{Bl}_{\mathfrak{m}_T} \text{Spec } T$ and $E_T = \mathbf{P}_{\kappa(T)}^1$. The blow-up is regular of dimension two. Every closed point lies over the closed point of T , by properness, and is a closed point of E_T , with a finite residue extension and local dimension two. The cdh blow-up square gives $Q_p(T) \simeq Q_p(B_T)$, since $Q_p(E_T) = Q_p(\kappa(T)) = 0$. The centre is a regular closed immersion, so the regular blow-up formula for M_p , together with cdh excision for F_p , gives the same cartesian square for $\mathcal{N}_p$. Its centre and exceptional curve have zero $\mathcal{N}_p$ -values by (24). Thus for either G ,

$$G(T) \simeq G(B_T) \simeq \bigoplus_{b \in E_T \text{ closed}} G(\mathcal{O}_{B_T,b}^h). \quad (26)$$

The components are restrictions, and each class has finitely many nonzero components.

The case $\dim R = 1$ has already been proved. Suppose $\dim R = 2$ and choose a nonzero class $\theta \in H^j G(R)$. Formula (26) supplies a closed point with a nonzero component. Applying the same formula repeatedly gives an infinite sequence of ordinary local quadratic transforms

$$R = T_0 \subsetneq T_1 \subsetneq T_2 \subsetneq \cdots \subset \text{Frac } R$$

at closed exceptional points. The selected images in $H^j G(T_i^h)$ are compatible and nonzero at every stage. Equivalently, these are the nonzero vertices along a branch of the finitely branching tree of components. The equivalence $G(T_i) \rightarrow G(T_i^h)$ ensures that the branch can be taken among the ordinary quadratic transforms; no finite-type model of R is required.

Put $V = \bigcup_i T_i$. The union is a valuation ring of $\text{Frac } R$ dominating R , by ([Lévy-Bruhl, 1957–1958](#), Lemma 5, printed pp. 19–9–19–10). That lemma is stated for arbitrary two-dimensional regular local rings and successive quadratic transforms with a common fraction

field. Its proof uses the general Lemma 3 and Property II in the same exposition. The algebraic or absolute hypotheses imposed in the later Property III are not hypotheses of Lemma 5. This citation refers to the published exposition and proof, rather than to an unconsulted locator in Abhyankar’s original article.

Henselization commutes with these filtered colimits along local maps:

$$V^h \simeq \operatorname{colim}_i T_i^h.$$

Indeed, a filtered colimit of henselian local rings along local maps is henselian by the finite-data simple-root criterion, and the universal property of henselization gives the displayed identity. The henselization of a valuation ring is a valuation ring ([The Stacks Project Authors, 2026](#), Tag 0ASK). It is mixed characteristic, because V dominates R and has fraction field $\operatorname{Frac} R$. Each T_i^h is regular noetherian, hence F -smooth and p -torsionfree; the filtered colimit is F -smooth and p -torsionfree. Finitarity gives

$$G(V^h) \simeq \operatorname{colim}_i G(T_i^h).$$

The image of θ is nonzero in this value: a class vanishes in a filtered colimit of homotopy groups only if it vanishes at some later stage, whereas the selected images are nonzero at every stage.

For $G = \mathcal{N}_p$, the henselian valuation comparison gives $G(V^h) = 0$. For $G = Q_p$, [proposition 3.5](#), using [theorem 3.4](#), gives $G(V^h) = 0$, since V^h is F -smooth. This contradicts the surviving class in either case. Therefore $\mathcal{N}_p(q)(R) = Q_p(q)(R) = 0$, proving both comparisons. The all-rank reduction makes a separate valuation-rank bound or equality case of Abhyankar’s inequality unnecessary here. $\square$

4.2 Regular models and excellent-scheme descent

We use resolution of integral excellent schemes in dimensions at most two: finite normalization for curves and ([Lipman, 1978](#), Theorem, p. 151) for surfaces. In dimension three we invoke Cossart–Piltant’s resolution theorem: a reduced separated noetherian quasi-excellent scheme of dimension at most three has a proper birational regular model, isomorphic over its regular locus ([Cossart and Piltant, 2019](#), arXiv v2, Theorem 1.1, p. 3). There is no residue-field or characteristic restriction, and no assertion that the model is obtained by blow-ups in regular centres is needed.

For a separated noetherian X , let Sch_X be its separated finite-type schemes. When X is excellent of finite dimension, for $d \geq 0$ define $\mathcal{W}_d(X)$ to consist of henselian valuation rings W mapping to an integral $Y' \in \operatorname{Sch}_X$ of dimension at most d , with generic point mapping to the generic point of Y' and $\operatorname{Frac} W$ algebraic over $\kappa(Y')$. A *trdeg-zero point* of Y is a map $\operatorname{Spec} W \rightarrow Y$ with W henselian valuation and $\operatorname{Frac} W$ algebraic over the residue field of the image of its generic point. It belongs to $\mathcal{W}_{\dim Y}(X)$ by taking the reduced closure of that image. There is no bound on rank or residue extension.

Proposition 4.3 (*F-smoothness from regular models*). *Let $d \leq 3$, invoking Cossart–Piltant’s resolution theorem when $d = 3$. For every mixed-characteristic $W \in \mathcal{W}_d(X)$, W is a filtered union of regular local rings of dimension at most d , dominated by W and essentially of finite type over X . Thus W is p -torsionfree, p -quasisyntomic and F -smooth, and the syntomic cofiber at W belongs to $D^{\geq q}$ for all $q \geq 0$.*

Proof. Choose Y' in the definition, put $K = \kappa(Y')$ and $L = \operatorname{Frac} W$, and choose an affine neighbourhood $U = \operatorname{Spec} R_U \subset Y'$ of the centre of W , over an affine $\operatorname{Spec} B \subset X$. It contains the image of $\operatorname{Spec} W$, and $R_U \rightarrow W$ is injective. For a finite subset $T \subset W$, set

$$A_T = (R_U[T])_{\mathfrak{m}_W \cap R_U[T]} \subset W.$$

Its unlocalized ring is an integral excellent finite-type B -algebra with fraction field $K(T)$ finite over K . The dimension formula for excellent schemes gives $\dim R_U[T] \leq \dim R_U \leq d$ ([The Stacks Project Authors, 2026](#), Tag 02IJ). The A_T are a directed family with union W .

Resolve $\text{Spec } R_U[T]$. In dimension one its finite normalization is regular; finiteness follows from excellence. In dimension two Lipman's theorem applies to the excellent surface. In dimension three use Cossart–Piltant's resolution theorem. Take the integral component of the regular model birational to this integral scheme. The valuative criterion lifts $\text{Spec } W$ to the proper model; at its centre c_T , the regular local ring $\mathcal{O}_{X_T, c_T}$ is a subring of W dominated by W , and contains A_T . Its dimension is at most d by the same dimension formula.

The regular local subrings so obtained can be made directed. Given two of them, choose finite sets of their generators as localizations of finite-type B -algebras and include these, as well as generators of R_U , in T . The ring $\mathcal{O}_{X_T, c_T}$ then contains both: every element inverted in either localization is outside $\mathfrak{m}_W$ and hence invertible in this local ring. Their union contains all A_T and equals W .

Each regular local ring is p -quasisyntomic and F -smooth by ([Bhatt and Mathew, 2023](#), arXiv v2, Theorem 4.15 and its proof). These properties pass to the filtered union; p -torsionfreeness also passes, or follows from the characteristic-zero fraction field of W . Apply (14) and shift from fibre to cofiber. $\square$

Theorem 4.4 (Comparison on excellent schemes of dimension at most three). *Let Y be an excellent noetherian scheme of dimension at most three. The dimension-three proof invokes Cossart–Piltant's resolution theorem. Then $Q_p(q)(Y) = 0$ for every prime p and every $q \geq 0$. Thus $F_p(q)(Y) \rightarrow A_p(q)(Y)$ is an equivalence. The assertion allows singularities, arbitrary residue fields, and arbitrary ramification.*

Proof. Fix p, q . Let $\text{Claim}(d)$ assert the vanishing for all excellent noetherian schemes of dimension at most d . For $d = 0$, nilinvariance of a cdh sheaf reduces to a finite disjoint union of fields. The equal-characteristic and p -unit cases of [proposition 3.1](#) kill them.

Assume $\text{Claim}(d-1)$, with $1 \leq d \leq 3$. We may first treat separated schemes: finite affine coverings and their intersections then extend the answer to arbitrary excellent noetherian schemes by Nisnevich descent. Each such intersection is an open subscheme of an affine scheme, hence is separated and has the same dimension bound. Apply [lemma 4.1](#) to Q_p and Y . It suffices to kill $Q_p(R)$ for excellent henselian local R of dimension d .

Nilinvariance and abstract blow-up excision for a closed cover reduce this ring to a domain. Explicitly, if the reduced scheme is a union of its finitely many irreducible components, their pairwise intersections have dimension less than d whenever a component has dimension d ; components of smaller dimension are already zero. Induction on the number of components splits $Q_p(R)$ as the sum of the d -dimensional domain values. If $p = 0$ or p is a unit, [proposition 3.1](#) finishes. Thus take a domain R with $0 \neq p \in \mathfrak{m}_R$, fraction field K of characteristic zero, and closed point s .

For every nonzero ideal $I \subset R$, its blow-up Y_I is integral, excellent, proper and birational over R . Its dimension is at most d by the dimension formula. The centre $Z_I = V(I)$ and its inverse image E_I have dimension less than d : the latter is an effective Cartier divisor when the centre is nonempty. Abstract blow-up excision and $\text{Claim}(d-1)$ give

$$Q_p(R) \simeq Q_p(Y_I). \quad (27)$$

The closed points of Y_I lie over s by properness and have residue fields finite over $\kappa(R)$. Universal catenarity and the birational dimension formula give height d at these points

and height less than d elsewhere. Hence lemma 4.1 gives

$$Q_p(R) \simeq \bigoplus_{y \in Y_I \text{ closed}} Q_p(\mathcal{O}_{Y_I, y}^h).$$

Suppose a nonzero class $\theta \in H^j Q_p(R)$ exists. Let $S(Y_I)$ consist of the closed points where its component is nonzero. These are nonempty finite sets. The blow-ups of all nonzero ideals form a directed system: blowing up IJ dominates the blow-ups of I and of J . Indeed an invertible product of nonzero ideals on an integral scheme makes each factor invertible, and the universal property of blow-ups applies. Birational dominating morphisms are unique.

Restriction induces maps $S(Y_J) \rightarrow S(Y_I)$ along every dominating morphism $Y_J \rightarrow Y_I$. They are well defined: the pullback of a class supported at a finite set has support in its inverse image, and its restriction at a point over y is the restriction of the component at y along the local map of henselizations. In particular a zero component cannot acquire a nonzero pullback component. This argument does not require surjectivity of these maps. The inverse limit of these nonempty finite sets is nonempty. One proof is to put the discrete topology on each finite set; in the compact product, the finitely many compatibility conditions in any finite collection can be satisfied by choosing a point at a common dominating blow-up. The closed conditions have the finite intersection property.

Choose a compatible family (y_I) and put

$$V = \operatorname{colim}_I \mathcal{O}_{Y_I, y_I} \subset K.$$

This is a local subring dominating R . It is a valuation ring: for $f = a/b \in K^\times$, with $a, b \in R \setminus 0$, the ideal (a, b) is principal at the chosen centre on its blow-up, so f or f^{-1} is in its local ring. The residue field of V is algebraic over $\kappa(R)$. Henselization commutes with filtered colimits of local rings along local maps: a map to a henselian local target factors uniquely through each henselization, and the filtered colimit of henselian local rings is henselian by the finite-data simple-root criterion. Thus

$$V^h = \operatorname{colim}_I \mathcal{O}_{Y_I, y_I}^h.$$

It is a valuation ring by (The Stacks Project Authors, 2026, Tag 0ASK), and its fraction field is algebraic over K , since henselization is ind-étale. Consequently $V^h \in \mathcal{W}_d(\operatorname{Spec} R)$ and is mixed characteristic. By finitariness, the image of θ in $H^j Q_p(V^h)$ is nonzero: along the selected branch every subsequent component is nonzero, so the class is never killed in the filtered colimit.

But proposition 4.3 makes V^h F-smooth, using excellent-surface resolution for $d \leq 2$ and Cossart–Piltant’s resolution theorem for $d = 3$. Proposition 3.5 then kills $Q_p(V^h)$, a contradiction. This proves Claim(d) and completes the induction. At no step is polynomial extension of an excellent d -dimensional ring used to produce a regular model. $\square$

4.3 The independent cdh comparison bound

The following argument concerns

$$C_p(q) = \operatorname{cofib}(F_p(q) \rightarrow \mathcal{E}_p(q)), \quad \operatorname{fib}(F_p(q) \rightarrow \mathcal{E}_p(q)) = C_p(q)[-1]. \quad (28)$$

It does not use valuation polynomial invariance or the preceding vanishing theorem for Q_p .

Lemma 4.5 (Trdeg-zero detection of cdh covers). *Let X be separated noetherian, and let $Y \in \operatorname{Sch}_X$. A sieve on Y is a cdh covering sieve if every trdeg-0 point of Y factors through a member of the sieve.*

Proof. By (Gabber and Kelly, 2015, arXiv v2, Theorems 2.3 and 2.6, and Table 1), spectra of henselian valuation rings give a conservative family of cdh points on Sch_X . Coverings have finite refinements because the schemes are noetherian. It suffices to lift every map $g : \text{Spec } W \rightarrow Y$ from a henselian valuation ring, of arbitrary rank and fraction field.

Put $y = g(\eta_W)$, and let K_0 be the algebraic closure of $\kappa(y)$ inside $L = \text{Frac } W$. The intersection $W_0 = W \cap K_0$ is a valuation ring. Choose an affine neighbourhood of the image of the closed point of W in Y ; it contains the whole image. Its coordinate map has image in $\kappa(y) \cap W \subset W_0$, so g factors through $\text{Spec } W_0$. The inclusion $W_0 \rightarrow W$ is local and W is henselian, so it extends to $W_0^h \rightarrow W$. The henselization is a valuation ring by (The Stacks Project Authors, 2026, Tag 0ASK), with fraction field algebraic over K_0 and thus over $\kappa(y)$. The resulting map $\text{Spec } W_0^h \rightarrow Y$ is a trdeg-0 point. A lift of it through the sieve supplies a lift of g . $\square$

Proposition 4.6 (Cdh detection with a uniform lower bound). *Let X be separated excellent noetherian of finite dimension, let $d \geq 0$, and let G be a finitary cdh sheaf of complexes on qcqs X -schemes. Suppose $G(U) \in D^{\geq m_0}$ for every $U \in \text{Sch}_X$, for one integer m_0 . Then the following hold.*

1. *If $G(W) \in D^{\geq n}$ for every $W \in \mathcal{W}_d(X)$, then $G(Y) \in D^{\geq n}$ for every $Y \in \text{Sch}_X$ with $\dim Y \leq d$.*
2. *If $H^a G(U) = 0$ for all $a < j$ and all such U of dimension at most d , then for every $Y \in \text{Sch}_X$ with $\dim Y \leq d$, a class in $H^j G(Y)$ is zero if and only if its pullback is zero at every trdeg-0 point of Y .*

Proof. Let $(H_{<j})$ denote the vanishing in the second assertion. It holds initially for $j \leq m_0$. Under $(H_{<j})$, prove the second assertion by induction on $e = \dim Y$ and on the number of its irreducible components. Nilinvariance permits reduction to reduced Y . For a closed cover by one component Y_1 and the union Y_2 of the others, cdh excision gives

$$H^{j-1}G(Y_1 \cap Y_2) \longrightarrow H^j G(Y) \longrightarrow H^j G(Y_1) \oplus H^j G(Y_2).$$

The first group is zero by $(H_{<j})$. Restricting the trdeg-0 point condition to the components therefore reduces to the integral case. At dimension zero the identity of its field is itself a trdeg-0 point, so the assertion is immediate.

Let Y be integral of dimension $e > 0$, and let $c \in H^j G(Y)$ have zero pullbacks at its trdeg-0 points. Finitarity says that each of these points factors through some finite-type affine scheme $U \rightarrow Y$ where c becomes zero: express its valuation ring as the union of finite-type subalgebras over an affine neighbourhood in Y , then use vanishing of the class in the filtered colimit. The sieve of all such morphisms is covering by lemma 4.5. A finite refinement has the form

$$\{V_k \longrightarrow Y' \longrightarrow Y\},$$

with $Y' \rightarrow Y$ a proper cdh cover and $\{V_k \rightarrow Y'\}$ a Nisnevich cover. This refinement is proposition C.1. Its reduced-component and flattening proof repairs the argument of (Suslin and Voevodsky, 2000, Proposition 5.9, pp. 38–39, and Section 12, p. 67) and applies to arbitrary noetherian bases.

The generic point has a lift with its residue field unchanged, since a proper cdh cover is completely decomposed. The reduced closure Y'_0 of that lift is proper and birational over Y , and has dimension at most e by excellence and the dimension formula. It is an isomorphism outside a nowhere dense closed $Z \subset Y$; write $E = Y'_0 \times_Y Z$. Both Z and E have dimension less than e . The abstract blow-up sequence gives

$$H^{j-1}G(E) \longrightarrow H^j G(Y) \longrightarrow H^j G(Y'_0) \oplus H^j G(Z).$$

Its first term is zero, and the component of c on Z is zero by induction on dimension. On Y'_0 , the pulled-back Nisnevich cover has Čech terms étale over Y'_0 , all of dimension at most e . The descent spectral sequence

$$E_1^{a,b} = H^b G(V'^{(a)}) \implies H^{a+b} G(Y'_0)$$

converges with finitely many terms contributing to each total degree, because $b \geq m_0$. The restriction of c to its $a = 0$ term is zero. For $a \geq 1$, the terms contributing to degree j have $b = j - a < j$ and vanish by $(H_{<j})$. Thus its restriction to Y'_0 is zero, and excision kills c . This proves the second assertion.

For the first assertion, every trdeg-0 point of a scheme of dimension at most d belongs to $\mathcal{W}_d(X)$. For $j < n$ its $H^j G(W)$ is zero. The second assertion therefore advances $(H_{<j})$ to $(H_{<j+1})$. Starting at m_0 and inducting to n proves the bound. If $n \leq m_0$ it already holds. $\square$

Lemma 4.7 (The comparison cofiber on bounded-dimensional points). *The $C_p(q)$ of (28) is a finitary cdh sheaf and satisfies $C_p(q)(U) \in D^{\geq -1}$ for every qcqs U . Let X be separated excellent noetherian of finite dimension, and let $d \leq 3$, invoking Cossart–Piltant’s resolution theorem when $d = 3$. For every $Y \in \text{Sch}_X$ of dimension at most d ,*

$$C_p(q)(Y) \in D^{\geq q-1},$$

and the restriction map

$$H^{q-1} C_p(q)(Y) \longrightarrow \prod_{y \in Y_{\mathbb{F}_p}} H^{q-1} C_p(q)(\kappa(y)) \quad \text{is injective.} \quad (29)$$

Here $Y_{\mathbb{F}_p}$ means all points where the residue characteristic is p , with no perfection assumption.

Proof. The sheaf and finitary assertions for $C_p(q)$ follow from those of $F_p(q)$ and étale cohomology; they are used in (Bachmann et al., 2025, arXiv v1, proof of Lemma 8.11(1)). The uniform bound follows because $F_p(q)$ has no negative cohomology by that source’s Lemma 7.31 and $\mathcal{E}_p(q)$ has no negative cohomology.

At a henselian valuation ring W , cdh sheafification does not change the finitary syntomic values, and the comparison factorization of (28) gives a triangle

$$T_q(W) \longrightarrow C_p(q)(W) \longrightarrow \text{cofib}(\mathcal{S}_p(q)(W) \longrightarrow \mathcal{E}_p(q)(W)), \quad T_q(W) \in D^{\geq q+1}. \quad (30)$$

Here $T_q(W) = \text{cofib}(\tau^{\leq q} \mathcal{S}_p(q)(W) \rightarrow \mathcal{S}_p(q)(W))$. This is the pointwise triangle in the proof of Lemma 8.11(1), which itself does not require that the Val bound hold at every valuation over a base.

For $W \in \mathcal{W}_d(X)$, the cases are as follows. If p is invertible, the right term is zero, hence $C_p(q)(W) \in D^{\geq q+1}$. If W is mixed characteristic, proposition 4.3 and (14) put the right term in $D^{\geq q}$ and hence $C_p(q)(W) \in D^{\geq q}$. If $p = 0$ in W , it is Cartier smooth and

$$C_p(q)(W) \simeq \Omega_{W, \log}^q[-(q-1)].$$

This is (Bachmann et al., 2025, arXiv v1, Example 5.9 and proof of Lemma 8.11(2)), or follows by taking the cofiber of $F_p(q)(W) = \Omega_{W, \log}^q[-q]$ to its zero generic-fibre étale term. All cases give $D^{\geq q-1}$. Proposition 4.6(1) proves the bound on Y .

For injectivity, let c in degree $q-1$ restrict to zero at every characteristic- p residue field. The bound just proved allows use of proposition 4.6(2). At a trdeg-0 point W where p is a

unit or W is mixed, its degree $q - 1$ group is already zero. If $p = 0$ in W , its restriction to $\text{Frac } W$ factors through the residue field of the image of its generic point, where the given class is zero. The map $\Omega_{W, \log}^q \rightarrow \Omega_{\text{Frac } W, \log}^q$ is injective. Precisely this unconditional valuation injectivity is applied in (Bachmann et al., 2025, arXiv v1, proof of Lemma 8.11(3), equation (8.12)); it follows from the torsionfreeness of valuation differential forms. Thus c dies at every trdeg-0 point, and bounded detection kills it. $\square$

Lemma 4.8 (A residue-preserving valuation). *Let R be regular local of dimension $e \geq 1$ with $0 \neq p \in \mathfrak{m}_R$. There is a valuation ring C of $\text{Frac } R$ dominating R with $\kappa(C) = \kappa(R)$. Its henselization $V = C^h$ is mixed characteristic, induces the same residue-field isomorphism, and has fraction field algebraic over $\text{Frac } R$.*

Proof. Choose a regular system of parameters $(x_1, \dots, x_e)$ and put $R_i = R/(x_1, \dots, x_i)$. For $1 \leq i \leq e - 1$, the localization of R_{i-1} at (x_i) is a DVR O_i with residue field $\text{Frac } R_i$. Begin with the DVR $C_e = R_{e-1}$, whose residue field is $\kappa(R)$. Descending on i , put

$$C_i = \{z \in O_i : z \bmod \mathfrak{m}_{O_i} \in C_{i+1}\}.$$

This is the composite valuation ring of O_i and the valuation ring C_{i+1} in its residue field (The Stacks Project Authors, 2026, Tag 088Z). It is a valuation ring of $\text{Frac } R_{i-1}$ and has the residue field of C_{i+1} . The image of R_{i-1} in $\text{Frac } R_i$ lies in $R_i \subset C_{i+1}$. Thus C_i contains R_{i-1} , and the map to its residue field is the residue map of R_{i-1} . It follows that $C = C_1$ dominates R and preserves its residue field. Henselization preserves the residue field and the valuation-ring property (The Stacks Project Authors, 2026, Tags 04GN and 0ASK); ind-etaleanness gives the fraction-field algebraicity. Since R is a domain and $p \neq 0$ in R , its fraction field and that of V have characteristic zero, and p remains in the maximal ideal. $\square$

Proposition 4.9 (The cdh Beilinson–Lichtenbaum bound). *Let Y be regular excellent noetherian, of dimension at most three, and suppose that p is a nonzerodivisor on Y . The dimension-three proof invokes Cossart–Piltant’s resolution theorem. Then, for every $q \geq 0$,*

$$\text{fib}(F_p(q)(Y) \longrightarrow \mathcal{E}_p(q)(Y)) \in D^{\geq q+1}. \quad (31)$$

Equivalently the comparison is an isomorphism in degrees at most $q-1$ and injective in degree q . All primes, including 2 and 3, arbitrary residue fields, and arbitrary ramification are allowed. For dimension at most two, the dimension-three resolution theorem is unnecessary.

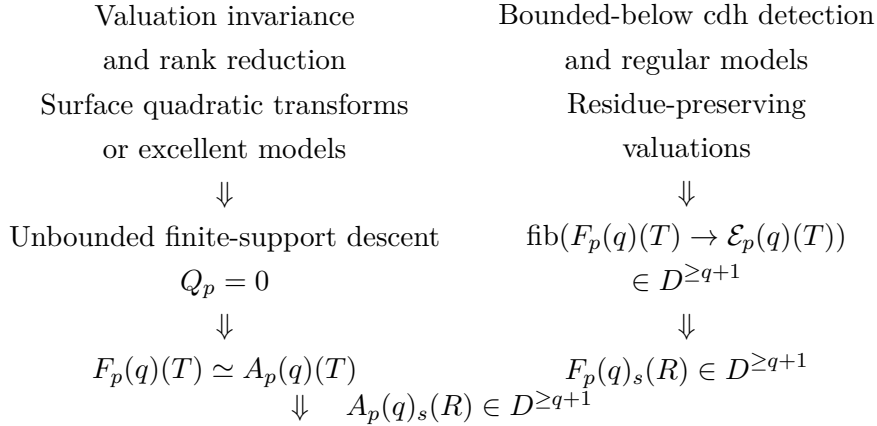
Proof. First assume Y separated and take $X = Y$ in lemma 4.7. We know that $C_p(q)(Y) \in D^{\geq q-1}$ and that (29) is injective. Let $c \in H^{q-1}C_p(q)(Y)$, and let x have characteristic- p residue field. The ring $R = \mathcal{O}_{Y,x}$ is a regular local domain of positive dimension, with $0 \neq p$ in its maximal ideal. Use lemma 4.8 to obtain $g : \text{Spec } V \rightarrow Y$ whose closed point is x and whose residue field is equal to $\kappa(x)$. Its fraction field is algebraic over the function field of the component of Y containing x ; hence $V \in \mathcal{W}_{\dim Y}(Y)$. Proposition 4.3 makes it F -smooth, with Cossart–Piltant’s resolution theorem used only when the dimension is three. The mixed-characteristic case of (30) gives $C_p(q)(V) \in D^{\geq q}$, so $g^*c = 0$. Restricting further to $\kappa(V) = \kappa(x)$ shows that $c|_{\kappa(x)} = 0$. This holds at every characteristic- p point; injectivity of (29) yields $c = 0$. Therefore $C_p(q)(Y) \in D^{\geq q}$. Shifting by $[-1]$ gives (31).

For nonseparated Y , take a finite affine covering. Every multiple intersection is an open subscheme of an affine scheme and thus is separated, regular, excellent, and satisfies the same dimension and p -torsionfreeness bounds. The comparison fibre satisfies Zariski descent. Since $D^{\geq q+1}$ is closed under limits, its bound on the Čech terms gives the bound on Y . $\square$

The equality of residue fields in [lemma 4.8](#) is essential: in $K = \mathbf{F}_p(t)$, the logarithmic differential $d \log t = dt/t$ is nonzero, but its image in $L = \mathbf{F}_p(u)$ with $t = u^p$ is $d \log(u^p) = 0$. The nonzerodivisor hypothesis is also essential to this bound; in characteristic p the logarithmic term lies one degree lower. The proof uses only $\mathcal{W}_d(Y)$, rather than the unrestricted valuation hypothesis of ([Bachmann et al., 2025](#), arXiv v1, Section 8.1). The proper part of descent is handled by abstract blow-up squares. Only the Nisnevich part uses a Čech nerve, whose terms preserve the dimension bound.

5 Integral purity and regular pairs

The mixed-characteristic cases of [Theorem A](#) use the following proof dependencies. The two branches are independent:



Here $T = R, U$, where $U = \text{Spec } R \setminus \{s\}$, and $\mathcal{E}_p(q)_s(R) = 0$. In the nonexcellent surface case, the right-hand bound follows from the comparison $M_p \simeq F_p$ in the first assertion of [theorem 4.2](#), together with the F-smooth Beilinson–Lichtenbaum bound for M_p . In the excellent case, the independent right branch uses the cofiber $C_p(q)$, whose uniform global bound is only $D^{\geq -1}$. On the selected bounded-dimensional points it first produces $D^{\geq q-1}$ and the residue restriction injection; on mixed-characteristic regular models the cofiber bound is $D^{\geq q}$, not $D^{\geq q+1}$. The latter bound belongs to the fiber $C_p(q)[-1]$. Neither branch uses the other’s conclusion.

Proposition 5.1 (The supported residue-characteristic bound). *Let R be regular noetherian local, with closed point s and $0 \neq p \in \mathfrak{m}_R$. Suppose $\dim R \leq 2$, or that $\dim R \leq 3$ and R is excellent. Then*

$$A_{p^m}(q)_s(R) \in D^{\geq q+1} \quad (q \geq 0, m \geq 1).$$

Proof. Dimensions at most two. First replace R by R^h . The finite-coefficient theories A_{p^m} are finitary Nisnevich sheaves ([Bachmann et al., 2025](#), arXiv v1, Theorem 1.1(3)). Supported Nisnevich excision and finitariness identify the supported values of M_p , F_p , and A_{p^m} on the two rings. Henselization preserves regularity, noetherianity, dimension, and the residue field.

Write $U = \text{Spec } R^h \setminus \{s\}$. The proof of [theorem 4.2](#) gives $\mathcal{N}_p(q)(U) = Q_p(q)(U) = 0$, since $\dim U \leq 1$. Together with that theorem on R^h , this identifies

$$M_p(q)_s(R^h) \simeq F_p(q)_s(R^h) \simeq A_p(q)_s(R^h).$$

Both $\text{Spec } R^h$ and U are regular and p -torsionfree, hence F-smooth. By ([Bouis, 2024](#), arXiv v3, Corollary 5.12), the complex

$$\mathcal{B}(q)(T) = \text{fib}(M_p(q)(T) \longrightarrow \mathcal{E}_p(q)(T))$$

lies in $D^{\geq q+1}$ for $T = \operatorname{Spec} R^h$ and $T = U$. This source bound requires p -torsionfreeness and F -smoothness, and imposes no excellence assumption. Moreover $\mathcal{E}_p(q)_s(R^h) = 0$, since $R^h[1/p] \subset U$ and $U[1/p] = \operatorname{Spec} R^h[1/p]$. Thus

$$A_p(q)_s(R^h) \simeq M_p(q)_s(R^h) \simeq \operatorname{fib}(\mathcal{B}(q)(R^h) \rightarrow \mathcal{B}(q)(U)) \in D^{\geq q+1}.$$

This proves the case $m = 1$. For $m > 1$, the exact coefficient sequences give fibre sequences

$$A_{p^{m-1}}(q)_s(R^h) \longrightarrow A_{p^m}(q)_s(R^h) \longrightarrow A_p(q)_s(R^h).$$

Since $D^{\geq q+1}$ is closed under extensions, induction on m proves the bound. Supported excision returns it to the original ring.

The excellent dimension-three case. Put $X = \operatorname{Spec} R$ and $U = X \setminus \{s\}$. The theorem [theorem 4.4](#) gives $Q_p(q)(X) = Q_p(q)(U) = 0$. Thus, for $T = X, U$,

$$\operatorname{fib}(A_p(q)(T) \rightarrow \mathcal{E}_p(q)(T)) \simeq \operatorname{fib}(F_p(q)(T) \rightarrow \mathcal{E}_p(q)(T)) \in D^{\geq q+1}$$

by [proposition 4.9](#). The bound on X uses the dimension-three resolution theorem; the bound on U uses its dimension-two clause. The two generic fibres coincide, so $\mathcal{E}_p(q)_s(R) = 0$. Taking the fiber of these two comparison fibers gives $A_p(q)_s(R) \in D^{\geq q+1}$. The coefficient sequences then give the same bound for every p^m . $\square$

For a field k of characteristic p ,

$$A_{p^m}(j)(k) = W_m \Omega_{k, \log}^j[-j] \quad (j \geq 0) \quad (32)$$

by ([Bachmann et al., 2025](#), arXiv v1, Theorem 1.1(5)). Hence the closed-point purity source in dimension d is concentrated in degree $q + d$, or is zero when $q < d$.

Lemma 5.2 (Purity at the residue prime). *Under the hypotheses of [proposition 5.1](#), the point-purity map modulo p^m is an equivalence in every weight and for every $m \geq 1$.*

Proof. If $1 \leq d \leq 2$, the supported bound and the source calculation give

$$\Delta_{p^m}(q) \in D^{\geq q+d-1}.$$

In particular the cofiber lower bound is q in dimension one. Start with $\Psi_{p^m}^0 = 0$ from [lemma 2.2](#). If $\Psi_{p^m}^q = 0$, then

$$\Delta_{p^m}(q)[2q - 1] = \operatorname{gr}^q \Psi_{p^m} \simeq \Sigma \Psi_{p^m}^{q+1}$$

is $(q-d+1)$ -connective. Thus $H^n \Delta_{p^m}(q) = 0$ for $n \geq q+d-1$. The interval $[q+d-1, q+d-2]$ is empty. Induction gives $\Delta_{p^m}(q) = \Psi_{p^m}^{q+1} = 0$ in every nonnegative weight.

Let $d = 3$. The supported bound gives $\Delta_{p^m}(q) \in D^{\geq q+1}$. By (2), values on R and its puncture have upper bounds $q + 3$ and $q + 2$, respectively. The source has degree $q + 3$. Consequently

$$\Delta_{p^m}(q) \in D^{[q+1, q+3]}.$$

Put $B_q = \Delta(q)[-1]$. The same integral cofiber underlies all reductions, so

$$B_q/p^m \in D^{[q+2, q+4]}, \quad \operatorname{gr}^q \Psi_{p^m} = (B_q/p^m)[2q], \quad \Psi_{p^m}^0 = 0.$$

Tail connectivity tends to infinity by [lemma 2.2](#), and the exact eigenweights of [proposition 2.5](#) give $(k^r - 1)d_r = 0$ for every k prime to p . Apply [lemma 2.6](#) with $\ell = p$, $\epsilon = 0$ and numerical window parameter $N = 4$. This parameter is the shifted cohomological bound, while the local ring has dimension three. Since $4 \leq p + 2$ for every prime, $B_q/p = 0$. The lemma uses only the reductions modulo p and p^2 at the endpoint, and its universal-coefficient argument includes $p = 2$. Coefficient extensions give $\Delta_{p^m}(q) = 0$ for every m . Negative weights vanish in both cases. $\square$

Lemma 5.3 (Integral purity on values). *For a regular noetherian local ring R with fraction field of characteristic zero, if $\dim R \leq 2$, or if $\dim R \leq 3$ and R is excellent, then*

$$A(q-d)(\kappa)[-2d] \xrightarrow{\sim} A(q)_s(R) \quad (q \in \mathbf{Z}).$$

Proof. The dimension-zero case is the identity. Let $d \geq 1$. The integral cofiber $\Delta(q)$ has zero rationalization by [proposition 2.7](#). For every prime different from the residue characteristic, Theorem [B](#) on values gives $\Delta(q)/\ell = 0$. If the residue characteristic is $p > 0$, then $p \neq 0$ in R and [lemma 5.2](#) gives $\Delta(q)/p = 0$. In residue characteristic zero every prime is invertible. Exactness of purity identifies these reductions with the cofibers of the corresponding coefficient maps. Thus multiplication by every prime is invertible on every $H^n \Delta(q)$. These groups are uniquely divisible and equal to their rationalizations, which are zero. Therefore $\Delta(q) = 0$. $\square$

5.1 Detection and smooth lifts

Lemma 5.4 (Supported continuity and the purity transformation). *Let X be noetherian of finite Krull dimension, let $i : Z \hookrightarrow X$ be a regular closed immersion, and let $f : X' \rightarrow X$ be a cofiltered limit of separated smooth finite-type maps $f_\alpha : X_\alpha \rightarrow X$ with affine dominant transition maps. Assume that X' is noetherian of finite Krull dimension. Put $Z_\alpha = Z \times_X X_\alpha$ and $Z' = Z \times_X X'$, and denote the induced maps by i_α, i' and $g : Z' \rightarrow Z$. For $E = H$ or H/n , where $H_T = s_0 KGL_T$, the canonical exceptional exchange map*

$$\chi_{i,f}(E) : g^* i^! E_X \longrightarrow i'^! E_{X'}$$

is an equivalence. With $N_{i'} \simeq g^ N_i$, the following square commutes:*

$$\begin{array}{ccc} g^*(\mathrm{Th}(-N_i) \otimes E_Z) & \xrightarrow{g^* \mathrm{pur}_i(E_X)} & g^* i^! E_X \\ \downarrow \simeq & & \downarrow \chi_{i,f}(E) \\ \mathrm{Th}(-N_{i'}) \otimes E_{Z'} & \xrightarrow{\mathrm{pur}_{i'}(E_{X'})} & i'^! E_{X'}. \end{array} \quad (\mathrm{SC})$$

Here the purity maps are the fundamental-class transformations, with the induced orientation for H/n . An equivalence of Cartesian coefficient objects transports this statement by naturality, as in [Section 1](#).

More explicitly, put $A_E(q) = A(q)$ for $E = H$ and $A_E(q) = A_n(q)$ for $E = H/n$. For a smooth test $V \rightarrow X'$ descending to compatible smooth tests $V_\alpha \rightarrow X_\alpha$, put $Z_{V_\alpha} = V_\alpha \times_X Z$ and $Z_V = V \times_X Z$. Then

$$\mathrm{colim}_\alpha A_E(q)_{Z_{V_\alpha}}(V_\alpha) \xrightarrow{\sim} A_E(q)_{Z_V}(V). \quad (\mathrm{SV})$$

The exchange in (SC) induces this canonical comparison on supported values. The conventions are $T^q E = E(q)[2q]$ and $\mathrm{map}(\mathbf{1}_V, E_V(q)) = A_E(q)(V)$. Thus (SV) has no additional $[2q]$.

Proof. We use three precise inputs. The absolute slice coefficient is Cartesian by ([Bachmann et al., 2025](#), arXiv version 1, Theorem 9.3(2)); its values and their shifts are ([Bachmann et al., 2025](#), Definition 4.15 and Remark 4.21). The presheaves $A(q)$ are finitary Nisnevich sheaves by ([Bachmann et al., 2025](#), Theorem 1.1(3)); negative weights are zero by its Remark 4.20. Finally, continuous evaluation on smooth tests, for every $F \in \mathbf{SH}(X)$, is

$$\mathrm{colim}_\alpha \mathrm{map}_{X_\alpha}(\Sigma^\infty(V_\alpha)_+, f_\alpha^* F) \xrightarrow{\sim} \mathrm{map}_{X'}(\Sigma^\infty V_+, f^* F). \quad (\mathrm{CE})$$

This is (Hoyois, 2015, arXiv version 5, Definition A.1 and Lemma A.7(1)): varying the suspension index in that lemma proves the assertion on every homotopy group, hence on mapping spectra. Its affine and dominance hypotheses are included in the statement above. Smooth finite-presentation tests descend, so these tests and their suspensions include a generating family.

Write $\Gamma_Z E_X = i_* i^! E_X$. Closed base change identifies

$$f^* \Gamma_Z E_X \simeq i'_* g^* i^! E_X.$$

Under this identification $i'_*(\chi_{i,f}(E))$ is the canonical map

$$f^* \Gamma_Z E_X \longrightarrow \Gamma_{Z'} E_{X'} \quad (\text{EX})$$

induced by the adjunction counit and the Cartesian identification of coefficients. These are proper base change and exceptional exchange from (Déglise et al., 2021b, arXiv version 3, Paragraphs 2.1.2(2),(5) and 2.1.9).

Apply (CE) to $F = \Gamma_Z E_X(q)$. At every finite stage f_α is smooth, so smooth exceptional base change identifies $f_\alpha^* \Gamma_Z E_X$ with $\Gamma_{Z_\alpha} E_{X_\alpha}$. Localization and smooth base change along $V_\alpha \rightarrow X_\alpha$ then identify its evaluation with

$$\text{fib}(A_E(q)(V_\alpha) \longrightarrow A_E(q)(V_\alpha \setminus Z_{V_\alpha})).$$

The same calculation identifies the evaluation of the target of (EX) on V with $A_E(q)_{Z_V}(V)$. The complement is quasi-compact, and its inverse system has affine transition maps and limit $V \setminus Z_V$. Finitarity and exactness of filtered colimits therefore give (SV). Reduction modulo n , being a cofiber, commutes with these fibers and filtered colimits. By the adjunction definition of exchange, the evaluation of (EX) is exactly this map of restriction fibers, rather than an unspecified isomorphism between their endpoints. Thus (EX) is an equivalence on all smooth suspension generators and all twists. It is an equivalence in $\mathbf{SH}(X')$; the fully faithful functor i'_* reflects equivalences, proving the assertion about χ .

It remains to identify the map, not just its two objects. The map f is flat, since it is a limit of smooth maps. Therefore the base-change square of i is Tor-independent and its normal bundle is the pulled-back normal bundle. The zero-excess case of (Déglise et al., 2021b, arXiv version 3, Proposition 3.2.8), applied to this square, identifies the pulled-back fundamental class with $\eta_{i'}$. This proposition allows an arbitrary base-change map; the closed immersions themselves are of finite presentation here. Tensoring that class with the coefficient, followed by the exchange map of Paragraph 2.1.10, gives precisely the transformation of Paragraph 2.5.1. Its transverse base-change square is Proposition 2.5.4(ii)(a), namely (SC). This proves the required compatibility also at the limit. We have proved invertibility of exchange for the named coefficients under the stated limit hypotheses; no exceptional-exchange assertion for arbitrary coefficients under arbitrary ind-smooth maps has been used. $\square$

Remark 5.5 (The local and henselian systems used here). An ordinary local scheme and its henselization are limits of affine Zariski and pointed affine *étale* neighborhoods, respectively. In each neighborhood discard the irreducible components not containing its marked point, then take an affine open neighborhood of that point. These choices are cofinal. Every transition is open and its image contains the marked point of the target; it therefore meets every irreducible component of the target and is dominant. Morphisms between these affine schemes are affine. This verifies the specific hypotheses of (CE) for the local and henselian systems, without excellence or a residue-field perfectness hypothesis. Neighborhoods at a point of a smooth test are again smooth over the original base.

Consequently (CE), which is coefficient-independent, gives the stalk identifications in lemma 5.6. For $v \in V$ over $t \in X$, the ordinary local map and the induced map of henselizations factor the pullback of an arbitrary object to $\mathrm{Spec} \mathcal{O}_{V,v}$ or $\mathrm{Spec} \mathcal{O}_{V,v}^h$ through the corresponding local scheme of X . Vanishing there makes the relevant stalk zero. For the Nisnevich assertion one applies this to all *étale* tests over each smooth V , which are themselves smooth over X . The needed finite-dimensional descent bounds are (Clausen and Mathew, 2021, arXiv version 3, Theorems 3.2, 3.12 and 3.18): Theorem 3.12 is the Zariski bound and Theorem 3.18 the Nisnevich bound. The local-detection argument and then the costalk induction in lemma 5.6 therefore apply as stated.

Lemma 5.6 (Local and henselian point-costalk detection). *For a noetherian scheme T of finite Krull dimension, an object $E \in \mathbf{SH}(T)$ is zero if its restriction to $\mathrm{Spec} \mathcal{O}_{T,t}$ is zero for every $t \in T$. The same conclusion holds if its restrictions to $\mathrm{Spec} \mathcal{O}_{T,t}^h$ are zero for every t .*

Let T be noetherian of finite Krull dimension and $E \in \mathbf{SH}(T)$. For $h_t : T_t = \mathrm{Spec} \mathcal{O}_{T,t}^h \rightarrow T$ and its closed point a_t , if $a_{t} h_t^* E = 0$ for every t , then $E = 0$.*

Proof. The coefficient-independent mapping-continuity formula (CE) and the local systems in remark 5.5 apply here. For V smooth over T and $v \in V$ over t , the Zariski stalk of the spectrum-valued sheaf $W \mapsto E(W)(q)$ at v is a continuous colimit of values on tests over $\mathrm{Spec} \mathcal{O}_{T,t}$, hence zero. A Zariski sheaf of spectra on the noetherian space V of finite dimension is hypercomplete, so $E(V)(q) = 0$ for all V, q . The smooth suspension spectra and their Tate shifts generate $\mathbf{SH}(T)$, so $E = 0$. The finite-dimensional hyperdescent bound is the Zariski case of (Clausen and Mathew, 2021, arXiv version 3, Theorems 3.2, 3.12 and 3.18).

For the henselian version, fix $v \in V$ over t . Henselization commutes with this local base change: $\mathcal{O}_{V,v}^h$ is the henselization at the corresponding point of $V \times_T \mathrm{Spec} \mathcal{O}_{T,t}^h$ (The Stacks Project Authors, 2026, Tag 08HU). That fibre product is smooth over the henselian local base. Its local henselization is a continuous limit of *étale* neighbourhoods of such smooth tests. Thus the vanishing of the base-restricted object makes the Nisnevich stalk $E(\mathrm{Spec} \mathcal{O}_{V,v}^h)(q)$ zero. Finite-dimensional Nisnevich hyperdescent, with the same cited bounds, gives $E(V)(q) = 0$. The generators again conclude $E = 0$.

Costalks. We prove by induction on n that $h_t^* E = 0$ for every t with $\dim \mathcal{O}_{T,t} \leq n$. Put $F_t = h_t^* E$ and $U_t = T_t \setminus \{t\}$. For $z \in U_t$, its image $u \in T$ is a proper generization of t , since the closed fibre of henselization is exactly $\mathrm{Spec} \kappa(t)$ (The Stacks Project Authors, 2026, Tag 07QM). Thus $\dim \mathcal{O}_{T,u} < \dim \mathcal{O}_{T,t}$. The local map $\mathcal{O}_{T,u} \rightarrow \mathcal{O}_{T,z}$ induces the factorization

$$\mathrm{Spec} \mathcal{O}_{T,z}^h \xrightarrow{b_z} \mathrm{Spec} \mathcal{O}_{T,u}^h \xrightarrow{h_u} T$$

by (The Stacks Project Authors, 2026, Tag 04GS). Consequently

$$F_t|_{\mathrm{Spec} \mathcal{O}_{T,z}^h} \simeq b_z^* h_u^* E = 0$$

by induction. Henselian-local detection on the noetherian puncture gives $F_t|_{U_t} = 0$. Localization and the original closed-costalk hypothesis now give $F_t \simeq a_{t*} a_t^! F_t = 0$. For local dimension zero the puncture is empty, which starts the induction. Finite dimension makes it well founded; henselian-local detection on T concludes. $\square$

Lemma 5.7 (Local injectivity over an infinite field). *Let F be an infinite field, perfect or not, $E \in \mathbf{SH}(F)$, $X \in \mathrm{Sm}_F$ irreducible with function field $F(X)$, and $x \in X$ any point. Then for all $n, q \in \mathbf{Z}$ the restriction*

$$\pi_n E(\mathrm{Spec} \mathcal{O}_{X,x})(q) \rightarrow \pi_n E(\mathrm{Spec} F(X))(q)$$

is injective.

Proof. The geometric input is [proposition D.7](#), the corrected presentation argument of (Colliot-Thélène et al., 1997, Theorem 3.1.1 and Sections 3.2–3.6). It supplies an étale support square with finite support over $\mathbf{A}^{d-1}$ for arbitrary marked points over the infinite field, without perfection or a separability hypothesis on their residue fields. We use this geometry and the following Nisnevich and $\mathbf{A}^1$ argument.

If $\dim X = 0$, then $\mathcal{O}_{X,x} = F(X)$. Let $d = \dim X \geq 1$.

Step 0 (reduction). Let $\alpha \in \pi_n E(\mathrm{Spec} \mathcal{O}_{X,x})(q)$ map to zero in $\pi_n E(\mathrm{Spec} F(X))(q)$. Filtered colimits are exact, so α is the image of some $\alpha_W \in \pi_n E(W)(q)$, with W an affine open neighbourhood of x , and α_W restricts to zero on a dense open $W \setminus Z$, with $Z \subset W$ closed of positive codimension. By the exact sequence $\pi_n E_Z(W)(q) \rightarrow \pi_n E(W)(q) \rightarrow \pi_n E(W \setminus Z)(q)$, $\alpha_W = u(\beta)$ for some $\beta \in \pi_n E_Z(W)(q)$, where u forgets supports. If $x \notin Z$, then $W \setminus Z$ is a neighbourhood of x and $\alpha = 0$. Assume $x \in Z$.

Step 1 (presentation). Choose $\varphi = (\psi, v)$, V and U from [proposition D.7](#), applied to W , Z and $t_1 = x$. Put $Z_U = Z \cap U$ and $T = \varphi(Z_U)$, which is closed in $\mathbf{A}_V^1$. The presentation identifies the inverse image of T in U with Z_U , so the square with rows $U \setminus Z_U \rightarrow U$ and $\mathbf{A}_V^1 \setminus T \rightarrow \mathbf{A}_V^1$ and vertical maps induced by φ is a distinguished Nisnevich square, so $\varphi^*: E_T(\mathbf{A}_V^1)(q) \rightarrow E_{Z_U}(U)(q)$ is an equivalence by Nisnevich descent. The scheme T is finite over V because $Z_U = Z \cap \psi^{-1}(V)$ is finite over V and maps isomorphically to T .

Step 2 (localization on V). Let $s = \psi(x) \in V$ and $V_s = \mathrm{Spec} \mathcal{O}_{V,s}$, an essentially smooth local F -scheme whose residue field $\kappa(s)$ contains F . Base change along $V_s \rightarrow V$ gives $U_s \rightarrow \mathbf{A}_{V_s}^1$ étale, T_s finite over V_s , and $Z_s = Z_U \times_V V_s$ mapping isomorphically onto T_s , with $\varphi_s^{-1}(T_s) = Z_s$. The square is a cofiltered limit of distinguished squares with affine transition maps, so by continuity again $\varphi_s^*: E_{T_s}(\mathbf{A}_{V_s}^1)(q) \simeq E_{Z_s}(U_s)(q)$.

Step 3 (a section missing T_s). The closed fibre of T_s over s is a finite subset of $\mathbf{A}_{\kappa(s)}^1$. As F is infinite, choose $c \in F$ such that the $\kappa(s)$ -rational point $t = c$ does not lie in it. The section $\sigma_c: V_s \rightarrow \mathbf{A}_{V_s}^1$, $t = c$, has $\sigma_c^{-1}(T_s)$ closed in the local scheme V_s and not containing its closed point, hence empty. By naturality of supports, $\sigma_c^* \circ u: E_{T_s}(\mathbf{A}_{V_s}^1)(q) \rightarrow E(V_s)(q)$ factors through $E_\emptyset(V_s)(q) = 0$, and σ_c^* on $E(\mathbf{A}_{V_s}^1)(q)$ is an equivalence, by $\mathbf{A}^1$ -invariance over the essentially smooth V_s and continuity. Hence $u: E_{T_s}(\mathbf{A}_{V_s}^1)(q) \rightarrow E(\mathbf{A}_{V_s}^1)(q)$ is null.

Step 4 (conclusion). φ_s^* commutes with u , so $u: E_{Z_s}(U_s)(q) \rightarrow E(U_s)(q)$ is null. The local scheme $\mathrm{Spec} \mathcal{O}_{X,x}$ maps to U_s , since $x \in U$ maps to s and a map from a local scheme sending the closed point to s factors through V_s . By naturality, α is the image of β under $E_Z(W) \rightarrow E_{Z_U}(U) \rightarrow E_{Z_s}(U_s) \xrightarrow{u} E(U_s) \rightarrow E(\mathrm{Spec} \mathcal{O}_{X,x})$, which is zero. $\square$

Proposition 5.8 (Detection by smooth function fields). *Let F be a field and $E \in \mathbf{SH}(F)$. If $E(\mathrm{Spec} L)(q) = 0$ for every $L = F(X)$ with $X \in \mathrm{Sm}_F$ irreducible and every $q \in \mathbf{Z}$, then $E = 0$.*

Proof. Let F be infinite. By [lemma 5.7](#), $\pi_n E(\mathrm{Spec} \mathcal{O}_{X,x})(q) = 0$ for every $X \in \mathrm{Sm}_F$, $x \in X$ and $n, q \in \mathbf{Z}$, since the connected components of X are irreducible. These are the stalks of the homotopy presheaves of the Zariski sheaf of spectra $U \mapsto E(U)(q)$ on the noetherian space X of finite Krull dimension. That sheaf is hypercomplete, so $E(X)(q) = 0$, as in the proof of [lemma 5.6](#). The $\Sigma^{a,b} \Sigma^\infty X_+$ generate $\mathbf{SH}(F)$, so $E = 0$.

Let F be finite, hence perfect. Every finitely generated extension of F is then separably generated, hence of the form $F(X)$ with X smooth. The value of E at such a field is the value there of its bigraded homotopy sheaves, since Nisnevich sheafification does not change values at fields. These sheaves are strictly $\mathbf{A}^1$ -invariant, and a morphism of strictly $\mathbf{A}^1$ -invariant sheaves over a perfect field is an isomorphism if it is so on all finitely generated

fields (Hoyois, 2015, arXiv version 5, Theorem 2.7). So all homotopy sheaves of E vanish, and $E = 0$ by (Hoyois, 2015, arXiv version 5, Theorem 2.3 and Corollary 2.4). $\square$

Lemma 5.9 (Smooth lifts and a fieldwise criterion for point purity). *Let $\mathrm{Spec} B \rightarrow \mathrm{Spec} A$ be a closed immersion of affine schemes. Smooth B -schemes have a Zariski basis of open subschemes lifting to smooth A -schemes, stable under further restriction. If R is regular noetherian local of dimension d , and U is an irreducible smooth $\kappa(R)$ -scheme with such a lift $\tilde{U}$, the henselization R_L of $\mathcal{O}_{\tilde{U}, \eta_U}$ is regular noetherian local of dimension d , with residue field $L = \kappa(U)$. If R is excellent, R_L is excellent. If $p \neq 0$ in R , then $p \neq 0$ in R_L .*

Let R be regular noetherian local, s its closed point and κ its residue field. For the intrinsic coefficient H , fix coefficients $\mathbf{Z}$ or $\mathbf{Z}/n$. If point purity on R_L is an equivalence on values in every weight for every smooth function field L/κ , then point purity on R is an equivalence in $\mathbf{SH}(\kappa)$.

Proof. Locally a smooth map factors as a standard étale map to affine space (The Stacks Project Authors, 2026, Tags 054L and 02GT). Write the chart as $B'[t]_h/(f)$, with B' an affine-space coordinate algebra over B , f monic and f' a unit. Lift B' to the corresponding algebra A' over A , lift the coefficients and h , and invert both the lifted h and the lifted f' ; the resulting standard étale algebra lifts the given chart. An open chart lifts by removing its closed complement in the closed fibre. The lifting construction is explicit and uses only the cited standard local structure. Put $S = \mathcal{O}_{\tilde{U}, \eta_U}$. Smoothness over the regular R gives regularity by (The Stacks Project Authors, 2026, Tag 07NF). The ring S is noetherian, $R \rightarrow S$ is flat local, and $S/\mathfrak{m}_R S = L$, since η_U is the generic point of the reduced closed fibre. In particular $\mathfrak{m}_S = \mathfrak{m}_R S$. If $t_1, \dots, t_d$ is a regular system of parameters of R , its image is a regular system of parameters of S and of S^h ; it generates the maximal ideals by the displayed equality and its henselian base change. The conormal frame, and therefore the dual normal frame, of the closed point is the pullback of this same frame over κ . Thus the flat dimension formula (The Stacks Project Authors, 2026, Tag 00ON) gives $\dim S = d$. For $R_L = S^h$, noetherianness, regularity and dimension preservation are (The Stacks Project Authors, 2026, Tags 06LJ, 06LN and 06LK).

If R is excellent, so is S by (The Stacks Project Authors, 2026, Tag 07QU). Its henselization is a G-ring by (The Stacks Project Authors, 2026, Tag 07QR). To check J-2, write S^h as a filtered colimit of étale S -algebras S_α . Every finite-type S^h -algebra B is finitely presented and descends to a finite-type S_α -algebra B_α . The latter is excellent. The map $B_\alpha \rightarrow B$ is ind-étale, and both rings are noetherian, so it is regular by (The Stacks Project Authors, 2026, Tag 07EP). At corresponding primes its local map is flat local; regularity ascends along this regular map and descends along its faithfully flat local map (The Stacks Project Authors, 2026, Tags 0H7S and 07NG). Consequently $\mathrm{Reg}(B)$ is the inverse image of $\mathrm{Reg}(B_\alpha)$ and is open. This proves J-2 for S^h . The regular local ring S^h is Cohen–Macaulay and universally catenary by (The Stacks Project Authors, 2026, Tags 00NQ and 00NM), hence excellent. Finally $R \rightarrow S \rightarrow S^h$ is flat local, using (The Stacks Project Authors, 2026, Tag 07QM) for the last map; it is faithfully flat and therefore does not kill p .

Put $E = H$ or H/n . Let $s = \mathrm{Spec} \kappa$, $d = \dim R$, and let

$$P_s = \mathrm{cofib}(\mathrm{Th}(-N_s) \otimes E_\kappa \xrightarrow{\mathrm{pur}_s(E_R)} s^! E_R).$$

Take an irreducible smooth κ -scheme U , shrink it to a chart admitting an affine smooth lift $\tilde{U} \rightarrow \mathrm{Spec} R$, and write $L = \kappa(U)$. Denote its closed-fibre immersion by $i_U : U \hookrightarrow \tilde{U}$ and its smooth projection by $p : U \rightarrow s$. At this finite smooth stage the actual transformation

fits into the square

$$\begin{array}{ccc} p^*(\mathrm{Th}(-N_s) \otimes E_\kappa) & \xrightarrow{p^* \mathrm{pur}_s} & p^* s^! E_R \\ \downarrow \simeq & & \downarrow \mathrm{Ex}^*! \simeq \\ \mathrm{Th}(-N_{i_U}) \otimes E_U & \xrightarrow{\mathrm{pur}_{i_U}} & i_U^! E_{\tilde{U}}. \end{array} \quad (\mathrm{F1})$$

This is transverse base change for the fundamental class and smooth exceptional base change, as in (SC). In particular it is a square of the given purity morphisms.

Choose a regular system of parameters of R . The induced frame of N_s pulls back to N_{i_U} and to the normal bundles in all the following restrictions. After twisting by (q) and mapping from the unit, the bottom arrow of (F1) becomes

$$A_E(q-d)(U)[-2d] \longrightarrow A_E(q)_U(\tilde{U}). \quad (\mathrm{F2})$$

Its cofiber is $P_s(U)(q)$ under the convention that (q) is the Tate twist alone. Indeed $E(q) = T^q E[-2q]$, so mapping into $E(q-d)[-2d]$ gives exactly the left side of (F2).

Let $\eta = \eta_U$ and $S = \mathcal{O}_{\tilde{U}, \eta}$. Take the filtered system of affine principal open neighborhoods W of η in $\tilde{U}$, after restricting to the component containing η ; regularity makes this component open and closed. Their closed fibres $U_W = U \cap W$ are cofinal in the dense opens of U : any such open can be obtained by deleting its closed complement in the closed fibre and then shrinking to an affine principal neighborhood of η . Thus their limit is $\mathrm{Spec} L$, whereas $\lim W = \mathrm{Spec} S$. Their complements have limit $\mathrm{Spec} S \setminus \{\mathfrak{m}_S\}$. Apply (SC) and (SV) to these restrictions.

Next write $S^h = R_L$ as the limit of pointed affine *étale* neighborhoods of $\mathrm{Spec} S$. They can be chosen with entire closed fibre $\mathrm{Spec} L$: remove the finitely many other points of that fibre and shrink about the chosen point. At every such stage the square with open complements is a Nisnevich distinguished square. Nisnevich excision identifies its supported value with $A_E(q)_{\mathfrak{m}_S}(S)$, and transverse base change identifies the purity arrows under this excision map. Finitarity passes this same square to S^h . Write γ_W , γ_S , and γ_{R_L} for the weight- q fundamental-class purity maps, with coefficient E , for $U_W \hookrightarrow W$, $\mathrm{Spec} L \hookrightarrow \mathrm{Spec} S$, and $\mathrm{Spec} L \hookrightarrow \mathrm{Spec} R_L$, respectively. We obtain the commutative diagram

$$\begin{array}{ccc} \mathrm{colim}_W A_E(q-d)(U_W)[-2d] & \xrightarrow{\mathrm{colim}_W \gamma_W} & \mathrm{colim}_W A_E(q)_{U_W}(W) \\ \downarrow \simeq & & \downarrow \simeq \\ A_E(q-d)(L)[-2d] & \xrightarrow{\gamma_S} & A_E(q)_{\mathfrak{m}_S}(S) \\ \downarrow = & & \downarrow \simeq \\ A_E(q-d)(L)[-2d] & \xrightarrow{\gamma_{R_L}} & A_E(q)_{\mathfrak{m}_{R_L}}(R_L). \end{array} \quad (\mathrm{F3})$$

Here $\mathfrak{m}_S = \mathfrak{m}_R S$, $\mathfrak{m}_{R_L} = \mathfrak{m}_R R_L$, and both residue fields are exactly L . The original d parameters give the pulled-back normal frame throughout; [lemma 5.9](#) gives $\dim S = \dim R_L = d$. No relative transcendence degree is added to this dimension.

Mapping spectra are exact, and filtered colimits commute with cofibers. Taking cofibers in (F3) therefore identifies $P_s(\mathrm{Spec} L)(q)$ with the cofiber of the actual closed-point purity map on R_L in weight q . The hypothesis makes all of these spectra zero. Every smooth function field has such a chart, so [proposition 5.8](#) gives $P_s = 0$. $\square$

5.2 Composition and the main purity statements

Proposition 5.10 (From point purity to regular pairs). *Let $i : Z \hookrightarrow X$ be a regular closed immersion between regular noetherian schemes of finite Krull dimension. Fix the coefficient*

H or H/n . If its point-purity map is invertible on every henselized local ring of X and Z , then its purity transformation along i is invertible.

Proof. Treat constant codimension c on the open-and-closed components of Z . Write $E = H$ or H/n and use the oriented normalization

$$\gamma_i : E_Z \longrightarrow i^! E_X(c)[2c], \quad P_i = \text{cofib}(\gamma_i).$$

Fix $z \in Z$, put $x = i(z)$, and write $X' = \text{Spec } \mathcal{O}_{X,x}^h$, $Z' = \text{Spec } \mathcal{O}_{Z,z}^h$. Henselization commutes with quotients ([The Stacks Project Authors, 2026](#), Tag 05WQ), so $Z' = Z \times_X X'$. Let h_X, h_Z be the projections and $i' : Z' \hookrightarrow X'$. The system in [remark 5.5](#) and [lemma 5.4](#) give a commutative diagram of the actual oriented transformations and their cofibers:

$$\begin{array}{ccccc} h_Z^* E_Z & \xrightarrow{h_Z^* \gamma_i} & h_Z^* i^! E_X(c)[2c] & \longrightarrow & h_Z^* P_i \\ \downarrow \simeq & & \downarrow \chi_{i,h_X}(E)(c)[2c] \simeq & & \downarrow \simeq \\ E_{Z'} & \xrightarrow{\gamma_{i'}} & i'^! E_{X'}(c)[2c] & \longrightarrow & P_{i'}. \end{array} \quad (\text{P1})$$

This is the specified coefficient exchange identifying the global and local defects.

Let $a : \text{Spec } \kappa(z) \hookrightarrow Z'$ be the closed point, put $b = i'a$, and write $e = \dim Z'$ and $d = \dim X' = e + c$. The conormal exact sequence, or equivalently the normal exact sequence

$$0 \longrightarrow N_a \longrightarrow N_b \longrightarrow a^* N_{i'} \longrightarrow 0,$$

identifies $[N_b] = [N_a] + [a^* N_{i'}]$. The multiplicative orientation identifies the associated Thom twists with the indicated Tate and cohomological shifts. Composition of the same fundamental classes, with the canonical Thom paths proved in [proposition D.3](#), is precisely the square

$$\begin{array}{ccc} E_{\kappa(z)}(-e)[-2e] & \xrightarrow{\text{pur}_a(E_{Z'})} & a^! E_{Z'} \\ \downarrow = & & \downarrow a^! \gamma_{i'} \\ E_{\kappa(z)}(-d)[-2d](c)[2c] & \xrightarrow{\text{pur}_b(E_{X'})(c)[2c]} & b^! E_{X'}(c)[2c]. \end{array} \quad (\text{P2})$$

In the right column we use the canonical composition identification $a^! i'^! \simeq b^!$. If both point-purity maps are invertible, then $a^! \gamma_{i'}$ is invertible. Since $a^!$ is exact, (P1) gives the specific costalk identification

$$a^! h_Z^* P_i \simeq \text{cofib}(a^! \gamma_{i'}) = 0. \quad (\text{P3})$$

Thus [lemma 5.6](#) gives $P_i = 0$. The construction uses regular noetherian local rings, the indicated finite-dimensional continuity, and the two point-purity hypotheses. Excellence and residue-field perfectness are not needed for this passage from points to pairs. $\square$

Proof of Theorem A. For every smooth function field L/κ , the lifted ring R_L in [lemma 5.9](#) has the same local dimension as R . The fraction field of R_L still has characteristic zero by faithful flatness; excellence persists when it is required. Thus [lemma 5.3](#) applies to every R_L in every weight. The fieldwise criterion gives the intrinsic object point-purity equivalence.

For the pair statement, all henselized local rings are regular noetherian with fraction fields of characteristic zero, by flatness over $\mathbf{Z}$. They satisfy the stated dimension bound, and remain excellent in the dimension-three case. The point assertion applies to both schemes. Now [proposition 5.10](#) proves the result. $\square$

Proof of Theorem B on objects and pairs. Every lifted R_L is regular noetherian of dimension d , and ℓ remains invertible. The value assertion already proved therefore holds on all of these rings in every weight, modulo each ℓ^m . The fieldwise criterion gives object point purity. For a regular pair, the henselized local rings of both schemes have dimension at most $\ell + 2$, and ℓ is invertible on them. Applying [proposition 5.10](#) gives the pair assertion. $\square$

6 Purity over regular DVR maps

Fundamental-class purity for Spitzweck’s coefficient is proved in the mixed-characteristic DVR setup of Frankland–Spitzweck’s Section 4 ([Frankland and Spitzweck, 2026](#), arXiv v4, Section 4, Proposition 8.3 and Corollary 8.4). The integral comparison of Section 1 and coefficient naturality identify this with purity for the intrinsic zero slice. In that setup, Corollary 8.4 gives its integral coefficient and prime-reduction module formulation; exactness further gives every finite reduction. This is a statement about the coefficient and its reductions, and does not assert purity for arbitrary modules.

For every mixed-characteristic DVR V , with residue field k and closed immersion $i_0 : \operatorname{Spec} k \hookrightarrow \operatorname{Spec} V$, Theorem A in dimension one gives the integral input

$$H_k(-1)[-2] \xrightarrow{\sim} i_0^! H_V.$$

This internal proof requires neither excellence nor perfection of k . We now extend this map along a regular ring map by explicit Popescu approximation and continuity, and combine it with perfect-field purity for Theorem C.

Lemma 6.1 (Special-fibre purity for a regular DVR algebra). *Let V be a mixed-characteristic DVR, let π be a uniformizer, and let R be a noetherian V -algebra such that $V \rightarrow R$ is a regular ring map. Put $X = \operatorname{Spec} R$, $D = \operatorname{Spec}(R/\pi)$, and let $i : D \hookrightarrow X$ be the special-fibre immersion. Then*

$$\operatorname{pur}_i(H_X) : H_D(-1)[-2] \xrightarrow{\sim} i^! H_X.$$

The assertion includes non-finite-type R and requires no excellence.

Proof. Popescu’s theorem gives a filtered presentation $R = \operatorname{colim}_\alpha R_\alpha$ with each R_α smooth over V ; see ([The Stacks Project Authors, 2026](#), Tag 07GC). At every stage smooth exceptional base change carries the DVR purity map above to purity for $D_\alpha = V(\pi) \subset X_\alpha = \operatorname{Spec} R_\alpha$. The normal line is trivialized by π . Thus, in every weight q , the supported Gysin map is an equivalence

$$\mathbf{Z}(q-1)^{\mathbb{A}^1}(D_\alpha)[-2] \xrightarrow{\sim} \mathbf{Z}(q)_{D_\alpha}^{\mathbb{A}^1}(X_\alpha).$$

Both R_α and R are flat over V . Every transition square for these divisors is consequently Tor-independent, even though the transition ring map need not be smooth: the resolution of R_α/π by multiplication by π stays exact after base change. Transverse base change for fundamental classes therefore makes these the compatible fundamental-class Gysin maps, not merely a collection of isomorphisms; use ([Déglise et al., 2021b](#), arXiv v3, Proposition 2.5.4 and Paragraph 4.3.1).

The source and the two terms defining the supported target commute with this filtered colimit. For the target these terms are the values on R_α and $R_\alpha[1/\pi]$. Filtered colimits of complexes commute with fibres. Finitarity ([Bachmann et al., 2025](#), arXiv v1, Theorem 1.1(3)) thus gives the same supported equivalence on X .

This argument also applies to every smooth affine X -scheme T . Indeed, a finite smooth presentation and its Jacobian data descend to some R_α , and its later models are smooth over V . Finite affine covers give the assertion for all smooth tests T . For these tests, mapping into the pushforward of the displayed purity transformation gives exactly

$$\mathbf{Z}(q-1)^{\mathbb{A}^1}(T \times_X D)[-2] \longrightarrow \mathbf{Z}(q)_{T \times_X D}^{\mathbb{A}^1}(T).$$

All its cohomological shifts are equivalences. Smooth suspension spectra and their Tate shifts generate $\mathbf{SH}(X)$, so $i_* \text{pur}_i(H_X)$ is invertible. Since i_* is fully faithful, $\text{pur}_i(H_X)$ is invertible. This proof uses smooth exchange at finite stages and finitary supported values; it does not assume an exceptional-exchange theorem for an arbitrary ind-smooth map. $\square$

Proof of Theorem C. The fibre $D = \text{Spec}(R/\pi)$ is regular of dimension $d-1$. The preceding lemma proves purity for $D \hookrightarrow \text{Spec } R$. Purity for $s \hookrightarrow D$ follows from [corollary D.6](#), applied over $\mathbf{F}_p$. The affine refinement and compact-test proof there supplies the first route of ([Dégliise et al., 2021a](#), arXiv v3, Appendix C, Theorem C.1) for the intrinsic transformation, without a finite-type hypothesis on D over the perfect field. The coefficient H is Cartesian by [proposition A.1](#). Composition of the fundamental classes gives point purity on R . The normal classes add, giving the twist $(-d)[-2d]$. This proves object purity, its value assertion, and every finite coefficient assertion. Formula [\(32\)](#) gives the logarithmic de Rham–Witt value in the statement.

If R is unramified, it is p -torsionfree and hence flat over $\mathbf{Z}_{(p)}$. Its generic fibre is regular over the perfect field $\mathbf{Q}$. Its closed fibre R/p is regular, since p is part of a regular system of parameters, and is geometrically regular over the perfect field $\mathbf{F}_p$. Therefore $\mathbf{Z}_{(p)} \rightarrow R$ is a local regular map, as required.

For the pair assertion, the structure maps of both schemes to $\mathbf{Z}_{(p)}$ are regular on affine charts: they are flat and their generic and special fibres are regular over perfect fields. The special-fibre lemma proves purity for $j_X : X_p \hookrightarrow X$ and $j_Z : Z_p \hookrightarrow Z$. The generic-fibre immersion $i_{\mathbf{Q}}$ has regular source and target. Flatness of Z makes the square defining $i_p : Z_p \hookrightarrow X_p$ Tor-independent, so i_p is a regular immersion. Its source and target are regular, and [corollary D.6](#) gives purity for $i_{\mathbf{Q}}$ over $\mathbf{Q}$ and for i_p over $\mathbf{F}_p$, in both cases for the intrinsic fundamental-class map.

The two factorizations $i \circ j_Z = j_X \circ i_p$ and composition show that $j_Z^! \text{pur}_i(H_X)$ is invertible: its composition with purity for j_Z equals the composite of purity for j_X and i_p , with the same normal classes and twists. The cone of $\text{pur}_i(H_X)$ has zero generic restriction and zero exceptional restriction to Z_p . The localization triangle on Z kills this cone. $\square$

The remaining supported-comparison boundary. For a general regular mixed-characteristic local ring, the bound $A_p(q)_s(R) \in D^{\geq q+1}$ is not supplied here beyond the low-dimensional or regular-DVR-map cases. A bound for M_p alone does not give this bound after $\mathbb{A}^1$ -localization. If such a supported bound is supplied independently in dimension $d \geq 2$, the shifted Adams argument has numerical parameter $N = d+1$ and gives residue- p value purity for $d \leq p+1$. In dimension one, the connectivity squeeze in [lemma 5.2](#) gives value purity. The invertible-prime argument proves purity in the range $d \leq \ell+2$; no optimal range for geometric purity is asserted. The nonexcellent surface argument does not supply valuation unions in dimension three; that case here retains its excellent regular-model and resolution inputs. No general ramified higher-dimensional integral assertion is inferred from these methods.

A Coefficient and slice comparisons

The intrinsic coefficient, its Cartesian structure, and its filtration bounds are common inputs to the three main theorems. We spell out the corrected deductions in (Bachmann et al., 2025, arXiv v1). The statements are unchanged. The corrections below concern the cited proofs and are supplied here, rather than taken from an author-issued erratum. The external zero-slice, prismatic, rigidity, and descent foundations named in those arguments remain cited inputs.

A.1 Rational Adams summands and the integral comparison

Proposition A.1 (The intrinsic absolute coefficient). *For every qcqs X with structure map $a_X : X \rightarrow \mathrm{Spec} \mathbf{Z}$, the canonical maps of the cdh-localized pullback construction and the intrinsic slice construction give*

$$L_{\mathbb{A}^1} \mathbf{Z}(*)^{\mathrm{cdh}}(X) \simeq \mathbf{Z}(*)^{\mathbb{A}^1, \mathrm{cdh}}(X) \simeq \mathbf{Z}(*)^{\mathbb{A}^1}(X),$$

with their specified multiplicative Chern-class maps. Moreover

$$a_X^* f_q KGL_{\mathbf{Z}} \simeq f_q KGL_X, \quad a_X^* s_q KGL_{\mathbf{Z}} \simeq s_q KGL_X, \quad s_0 \mathbf{1}_X \xrightarrow{\sim} s_0 KGL_X.$$

The rational zero slice is the weight-zero Adams summand $H_{X, \mathbf{Q}} \simeq KGL_X^{(0)}$. For $\dim_v X \leq a$, the intrinsic value filtration is $(q - a)$ -connective in degree q .

Proof. First correct the rational argument of (Bachmann et al., 2025, Corollaries 4.51–4.52). Choose the Adams operation ψ^2 . On distinct rational weights i, j its scalars satisfy $2^i \neq 2^j$. Thus $\psi^2 - 2^j$ is invertible on the complement of weight j in Riou’s rational decomposition. For the finite gamma filtration occurring there, $2^i - 2^{-j} \neq 0$ for $i \geq 0, j > 0$ gives invertibility by finite induction on the filtration. The choice $k = 1$ in the printed general assertion would give the zero operator and cannot be used. This is a coefficient-rational operation; no inversion of 2 on the base scheme is required.

In the factorization over a perfect field k in the proof of Corollary 4.52(1), rationalize the upper-left object. The horizontal equivalence supplied by the zero-slice input is

$$(s_0 \mathbf{1}_k)_{\mathbf{Q}} \xrightarrow{\sim} (s_0 KGL_k)_{\mathbf{Q}},$$

rather than an equivalence from an integral object to a rational one. The intermediate inclusion of the weight-zero summand is split. Its composite with this equivalence is an equivalence, so that split inclusion is itself an equivalence and the other rational zero-slice summands vanish. Slice exhaustivity gives the effectivity used there. Effectivity is tested over the finite-dimensional base $\mathrm{Spec} \mathbf{Z}$ and then pulled back; it is not tested directly on an arbitrary infinite-dimensional X . The corrected rational route gives the rational part of the projective-bundle argument of Theorem 9.13.

The finite-coefficient route to that projective-bundle theorem is [proposition A.2](#) below. Together these routes prove the canonical integral projective-bundle formula used in Theorem 9.13: its cofiber is rationally zero and zero modulo each prime, so multiplication by every prime is invertible on its torsion homotopy groups, forcing the cofiber to vanish. The categorical comparison of Corollary 3.34 and Remark 4.28 then gives Theorem 9.1, the first equivalence in the statement. That bridge begins with pullback of the fixed zero slice over $\mathbf{Z}$, before proving its equality with intrinsic zero slices; it does not assume the desired Cartesian assertion.

To deduce Corollary 9.2, effective tests are $M_X(Y) \otimes T^r$ with $r \geq 1$, where Y is smooth over X and $M_X(Y) = \Sigma^\infty Y_+$. Tensor invertibility gives

$$\mathrm{map}(M_X(Y) \otimes T^r, E) \simeq \mathrm{map}(M_X(Y), E \otimes T^{-r}).$$

They therefore test negative cohomological weights. Those weights of the cdh theory vanish and remain zero under $L_{\mathbb{A}^1}$. The signs in the two testing ranges of the printed proof must accordingly be $r \geq 1$ and $q < 0$. This proves the stated positive-effectivity vanishing.

Bott periodicity reduces Theorem 9.3 to the zero cover. Apply $f_0 a_X^*$ to successive negative-slice quotient sequences. The preceding negative-weight vanishing kills their fibres; exhaustivity and filtered-colimit compatibility kill $f_0 a_X^* f_{<0} KGL_{\mathbf{Z}}$. Taking cofibers gives the zero-cover comparison and then all the cover and slice comparisons by Bott periodicity. This also identifies the multiplicative value presheaves.

For the unit comparison, the fibre of $\mathbf{1}_{\mathbf{Z}} \rightarrow kgl_{\mathbf{Z}}$ is 1-effective. Pullback preserves that property, and kgl is absolute by Corollary 9.6. Corollary 9.7 therefore gives the stated zero-slice unit equivalence, with its E_∞ structure. Finally, in the Suslin realization for Corollary 9.5, the r th term has valuative dimension at most $a + r$ and filtration connectivity $q - a - r$. The new normalized term of its skeletal realization is shifted by r , giving the uniform bound $q - a$. Suspending before this calculation handles a negative value of $q - a$. This proves the intrinsic filtration bound. $\square$

A.2 The finite-coefficient projective-bundle route

Here p is a prime and $u : \mathrm{Spec} \mathbf{Z}[1/p] \hookrightarrow \mathrm{Spec} \mathbf{Z}$ is the open immersion. All extension-by-zero functors in this subsection are those of the presheaf constructions in the cited source. We use its graded Kato motivic theory F_p^{Kato} , with coherent first-Chern-class action, from (Bachmann et al., 2025, arXiv v1, Definition 9.22) initially in the stable category of graded theories with coherent Chern-class action. Projective-bundle invariance is a property to be proved there, so the premature c, pbf subscript in that definition is read as c .

Proposition A.2 (The corrected Kato projective-bundle argument). *The finite-coefficient route of (Bachmann et al., 2025, arXiv v1, Theorem 9.13) gives the canonical map*

$$L_{\mathbb{A}^1} \mathbf{Z}/p(q)^{\mathrm{cdh}}(X) \oplus L_{\mathbb{A}^1} \mathbf{Z}/p(q-1)^{\mathrm{cdh}}(X)[-2] \xrightarrow{\pi^* + c_1(\mathcal{O}(1)) \cup \pi^*} L_{\mathbb{A}^1} \mathbf{Z}/p(q)^{\mathrm{cdh}}(\mathbf{P}_X^1)$$

as an equivalence for all qcqs X and all integer q .

Proof. Begin with the K -theory argument in Lemma 9.31(3). Since $K^{\mathrm{Kato}} = \mathrm{cofib}(K \rightarrow K^{\mathrm{Sel}})$, applying the exact finite-spectrum quotient $/(p, v^N)$ gives, on p -inverted schemes,

$$K^{\mathrm{Kato}}/(p, v^N) \simeq (K/(p, v^N))[1].$$

Here v is the suitable positive-degree v_1 self-map, whose action is invertible on K^{Sel}/p . The missing suspension in the printed identity must be retained. The boundary is a map of connective- K -theory modules, so it transports the canonical projective-bundle map

$$(a, b) \longmapsto \pi^* a + [\mathcal{O}(-1)] \pi^* b.$$

Suspension and finite cofibers preserve that map. Thus Lemma 9.31(2) gives the same formula for $L_{\mathrm{cdh}} u_! u^* K/(p, v^N)$. On finite-type $\mathbf{Z}$ -models its underlying theory is bounded below, and the fibres of its v -adic quotients are increasingly connective positive shifts. Their inverse limit is zero, so taking limits gives the formula for $L_{\mathrm{cdh}} u_! u^* K/p$. Finitarity

extends this conclusion to the qcqs domain. No algebra structure on a Moore spectrum is used.

The next local step is Lemma 9.34. Put

$$\Lambda(q) = L_{\text{cdh}} u_! \tau^{\leq q} R\Gamma_{\text{ét}}(-, \mu_p^{\otimes q}).$$

For a henselian rank-one mixed-characteristic valuation ring V containing ζ_p , that lemma proves $\Lambda(q)(\mathbf{P}_V^1) = 0$. The filtration used in its proof is

$$B_q = (L_{\text{cdh}} u_! \text{Fil}_{\text{cdh}}^q KH/p)(\mathbf{P}_V^1), \quad B_0 = 0.$$

The induction must start at $q = 0$, rather than $q > 0$: $\Lambda(-1) = 0$ directly because étale cohomology is nonnegative. With $k_q = H_{\text{ét}}^q(-, \mathbf{Z}/p)$, the truncation and filtration triangles give

$$B_{q+1}[1] \simeq (L_{\text{cdh}} u_! k_q)(\mathbf{P}_V^1)[q].$$

The local filtration bound and $\dim_v \mathbf{P}_V^1 = 2$ put the left side in $D^{\leq -q}$; the right side is in $D^{\geq -q}$. Their common possible group is $H_{\text{cdh}}^0(\mathbf{P}_V^1, u_! k_q)$.

This group is zero by the valuation-point argument: at valuations with noninvertible p the extension by zero vanishes, while the other restrictions factor through the generic projective line. Since V has rank one, $V[1/p] = \text{Frac } V$. Over that characteristic-zero field the source's homotopy-invariant étale-cohomology presheaf with transfers has injective restriction at the zero section. The class has zero restriction from the V -value, so it vanishes at all valuation points. Consequently $B_{q+1} = 0$, including the initially missing B_1 case, and the induction proves the required Λ -vanishing. The occurrences of H^n and k_n in this paragraph of the source mean H^q and k_q .

For the root-of-unity descent of Lemma 9.37(2), when $p \in V^\times$ the correct algebra is

$$V \otimes_{\mathbf{Z}[1/p]} \mathbf{Z}[1/p, \zeta_p].$$

It is finite étale over V , a product of henselian valuation rings, and a C_{p-1} -torsor. Étale descent and averaging over this prime-to- p group give the required canonical fixed-point comparison; for $p = 2$ the group is trivial. There is no $\mathbf{Q}$ -algebra hypothesis on V . The tensor product with $\mathbf{Q}$ in the separate mixed-characteristic fraction-field branch is valid only after passing to that fraction field. The same averaging is compatible with filtered colimits on the p -torsion modules.

These corrected inputs give Lemma 9.38, the projective-bundle formula for $L_{\text{cdh}} u_! u^* F_p^{\text{Kato}}$. To reach all of F_p^{Kato} , insert the localization sequence of Lemma 9.25(1),(2):

$$L_{\text{cdh}} u_! u^* F_p^{\text{Kato}} \longrightarrow F_p^{\text{Kato}} \longrightarrow i_* i^* F_p^{\text{Kato}}, \quad i : \text{Spec } \mathbf{F}_p \hookrightarrow \text{Spec } \mathbf{Z}.$$

The last term has the independent characteristic- p formula of that lemma. The projective-bundle comparison is a natural transformation of exact functors in the Chern-action category; the fibre sequence gives its formula for the middle term with the same maps. Termwise Suslin localization preserves it: $\mathbf{P}_X^1 \times \Delta^r = \mathbf{P}_{X \times \Delta^r}^1$, the first Chern class pulls back, and realization commutes with finite sums and shifts. Finally Example 9.21 supplies the formula for the middle term of the Chern-compatible fibre sequence (9.41),

$$L_{\mathbb{A}^1} \mathbf{Z}/p(*)^{\text{cdh}} \longrightarrow L_{\mathbb{A}^1} L_{\text{cdh}} \mathbf{Z}/p(*)^{\text{syn}} \longrightarrow L_{\mathbb{A}^1} F_p^{\text{Kato}}.$$

Exactness gives the asserted formula for the first term. This is the missing localization inference in the final proof of Theorem 9.13. $\square$

A.3 The Chern normalization and syntomic inputs

Fix a prime p and $m \geq 1$. For the prime-to-characteristic comparison in (Bachmann et al., 2025, arXiv v1, Proposition 6.8, Step 2), the relevant Bockstein kernel over a separably closed field k of characteristic different from p is $K_2(k)/p^m$. Every element of $k^\times$ has a p^m th root, and Matsumoto’s symbolic presentation gives

$$\{c^{p^m}, b\} = p^m \{c, b\}.$$

Thus $K_2(k)$ is p^m -divisible. The sequence

$$0 \longrightarrow K_2(k)/p^m \longrightarrow K_2(k; \mathbf{Z}/p^m) \longrightarrow K_1(k)[p^m] \longrightarrow 0$$

makes the two Bockstein arrows in the comparison diagram isomorphisms. Since $KH = K$ on fields, the same applies to its KH arrow. This corrects the printed invocation of divisibility of $K_1(k)$ in place of the required kernel calculation and preserves the determinant and first-Chern-class normalization of the canonical comparison.

For residue-prime coefficients the comparison used here is always the truncated one,

$$F_p(q) = L_{\text{cdh}} \tau^{\leq q} \mathcal{S}_p(q),$$

as in (Bachmann et al., 2025, arXiv v1, Theorem 7.16). Together with the finite-coefficient pullback of (Bouis, 2024, arXiv v3, Proposition 3.26 and Corollary 3.27), it gives the natural triangle

$$M_p(q) \longrightarrow \mathcal{S}_p(q) \longrightarrow L_{\text{cdh}} \tau^{> q} \mathcal{S}_p(q)$$

of that source’s Theorem 5.10. In particular the full untruncated identification printed in its Theorem 3.13(3) is not used. For a p -torsionfree F-smooth scheme, this triangle and (Bhatt and Mathew, 2023, arXiv v2, Proposition 5.2) give the comparison fibre bound $D^{\geq q+1}$ used in proposition 5.1. The finite-coefficient $\mathbb{A}^1$ comparison follows from the same pullback: after $\mathbb{A}^1$ -localization the BMS term is zero, as in (Bouis, 2024, arXiv v3, Lemmas 6.1 and 6.4 and Theorem 6.5). These uses do not require the full integral condensed completeness theorem.

There is one additional smooth-DVR input in the construction of the canonical syntomic comparison, (Bachmann et al., 2025, arXiv v1, Theorem 5.58). The arbitrary-residue-field branch of (Bhatt and Mathew, 2023, arXiv v2, Example 5.9) uses the local Bloch–Kato residue calculation, rather than dropping the perfect-residue hypothesis of that paper’s separate semistable Theorem 5.8. Bloch–Kato use absolute special-fibre differentials and logarithmic de Rham–Witt groups over the prime field; their local henselian DVR theorem and nearby-cycle calculation impose no perfection condition (Bloch and Kato, 1986, Sections 1.3 and 5, Theorem 5.12, and Sections 6.1–6.6).

More explicitly, at a strict local smooth-DVR ring R the base uniformizer π is prime because R/π is regular local. Hence $R[1/p]^\times = \pi^{\mathbf{Z}} R^\times$. Steinberg relations reduce repeated uniformizer entries to at most one. A principal-unit symbol with that entry lies in the subgroup generated by unit symbols: use $\{1 + \pi a, -\pi a\} = 0$ for a unit a ; for a nonunit a , factor

$$1 + \pi a = \frac{1 + \pi a}{1 + \pi(a+1)} (1 + \pi(a+1)),$$

whose two factors have the required form with unit coefficients. Thus the principal-unit subgroup U^1 lies in the subgroup B generated by unit symbols. Bloch–Kato’s Section 6.6 identifies the quotient of degree- q nearby cycles by B with $W_m \Omega_{\log}^{q-1}$. The uniformizer-symbol projection is the canonical residue; changing π changes it by a unit symbol.

Bhatt–Mathew’s Theorem 1.8 identifies B with the syntomic image, giving precisely the truncated smooth-DVR residue comparison used by BEM. This argument concerns smooth DVR schemes, where π is prime; it is not a proof of the general semistable crossing case.

The remaining foundations in these deductions are the named rational zero-slice and Adams comparisons, the BMS/prismatic construction and rigidity, the characteristic- p cdh-syntomic formula, the Selmer and height-one telescope inputs in BEM’s own Kato argument, and its categorical periodization and cdh bridge. The ordinary- K replacement in [section B](#) addresses a different projective-line argument in Bouis. Neither appendix asserts a filtered Selmer/formal comparison for arbitrary valuation models.

B The finite-coefficient projective-line formula

The projective-line formula used in [lemma 3.3](#) and [theorem 3.4](#) needs its specified first Chern class, since restriction must kill the second summand. We prove the finite-coefficient formula through the ordinary motivic filtration on K -theory. This replaces the final Selmer-filtration argument in ([Bouis, 2025](#), arXiv v1, Proposition 4.15); it does not establish the filtered Selmer comparison asserted in ([Bouis, 2024](#), arXiv v3, Proposition 3.16). Throughout this appendix, p is prime and $m \geq 1$.

Remark B.1 (The filtered source comparison). The all-animated-ring identification in ([Bouis, 2024](#), arXiv v3, Proposition 3.16) is false. Take the smooth $\mathbf{Z}$ -algebra $A = \mathbf{Z}[1/p]$. The p -completed left Kan extension of the étale-sheafified classical filtration restricts to that filtration on A . Its weight-zero graded piece has $H^0 = \mathbf{Z}_p$, or $\mathbf{F}_p$ after reduction modulo p , because the classical weight-zero complex is the constant sheaf on the connected scheme $\mathrm{Spec} A$. In contrast, $\mathrm{Fil}_{\mathrm{BMS}} TC(A; \mathbf{Z}_p) = 0$: it depends only on the derived p -completion of A , which is the zero ring ([Bouis, 2024](#), arXiv v3, Remark 2.16).

The ordinary filtered trace used here is instead the map of ([Bouis, 2024](#), arXiv v3, Theorem 3.15), constructed from ([Annala et al., 2024](#), arXiv v1, Lemmas 6.10–6.11 and Proposition 6.12). Bouis’s Construction 3.17 uses the adjunction counit as a map, without asserting that it is an isomorphism ([Bouis, 2024](#), arXiv v3, Construction 3.17). This map supplies the lisse module and Chern-class construction in the proof below. At A its BMS target is simply zero; existence of that map is compatible with the counterexample. The finite-coefficient argument therefore requires neither the false identification nor a filtered Selmer/formal comparison.

Lemma B.2 (The complete finite-coefficient motivic filtration). *For a qcqs scheme Y put*

$$\mathcal{T}^q(Y) = \mathrm{Fil}_{\mathrm{mot}}^q K(Y)/p^m, \quad \mathcal{M}_q(Y) = \mathbf{Z}/p^m(q)^{\mathrm{mot}}(Y).$$

This decreasing filtration is $\mathbf{N}$ -indexed, with $\mathcal{T}^q = K/p^m$ for $q \leq 0$ and $\mathrm{gr}^q \mathcal{T} = \mathcal{M}_q[2q]$. If $\dim_v Y \leq a < \infty$, then

$$\mathcal{T}^q(Y) \in \mathrm{Sp}_{\geq q-a-2}, \quad \lim_q \mathcal{T}^q(Y) = 0.$$

Proof. The finite-coefficient pullback of ([Bouis, 2024](#), arXiv v3, Proposition 3.26) identifies $\mathcal{T}^q$ with

$$(\mathrm{Fil}_{\mathrm{cdh}}^q KH/p^m) \times_{L_{\mathrm{cdh}}(\mathrm{Fil}_{\mathrm{BMS}}^q TC/p^m)} (\mathrm{Fil}_{\mathrm{BMS}}^q TC/p^m).$$

The three filtrations are $\mathbf{N}$ -indexed, by that source’s Remark 2.22 and Theorem 3.13(1). At $q = 0$ their pullback is the unfiltered trace square for K/p^m . Exactness of reduction and of taking graded pieces gives the stated grades.

On affines the BMS term modulo p is $(q-1)$ -connective, by (Bouis, 2024, arXiv v3, Lemma 2.21). Finite coefficient sequences give the same bound modulo p^m . Zariski and cdh homotopy dimension give the sufficient $(q-a-1)$ bound for its sheafification on Y ; the cdh KH term is $(q-a)$ -connective by (Bachmann et al., 2025, arXiv v1, Theorem 7.12(3)). The fibre description of a pullback gives the stated, possibly nonsharp, $(q-a-2)$ bound. It tends to infinity and proves that the inverse limit is zero. This is the finite-coefficient route singled out in (Bouis, 2024, arXiv v3, Remark 3.29). No integral or rational completeness assertion is needed. $\square$

For a graded theory $\mathcal{B}_*$ equipped with the first Chern class of $\mathcal{O}(1)$, write

$$P_q(\mathcal{B}; Y) = \text{cofib} \left(\mathcal{B}_q(Y) \oplus \mathcal{B}_{q-1}(Y)[-2] \xrightarrow{\pi^* + c_1(\mathcal{O}(1)) \cup \pi^*} \mathcal{B}_q(\mathbf{P}_Y^1) \right).$$

When a comparison preserves this class, taking P_q preserves its fibre and cofiber sequences. This follows by taking cofibers of a natural transformation between exact functors.

Lemma B.3 (The syntomic-tail defect on a valuation ring). *Let V be a rank-one henselian valuation ring of mixed characteristic $(0, p)$. Set*

$$\mathcal{S}_q = \mathbf{Z}/p^m(q)^{\text{syn}}, \quad \mathcal{C}_q = L_{\text{cdh}} \tau^{>q} \mathcal{S}_q, \quad J_q(Y) = R\Gamma_{\text{ét}}(Y, j_{Y!} \mu_{p^m}^{\otimes q}),$$

where $j_Y : Y[1/p] \hookrightarrow Y$. For $q \geq 0$, the canonical projective-line cofiber satisfies

$$P_q(\mathcal{C}; V) \simeq E_q, \quad E_q = (L_{\text{cdh}} \tau^{>q} J_q)(\mathbf{P}_V^1) \in D^{[q+1, q+1]}.$$

Proof. The first half of (Bouis, 2025, arXiv v1, Proposition 4.15) proves the projective-line formula for the cdh-sheafified BMS tails

$$\mathcal{B}_q = L_{\text{cdh}} \tau^{>q} \mathbf{Z}/p^m(q)^{\text{BMS}}$$

at V , with their induced Chern maps. The reduction used there is valid for nondiscrete V as well: apply Remark 4.14, rather than just the DVR wording of Corollary 4.13. Since $\sqrt{pV} = \mathfrak{m}_V$, the reduction is to $\kappa(V)$. The nil ideal $\mathfrak{m}_V/pV$ need not be nilpotent, but it is the filtered union of finitely generated nil ideals, each nilpotent. Finitarity extends nilpotent invariance to this quotient. The remaining inputs are the characteristic- p cdh motivic projective-line formula (Bachmann et al., 2025, arXiv v1, Corollary 8.23) and the characteristic- p cdh-syntomic formula (Elmanto and Morrow, 2026, arXiv v2, Theorem 5.21). Their Chern maps commute with reduction to the residue field.

Lemma 4.8 of Bouis (2025) gives, after truncation and cdh sheafification, the natural triangle from the étale extension-by-zero tail to $\mathcal{C}_q$ and then to $\mathcal{B}_q$. Its stalk argument uses henselian local rings: if p is a unit the BMS term is zero, and otherwise the ring is p -henselian and the syntomic-to-BMS comparison is an equivalence. There is no further valuation hypothesis in this step.

At V , p -henselian rigidity gives $J_q(V) = 0$, so the syntomic and BMS source values agree. Applying P_q to the triangle, the étale-tail source terms are zero, its target is E_q , and $P_q(\mathcal{B}; V) = 0$. This gives the displayed equivalence with the canonical induced map. At $q = 0$ its lower source terms are zero: $\mathcal{M}_{-1} = 0$ by $\mathbf{N}$ -indexing, and (Bhatt and Lurie, 2022, arXiv v1, Remark 8.4.4 and Example 8.4.5) identify negative-weight full syntomic cohomology with J_{-1} . Thus $\mathcal{S}_{-1}(V) = 0$, and the negative BMS term is zero as well.

The lower bound on E_q is its truncation bound. For the upper bound, $\dim_v \mathbf{P}_V^1 = 2$. Hence $(L_{\text{cdh}} \tau^{\leq q} J_q)(\mathbf{P}_V^1)$ lies in $D^{\leq q+2}$. The full value $J_q(\mathbf{P}_V^1)$ is zero: proper projective-line cohomology and the projection formula reduce it to the zero J_q values on V . The

proper-base-change calculation is also part of (Bhatt and Lurie, 2022, arXiv v1, proof of Theorem 9.1.1). The truncation triangle consequently puts E_q in $D^{\leq q+1}$. This is the argument of (Bouis, 2025, arXiv v1, Lemma 4.9). It uses neither a finite residue cohomological dimension nor henselianity of $\mathbf{P}_V^1$ or its affine charts. $\square$

Lemma B.4 (Vanishing of a complete projective-line defect). *Under the hypotheses of lemma B.3, $E_q = 0$ and $P_q(\mathcal{C}; V) = 0$ for every $q \geq 0$.*

Proof. The filtered Chern map is

$$u^q : \mathcal{T}^q(V) \oplus \mathcal{T}^{q-1}(V) \longrightarrow \mathcal{T}^q(\mathbf{P}_V^1).$$

It comes from the filtered lisse connective- K -theory module structure of (Bouis, 2025, arXiv v1, Construction 4.3 and Remark 4.4). Its underlying map is

$$(a, b) \longmapsto \pi^* a + (1 - [\mathcal{O}(-1)])\pi^* b.$$

The basis $\{1, 1 - [\mathcal{O}(-1)]\}$ differs from $\{1, [\mathcal{O}(-1)]\}$ by an integral matrix of determinant -1 . The ordinary K -theory projective-line formula therefore makes u^0 an equivalence, including when $p = 2$. Here $\mathcal{T}^{-1}(V) = K(V)/p^m$, whereas its negative graded piece is zero.

Set $D^q = \text{cofib}(u^q)$. By lemma B.2, D is complete and $D^0 = 0$: both V and $\mathbf{P}_V^1$ have finite valuative dimension, and finite cofibers commute with limits in spectra. On graded pieces the second map is multiplication by $c_1^{\text{mot}}(\mathcal{O}(1))$, with the positive sign fixed by $1 - [\mathcal{O}(-1)]$. Consequently

$$\text{gr}^q D = P_q(\mathcal{M}; V)[2q].$$

The natural triangle

$$\mathcal{M}_q \longrightarrow \mathcal{S}_q \longrightarrow \mathcal{C}_q$$

is (Bouis, 2024, arXiv v3, Theorem 5.10). Its Chern compatibility follows from Construction 5.8 there and (Bachmann et al., 2025, arXiv v1, Remark 5.41), which compares the trace class with the prismatic logarithm. This graded compatibility needs no filtered Selmer identification. Full syntomic projective-line invariance with this same class is (Bhatt and Lurie, 2022, arXiv v1, Theorem 9.1.1). Exactness of P_q and lemma B.3 give

$$\text{gr}^q D \simeq E_q[2q - 1]. \tag{33}$$

The shift is $[2q - 1]$. Since E_q has cohomology only in degree $q + 1$, the grade can have homotopy only in degree $q - 2$.

Inductively suppose $D^q = 0$. The grade triangle identifies $D^{q+1} \simeq \Omega \text{gr}^q D$, whose only possible homotopy group is in degree $q - 3$. For $r \geq q + 1$, the grade $\text{gr}^r D$ is supported in homotopy degree $r - 2 \geq q - 1$. Its groups in degrees $q - 2$ and $q - 3$ are zero, so

$$\pi_{q-3} D^{r+1} \xrightarrow{\sim} \pi_{q-3} D^r.$$

The only possible group of D^{q+1} is thus the inverse limit of these groups. The Milnor exact sequence has the surjection

$$\pi_{q-3}(\lim_r D^r) \longrightarrow \lim_r \pi_{q-3} D^r \longrightarrow 0.$$

Completeness makes its source zero. Hence $D^{q+1} = 0$, $\text{gr}^q D = 0$, and $E_q = 0$ by (33). The induction starts with $D^0 = 0$ and applies to every $q \geq 0$. In particular it rules out a nonzero constant tail invisible to the graded pieces. $\square$

Proposition B.5 (The canonical finite-coefficient formula). *For every qcqs Y , every integer q , and every $m \geq 1$, the map*

$$\mathcal{M}_q(Y) \oplus \mathcal{M}_{q-1}(Y)[-2] \xrightarrow{\pi^* + c_1^{\text{mot}}(\mathcal{O}(1)) \cup \pi^*} \mathcal{M}_q(\mathbf{P}_Y^1)$$

is an equivalence. For every qcqs Y the two restrictions

$$j_Y^*, \text{pr}_Y^* \circ \infty_Y^* : \mathcal{M}_q(\mathbf{P}_Y^1) \longrightarrow \mathcal{M}_q(\mathbf{A}_Y^1)$$

agree, where here j_Y denotes the affine chart.

Proof. The reduction in (Bouis, 2025, arXiv v1, Proposition 4.17) applies to the cofiber $P_q(\mathcal{C}; -)$. Its source is a finitary cdh sheaf with henselian v-excision, using Lemma 4.11 there for nonnegative weights; its projective-line target preserves this property by (Elmanto et al., 2021, arXiv v3, Lemma 3.3.7). Thus the cofiber has the same properties. Over the finite-valuation-dimensional base $\text{Spec } \mathbf{Z}$, the cdh valuation criterion and henselian v-excision, in the form of (Bouis, 2025, arXiv v1, Lemma 4.10), reduce its vanishing to rank-at-most-one henselian valuation rings. The characteristic- p case uses the independent characteristic- p formulas above. The p -inverted case uses the triangle with truncated cdh motivic and full étale values of (Bachmann et al., 2025, arXiv v1, Corollary 7.20) and their projective-line formulas. The mixed-characteristic case is lemma B.4, which holds for all the valuation rings in this reduction, without F-smoothness.

For the exceptional lower summand at $q = 0$, the globalization of (Bhatt and Lurie, 2022, arXiv v1, Example 8.4.5, Variant 8.4.7, and Section 8.4) gives

$$\mathcal{S}_{-1}(Y) = J_{-1}(Y), \quad L_{\text{cdh}} \tau^{>-1} \mathcal{S}_{-1}(Y) = \mathcal{S}_{-1}(Y).$$

The second equality uses the nonnegative finite-coefficient étale extension-by-zero complex and its cdh descent, as supplied by (Bhatt and Mathew, 2021, arXiv v4, Proposition 5.2). Its henselian v-excision follows from the same rigid étale description. Thus the required negative-weight comparison is the identity triangle

$$0 \longrightarrow \mathcal{S}_{-1} \xrightarrow{\text{id}} \mathcal{C}_{-1}.$$

There is no application of Bouis’s Theorem 5.10 in weight -1 . This verifies the weight-zero interface of the reduction.

It follows that $P_q(\mathcal{C}; Y) = 0$ on all qcqs Y . The triangle $\mathcal{M} \rightarrow \mathcal{S} \rightarrow \mathcal{C}$, full syntomic projective-line invariance, and exactness of P_q give the asserted motivic formula. The negative-weight motivic values are zero. The reduction is over $\text{Spec } \mathbf{Z}$; it imposes no dimension bound on the final qcqs scheme Y .

Finally $\mathcal{O}(1)$ becomes trivial on the affine chart and on the infinity section. Both restrictions kill its first Chern class, and both restrict the first summand by projection. This proves the equality of the specified maps. $\square$

C Proper cdh refinements and value-level Milnor excision

The two descent inputs in this appendix have different roles. Proper-then-Nisnevich refinement is used to retain the dimension bound in proposition 4.6. Milnor excision of the intrinsic coefficient values is used in the arbitrary-rank reduction proposition 3.5.

Proposition C.1 (Proper-then-Nisnevich refinement). *Let S be noetherian. Every cdh covering sieve of a separated finite-type S -scheme Y contains a refinement*

$$\{U_i \longrightarrow Y' \longrightarrow Y\},$$

where $Y' \rightarrow Y$ is proper and completely decomposed and $\{U_i \rightarrow Y'\}$ is a finite Nisnevich covering.

Proof. We use the noetherian induction and tower reduction of (Suslin and Voevodsky, 2000, Proposition 5.9, pp. 38–39, and Section 12, p. 67). Section 12 extends this reduction to arbitrary noetherian bases. Its only additional geometric step concerns a tower

$$T \longrightarrow U \longrightarrow Y$$

with $T \rightarrow U$ proper completely decomposed, $U \rightarrow Y$ a finite Nisnevich covering, and Y integral. The connected components of U need not be integral; the following construction uses its irreducible components instead.

The étale Y -scheme U is reduced and noetherian. For each irreducible component U_α , choose a residue-preserving lift of its generic point through $T \rightarrow U$, and let T_α be its reduced closure in T . The map $T_\alpha \rightarrow U_\alpha$ is proper birational and is an isomorphism on a dense open. Removing the component intersections and the exceptional loci gives a closed nowhere dense $E \subset U$ such that

$$T_0 = \left(\coprod_{\alpha} T_{\alpha} \right) \coprod (T \times_U E) \longrightarrow U$$

is proper completely decomposed and is an isomorphism over $U \setminus E$. It refines $T \rightarrow U$.

Every point of U over the generic point of Y is generic in U , since $U \rightarrow Y$ is flat of relative dimension zero. Thus the constructible image of E omits the generic point of Y . Its closure is proper: a dense constructible subset of an irreducible noetherian space contains its generic point. There is a dense open $D \subset Y$ with $T_{0,D} = U_D$.

Apply platication, in its D -admissible form, to $T_0 \rightarrow Y$ (The Stacks Project Authors, 2026, Tags 080X and 087E). It gives a blow-up $Y' \rightarrow Y$ outside D with strict transform T'_0 flat over Y' . The scheme Y' is integral. Put $U' = U \times_Y Y'$. The map $T'_0 \rightarrow U'$ is proper, since its source is a closed subscheme of the base change of the proper map $T_0 \rightarrow U$. It is flat: its graph in $T'_0 \times_{Y'} U'$ is a section of an étale map and hence an open immersion, while projection to U' is a base change of a flat map. It is an isomorphism over the inverse image of D , which meets every irreducible component of U' .

A proper flat finitely presented morphism generically an isomorphism on every component is finite (The Stacks Project Authors, 2026, Tag 0B48). Its algebra is locally free, with locally constant rank one because every component meets the indicated dense open. A unital rank-one algebra is the base ring: the unit is a basis on every fibre, and Nakayama's lemma makes the unit map an isomorphism locally. Consequently $T'_0 \simeq U'$. This supplies a map $U' \rightarrow T$, so the original tower has a Nisnevich refinement after pullback to Y' .

The blow-up need not itself be completely decomposed over its centre. On $Y \setminus D$, use the noetherian induction hypothesis to obtain a proper completely decomposed cover with Nisnevich refinement. Its disjoint union with Y' is a proper completely decomposed cover of Y : Y' supplies the lifts on D , and the added cover supplies them on the complement. The preceding construction and that refinement give the required finite Nisnevich cover. This finishes the source's induction, without excellence, resolution, or an integrality assertion about connected étale components. $\square$

Proposition C.2 (Milnor excision for the intrinsic values). *For every weight q , the canonical value presheaf*

$$Y \longmapsto A(q)(Y) = \text{map}(\mathbf{1}_Y, H_Y(q))$$

satisfies Milnor excision on qcqs schemes. Its finite reductions do as well. For the original localization unit $F_p(q) \rightarrow A_p(q)$, its cofiber $Q_p(q)$ satisfies the same excision; in particular the maps in (15) are the usual restriction maps.

Proof. Retain the Cartesian coefficient over $\text{Spec } \mathbf{Z}$ and the finitary cdh value presheaf provided by (Bachmann et al., 2025, arXiv v1, Theorem 1.1(3), Definition 4.15, and Theorem 9.3(2)), with the comparison argument in section A. For a finite-rank valuation ring V and a prime $\mathfrak{p}$, the geometric argument of (Elmanto et al., 2020, arXiv v2, Sections 3–4, especially Proposition 4.6) gives the canonical restriction square in large **SH**. Its relevant opens are quasi-compact: $\text{Spec } V$ is a finite chain of points. The localization argument used there therefore preserves compact objects on these closed pieces. Only this finite-rank case is required.

Mapping the compatible units to the pulled-back absolute object $H(q)$ in that categorical pullback gives the cartesian square

$$A(q)(V) \simeq A(q)(V_{\mathfrak{p}}) \times_{A(q)(\kappa(\mathfrak{p}))} A(q)(V/\mathfrak{p}).$$

The comparison is induced by restriction, so this assertion identifies the maps as well as the objects.

For arbitrary-rank V over $\mathbf{Z}$, write it as a filtered union of finite-rank valuation subrings V_{α} using (Elmanto et al., 2021, arXiv v3, Lemma 2.3.9). With $\mathfrak{p}_{\alpha} = \mathfrak{p} \cap V_{\alpha}$, finite numerators, denominators, and relations give

$$V_{\mathfrak{p}} = \text{colim}_{\alpha} (V_{\alpha})_{\mathfrak{p}_{\alpha}}, \quad V/\mathfrak{p} = \text{colim}_{\alpha} V_{\alpha}/\mathfrak{p}_{\alpha}, \quad \kappa(\mathfrak{p}) = \text{colim}_{\alpha} \kappa(\mathfrak{p}_{\alpha}).$$

Finitarity and commutation of filtered colimits with finite limits in spectra pass the canonical square to V . This proves henselian v-excision in particular.

Now apply (Elmanto et al., 2021, arXiv v3, Theorem 3.3.4, implication (1) $\Rightarrow$ (2)) with base $\text{Spec } \mathbf{Z}$ and target Sp . The base has finite valuative dimension and the target is compactly generated. This proves Milnor excision of $A(q)$. No cotruncation hypothesis from the different implication (1) $\Rightarrow$ (3) is used.

Finally retain the separate Milnor-excision theorem for $F_p(q)$, (Bachmann et al., 2025, arXiv v1, Theorem 7.21), and its given localization unit to $A_p(q)$. Derived reduction modulo p and pointwise cofibers preserve cartesian squares in a stable category. Naturality of that same unit therefore gives Milnor excision of $Q_p(q)$, with precisely the maps used in (15). The statement in $D(\mathbf{Z}/p)$ is detected by the conservative exact forgetful functor to spectra. $\square$

The value-level argument avoids the unrestricted compact-pushforward sentence on p. 4 of (Elmanto et al., 2020, arXiv v2). Localization alone does not show that every closed immersion of qcqs schemes preserves compact objects when its open complement is not quasi-compact. No assertion at that generality is needed for the finite-rank calculation or its finitary extension above.

D Classical purity, composition, and field presentation

We give the versions of three classical arguments needed here: KGL dévissage, composition of regular-immersion classes, and purity over a perfect field without a finite-type hypothesis. The final subsection records the geometric presentation used over an infinite field, including an imperfect one. These are repairs of the indicated source proofs, rather than author-issued errata.

D.1 The KGL class and dévissage

Proposition D.1 (*KGL purity and the proper-retraction argument*). *For a closed immersion $i : Z \hookrightarrow X$ between regular noetherian schemes of finite dimension, the Cisinski–Déglise class induces the K -theoretic dévissage equivalence and KGL purity. Under the intrinsic fundamental-class comparison, the same holds for $\text{pur}_i(KGL_X)$.*

Proof. In (Cisinski and Déglise, 2019, arXiv v4, Proposition 13.6.1, Step 1), restrict the retraction $r : X \rightarrow Z$ to be proper. Proper pushforward on coherent K -theory then gives $r_* i_* = \text{id}$. On these regular finite-dimensional schemes coherent bounded complexes are perfect, so this is the required K -theory pushforward. Quillen dévissage identifies forgetting supports $K_*^Z(X) \rightarrow K_*(X)$ with i_* ; it is therefore split injective in this case. The projection formula

$$i_*(a) = i_*(1) \cdot r^*(a)$$

then proves the equality of the class and the dévissage map tested in that step. No flatness of r is required for pullback of perfect complexes.

The reduction in Step 2 uses the zero section $Z \hookrightarrow \mathbf{P}(N_i \oplus \mathbf{1})$. Its projection to Z is projective, hence supplies exactly the proper retraction just proved. The deformation scheme is regular: it is the appropriate open in the blow-up of $\mathbf{A}_X^1$ along $0 \times Z$. In local regular-sequence coordinates its chart is

$$R[t, u_1, \dots, u_c]/(tu_h - a_h),$$

with the strict transform $\mathbf{A}_Z^1$ cut out by the u_h . The fibres at $t = 0$ and $t = 1$ give cartesian transverse squares; their Koszul resolutions remain resolutions after pullback. Compatibility of the classes under these pullbacks is (Cisinski and Déglise, 2019, arXiv v4, Theorem 13.5.2 and Remark 13.5.5(2)). Ordinary homotopy invariance and supported dévissage make the two restriction maps isomorphisms. Equality for the projective-bundle section consequently returns first to the middle deformation immersion and then to i .

Paragraph 13.6.2 and Theorem 13.6.3 of that source test this class on smooth generators and their Bott twists. Smooth schemes lift locally across a closed immersion; local covers and descent suffice. The class is therefore a purity equivalence. The comparison with the intrinsic class is the one in (Déglise et al., 2021b, arXiv v3, Paragraph 4.4.3 and footnote 12), with orientation as in (Déglise, 2018, Example 4.1.9(1) and Proposition 4.2.2). These comparisons identify the actual morphism used here. $\square$

Remark D.2 (Purity and forgetting supports). The properness of the retraction matters. For the zero section $i : \text{Spec } k \hookrightarrow \mathbf{A}_k^1$, purity and dévissage give $K_0^{\{0\}}(\mathbf{A}_k^1) = K_0(k) = \mathbf{Z}$. The image of its generator after forgetting supports is $[\mathcal{O}_0]$. The exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{A}_k^1} \xrightarrow{t} \mathcal{O}_{\mathbf{A}_k^1} \longrightarrow \mathcal{O}_0 \longrightarrow 0$$

gives $[\mathcal{O}_0] = 0$ in ordinary K_0 . Thus purity identifies a supported object without asserting injectivity of forgetting supports or generic-point effacement. The scheme projection is a retraction here, but is not proper.

D.2 Strict double deformation and composition

Proposition D.3 (Composition with the canonical normal twists). *Let $Z \xrightarrow{i} Y \xrightarrow{k} X$ be regular closed immersions between regular noetherian schemes of finite dimension. The intrinsic fundamental classes satisfy*

$$\eta_i \cdot \eta_k = \eta_{ki},$$

with the Thom identification induced by

$$0 \longrightarrow N_i \longrightarrow N_{ki} \longrightarrow i^* N_k \longrightarrow 0.$$

Consequently their purity transformations compose with this same identification.

Proof. Use the double deformation with strict centre $D_Z Y \hookrightarrow D_Z X$. This is the centre required in (Déglise et al., 2021b, arXiv v3, Paragraph 3.2.19); it is not the cartesian inverse image $D_Z X \times_X Y$. Locally write $X = \text{Spec } R$, with Y cut out by a regular sequence $a_1, \dots, a_r$ and Z by the further sequence $b_1, \dots, b_e$ modulo (a) . The combined tuple is regular. The first deformation and its strict centre are

$$B = R[t, u_1, \dots, u_r, v_1, \dots, v_e] / (tu_h - a_h, tv_j - b_j), \quad D_Z Y = \text{Spec}(B / (u_1, \dots, u_r)).$$

These are the regular-sequence presentations of the extended Rees algebras. The equations form a regular sequence: away from $t = 0$ they eliminate variables, and at $t = 0$ adjoining t changes them to (t, a, b) . For the centre, adjoining the u_h changes the sequence to $(u, a, tv - b)$. Regularity of a and of b modulo a , and permutability of regular sequences in the relevant noetherian local rings, prove that the u_h are regular in B .

Deform along this centre, using outer parameter s . The coordinate ring is

$$C = R[t, s, w_1, \dots, w_r, v_1, \dots, v_e] / (tsw_h - a_h, tv_j - b_j).$$

The parameter s is a nonzerodivisor by deformation, and t remains a nonzerodivisor on the $s = 0$ fibre by regularity of b modulo a . The axes are transverse. Put $E = N_Z Y$, $F = N_Y X|_Z$, and $N = E \times_Z F$. The correct fibre descriptions are

$$\begin{array}{l|l} s = 0 & D(N_Y X, N_Y X|_Z) \\ t = 0 & D(N_Z X, N_Z Y) \\ s = t = 0 & N \\ st \neq 0 & X \times \mathbf{G}_m^s \times \mathbf{G}_m^t. \end{array}$$

In particular, the first two axis descriptions in the source's table on p. 27 must be exchanged.

These identifications are intrinsic. If the generators change by $a' = Ma$ and $b' = Pb + Qa$, their deformation coordinates change by

$$w' = Mw, \quad v' = Pv + sQw.$$

At $s = 0$ these identify the normal bundle of $D_Z Y$ in $D_Z X$ with the pullback of $N_Y X$. At $t = 0$ they describe the deformation of E in $N_Z X$; at the origin the mixed term vanishes, giving the two canonical projections of N . Thus the construction glues without a chosen splitting of the normal exact sequence.

It remains to compare the specialization maps. In the sphere-valued bivariant formalism, interchanging classes with suspension degrees $-m$ and $-n$ contributes $(-1)^{mn}$ in addition to the specified permutation path of their K -theory Thom twists. This is the mapping-spectrum symmetry rule; the exponent $m + n$ printed in (Déglise et al., 2021b, arXiv

v3, Paragraph 2.2.13) must be replaced by mn . Each specialization symbol $\{s\}$ or $\{t\}$ has degree -1 and zero K -twist, so their interchange has sign -1 . The two localization boundaries anticommute by the stable 3-by-3 boundary rule. The two signs cancel. This proves the commuting specialization square of Lemma 3.2.20 with the corrected axes and their actual normal projections.

The other squares in Theorem 3.2.21(iii) use smooth Gysin compatibility and transverse base change. Their Thom identifications are the path in $K(Z)$ supplied by the normal exact sequence above, with its indicated permutation when needed. Combining those squares gives the asserted canonical composition identity and then the transformation identity of Theorem 3.3.2. No extra motivic symmetry sign is inserted after these Thom paths have been specified. For the pairs used here, take X as the ambient base; all the deformations are finite type over it. No finite-type hypothesis over $\mathbf{Z}$, excellence, or residue-field perfection is added. $\square$

D.3 Affine approximation over a perfect field

Proposition D.4 (Affine transverse approximation of a regular pair). *Let k be perfect, let A be a regular noetherian k -algebra of finite dimension, and let $I = (f_1, \dots, f_c)$ be a regular sequence with A/I regular. The pair $(\operatorname{Spec} A, \operatorname{Spec} A/I)$ is a cofiltered limit of smooth affine k -pairs of codimension c , with affine transition maps and cartesian transverse squares.*

Proof. Since k is perfect, $k \rightarrow A$ is regular. Popescu approximation gives A as a filtered colimit of smooth finitely presented k -algebras. Choose a smooth stage R_0 with lifts of the f_h . Let $\mathcal{D}$ be the category of smooth finitely presented k -algebras under R_0 and over A . Finite diagrams and relations descend along the presentation, so $\mathcal{D}$ is filtered and has colimit A .

For $R \in \mathcal{D}$ let $J = (f_1, \dots, f_c)R$ and $W = \operatorname{Spec}(R/J)$. Delete the closed locus where W is not smooth over k or has codimension different from c . On its smooth locus the codimension is locally constant, since both W and $\operatorname{Spec} R$ are smooth. This locus is therefore closed in $\operatorname{Spec} R$; let U be its complement. The image of $\operatorname{Spec} A$ lies in U . At a point over I , (Déglise et al., 2021a, arXiv v3, Lemma C.7) applies to the local regular rings $R_{\mathfrak{p}} \rightarrow A_{\mathfrak{q}}$ and the same c generators: their cotangent classes map to c independent parameter classes. It follows that $(R/J)_{\mathfrak{p}}$ is regular of codimension c . Perfectness makes this the smooth locus required above.

Choose $h_1, \dots, h_l$ cutting out $\operatorname{Spec} R \setminus U$. Their images generate the unit ideal in A ; write $\sum d_j h_j = 1$ there. The affine algebra

$$R' = R[z_1, \dots, z_l] / (\sum z_j h_j - 1), \quad z_j \mapsto d_j,$$

maps to A and lands in U . On $D(h_j)$ its map to R is an affine-space bundle, obtained by solving for z_j . Thus both R' and R'/JR' are smooth over k , and the closed immersion has codimension c . When $U = \operatorname{Spec} R$, one can simply use $R' = R$.

Let $\mathcal{D}_{\text{good}}$ be the full subcategory of these good smooth affine pairs, always using the chosen tuple from R_0 . Every object of $\mathcal{D}$ maps to a good one. A finite diagram of good objects first has a cocone in $\mathcal{D}$, then a good refinement; parallel arrows can be equalized before that refinement. The same argument in the relevant overcategories proves cofinality. Hence $\mathcal{D}_{\text{good}}$ is filtered with colimit A .

The Koszul complex on the fixed tuple is a resolution along each closed subscheme; off it, it is contractible because the tuple generates the unit ideal. Pullback to another good stage or to A remains a resolution. All the closed-pair squares are therefore cartesian and Tor-independent, with the pulled-back normal bundles. Both members of every pair are

affine, so transition maps are affine. This is the affine refinement needed for the local use of (Déglise et al., 2021a, arXiv v3, Proposition C.8). Removing singular loci alone would not verify that hypothesis on transitions. $\square$

Lemma D.5 (Purity through an affine transverse limit). *Let (X_α, Z_α) be an affine-transition system of noetherian finite-dimensional cartesian transverse regular closed pairs, with noetherian finite-dimensional limit (X, Z) . Let E be a Cartesian absolute coefficient in **SH**. If every stage purity transformation is an equivalence, then so is the limit transformation.*

Proof. Write i_α, i for the closed immersions and $G_\alpha = \mathrm{Th}(-N_{i_\alpha}) \otimes E_{Z_\alpha}$, $G = \mathrm{Th}(-N_i) \otimes E_Z$. For compact $P \in \mathbf{SH}(Z)$, continuity descends it to some compact P_α ; write P_β for its later pullbacks. Closed pushforward preserves compact objects in this noetherian setting (Cisinski and Déglise, 2019, arXiv v4, Proposition 4.2.9). Proper base change identifies i_*P with the pullback of $i_{\alpha*}P_\alpha$. Continuity of maps with compact source, (Déglise et al., 2021a, arXiv v3, Paragraph A.10(1),(2), Definition A.2, and Examples A.4 and A.9), gives

$$\begin{aligned} \mathrm{map}_Z(P, i^!E_X) &\simeq \mathrm{map}_X(i_*P, E_X) \\ &\simeq \mathrm{colim}_{\beta \geq \alpha} \mathrm{map}_{X_\beta}(i_{\beta*}P_\beta, E_{X_\beta}) \\ &\simeq \mathrm{colim}_{\beta \geq \alpha} \mathrm{map}_{Z_\beta}(P_\beta, i_\beta^!E_{X_\beta}). \end{aligned}$$

The analogous colimit computes $\mathrm{map}_Z(P, G)$. Transverse compatibility identifies the purity map between them with the colimit of the stage purity maps. Each is an equivalence, so compact generation proves limit purity. The coefficient E need not be compact, dualizable, or an algebra. In particular, this argument does not use a unit or multiplication on an arbitrary absolute object, as a bivariant-group formulation of the printed Lemma C.4 would require. $\square$

Corollary D.6 (Integral purity over a perfect field). *For a Cartesian absolute coefficient E in **SH**, regular closed pairs of regular noetherian finite-dimensional schemes over a perfect field have intrinsic integral purity. There is no finite-type hypothesis on their structure maps.*

Proof. Purity is Zariski-local on its closed target. Locally a regular pair is affine with ideal generated by a regular sequence of fixed length. Apply proposition D.4. Every stage is a pair smooth over the perfect field. Smooth-pair purity (Déglise et al., 2021b, arXiv v3, Proposition 4.3.10(i)) holds for the pullback of any coefficient over that field. Then lemma D.5 proves the given intrinsic transformation is an equivalence. This proves the first route of (Déglise et al., 2021a, arXiv v3, Appendix C, Theorem C.1) in precisely the scope used here, without its second proof. $\square$

D.4 Geometric presentation over an infinite field

Proposition D.7 (The support presentation). *Let k be infinite, let W be smooth affine irreducible of dimension $d \geq 1$, let $t_1, \dots, t_l$ be points of W , and let $Z \subset W$ be closed of positive codimension. There are $\varphi = (\psi, v) : W \rightarrow \mathbf{A}^{d-1} \times \mathbf{A}^1$, an open $V \subset \mathbf{A}^{d-1}$, and an open $U \subset \psi^{-1}(V)$ containing the marks, such that $Z \cap U = Z \cap \psi^{-1}(V)$ is finite over V , $\varphi|_U$ is étale, and it maps $Z \cap U$ isomorphically to a closed subscheme $T \subset \mathbf{A}_V^1$ with $\varphi^{-1}(T) \cap U = Z \cap U$. The field k need not be perfect.*

Proof. Use the projection and separation construction in (Colliot-Thélène et al., 1997, Theorem 3.2.2 and Sections 3.2–3.4). Lemma 3.2.1 first replaces arbitrary marks by suitable closed specializations and reduces to a divisor by prime avoidance. The generic linear projection makes that divisor finite over $\mathbf{A}^{d-1}$, and the last coordinate separates the relevant finite fibres. The parameter conditions for étaleness and separation define a nonempty open over an algebraic closure. As in Lemma 3.4.3, descend this open through a finite normal extension by intersecting conjugates, then through its purely inseparable part by powers of defining equations. Infinitude of k supplies a rational parameter in the resulting nonempty open. Étaleness and radiciality on the finite set descend faithfully flatly. Their conjunction makes the relative residue extensions trivial: they are both separable and purely inseparable. It does not assert that a marked residue field is separable over k .

We supply the two shrinking steps. Put $A_0 = k[u_1, \dots, u_{d-1}]$, $B = k[Z]$, and let $\mathfrak{a}$ be the intersection of the finitely many marked maximal ideals in A_0 . The map

$$(A_0/\mathfrak{a})[u_d] \longrightarrow B/\mathfrak{a}B$$

is surjective, which is all that is needed in Lemma 3.5.1. Indeed, the corresponding finite-fibre morphism is finite, unramified on the chosen étale locus, and radicial by the finite-set separation. An unramified universally injective morphism has diagonal a surjective open immersion and is a monomorphism; a finite monomorphism is a closed immersion. The cokernel of $A_0[u_d] \rightarrow B$ is finite over A_0 . Nakayama at the marked maximal ideals and prime avoidance give an f outside their union annihilating it. After restricting to $V = D(f)$, the support therefore maps by a closed immersion to $\mathbf{A}_V^1$. The displayed fibre map need not be an isomorphism of rings, contrary to the wording in the source.

Further shrink V away from the finite image of the non-étale locus on the support; this keeps the marks and makes φ étale along the entire support. Write T for this support in $\psi^{-1}(V)$, identified with its image in $\mathbf{A}_V^1$. The étale locus $U_{\text{ét}}$ of φ contains T . The map

$$U_{\text{ét}} \cap \varphi^{-1}(\varphi(T)) \longrightarrow \varphi(T)$$

is étale and its section T is open. Thus T is open in the entire inverse image as well. Its complement there is closed in $\psi^{-1}(V)$, since $\varphi(T)$ is closed. Remove that complement and intersect with $U_{\text{ét}}$. The separation condition in Theorem 3.2.2(4) ensures that none of the marks outside the support is lost. This gives the required U and exact inverse-image identity. It is unnecessary to claim étaleness on unwanted inverse-image components, as in the stronger wording of Lemma 3.6.1. These shrinking arguments finish Theorem 3.1.1 with its original infinite-field scope. $\square$

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